

Suffix-Invariant Programmable PRFs and Applications to Stacked Garbling

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Abstract

Garbled circuits are a fundamental primitive in cryptography. While the size of garbled circuits in Yao’s original scheme grows linearly with the circuit size, a recent line of work on stacked garbling (SGC) [Heath-Kolesnikov, CRYPTO’20] has achieved near-sublinear size for *branching computations*, based only on one-way functions. Specifically, these schemes achieve garbled size growing only with the size of a single branch and the *total input length* to all the branches. Due to the latter dependence, these results are best suited to “small” input settings.

We present a stacked garbling scheme for “large” input settings based on one-way functions. The garbled size in our scheme grows only with the size of a single branch and its input length (up to logarithmic factors), plus an additive term in the number of branches (as in prior SGC).

To obtain our result, we uncover a connection between stacked garbling and the notion of (adaptive) programmable pseudorandom functions (apPRFs) [Boneh-Lewi-Wu, PKC’17]. While existing apPRF constructions either rely on stronger assumptions (e.g., learning with errors or indistinguishability obfuscation) or incur noticeable security losses under weaker assumptions, we identify a relaxed notion of suffix-invariant programmable PRFs (sipPRFs) that suffices for our result, and establish its feasibility based on OWFs. Interestingly, we build on techniques from the SGC literature to construct sipPRFs with our desired efficiency, and then apply sipPRFs back to SGC to obtain our main result.

Along the way, as an additional result of independent interest, we also provide the first construction of (adaptive) programmable PRFs for polynomial-size domains based on one-way functions.

Keywords: Garbled Circuit · Stacked Garbling · Programmable PRF

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Contents

1	Introduction	1
1.1	Our Results	1
1.2	Related Work	3
2	Technical Overview	4
2.1	Suffix-Invariant Programmable PRFs (sipPRFs)	4
2.2	Stacked Garbling: A Concise Overview	6
2.3	Demultiplexers from Programmable PRFs for Stacked Garbling	6
2.4	Black-Box Connections Between napPRFs and Two-Party DPFs	8
2.5	Constructing Short-Key Offline-Online sipPRFs	9
3	Preliminaries	13
3.1	Notation	13
3.2	Distributed Point Functions	13
3.3	Pseudorandom Functions	14
3.4	Programmable PRFs	14
4	Definitions of Suffix-Invariant Programmable PRFs	15
4.1	Baseline Non-Adaptive Programmability	15
4.2	Non-Adaptive vs. Adaptive Programmable PRFs	17
4.3	Suffix-Invariant Programmable PRFs	17
5	Relationship Between napPRFs and Two-Party DPFs	18
5.1	Individual Pseudorandomness for DPFs	18
5.2	napPRF $\Rightarrow$ DPF	18
5.3	DPF $\Rightarrow$ napPRF	19
5.4	Programmable DPF $\Rightarrow$ apPRF	20
6	Toward Short-Key Offline-Online sipPRFs	21
6.1	Garbling-Based Short-Key Compiler	21
6.2	Short-Key sipPRFs via Recursively Stacked Offline Keys	22
7	Improving SGC via Short-Key Offline-Online sipPRFs	26
7.1	Off-the-Shelf Use of Our Offline-Online sipPRF	26
7.2	Tailoring Our Offline-Online sipPRF for Better Integration	28
A	Pseudorandomness of Master Evaluation Procedures	34
A.1	Superpolynomial-Size Domains	34
A.2	Polynomial-Size Domains	35
B	Deferred Proofs	35
B.1	Proof of Lemma 2	35
B.2	Proof of Theorem 5	38
B.3	Proof of Theorem 6	39
B.4	Proof of Theorem 7	40

B.5	Proof of Theorem 8	40
C	Formalism of Garbling-Based Short-Key Compiler	45
C.1	Garbling Schemes	45
C.2	Formalism	46
D	Tailored Offline-Online sipPRF for LogStack	47
E	The Improved LogStack Garbling Scheme	50

1 Introduction

Garbled circuits (GCs) [Yao86] are a fundamental primitive in cryptography. In Yao’s classical construction based on one-way functions (OWFs), given a Boolean circuit $\mathcal{C} : \{0, 1\}^I \rightarrow \{0, 1\}$, a *garbler* can generate a garbled circuit $\tilde{\mathcal{C}}$ together with short input encodings that associate *garbled labels* with each input wire. Given $\tilde{\mathcal{C}}$ and the labels corresponding to an input $x \in \{0, 1\}^I$, an *evaluator* can compute an output label that decodes to $\mathcal{C}(x)$, while learning nothing beyond $\mathcal{C}(x)$. Standard instantiations of GCs have size $\mathcal{O}(|\mathcal{C}| \cdot \lambda)$ and evaluation uses I λ -bit input labels, where λ is the security parameter.

While GCs provide a generic mechanism for securely evaluating *circuits*, many real-world computations are more naturally expressed as programs with data-dependent control flow. Among these, a particularly notable class is *branching computation*, where an intermediate step computes an n -bit selector $\alpha \in \{0, 1\}^n$ and then executes one out of $B = 2^n$ candidate subcircuits (or branches). Concretely, let $\mathcal{C}_0, \dots, \mathcal{C}_{B-1}$ be subcircuits, each of size C and with a common interface $\mathcal{C}_j : \{0, 1\}^I \rightarrow \{0, 1\}$, and consider computing $y \leftarrow \mathcal{C}_\alpha(x)$. A naïve approach evaluates all B candidate circuits, then uses α to *multiplex* (i.e., select) the output of circuit $\mathcal{C}_\alpha$. When garbled, this computation has size $\mathcal{O}(BC\lambda)$ bits.

Stacked garbling (SGC) [HK20a, HK21] provides smaller garbled encodings for this problem. At a high level, SGC observes that the garbler can *stack* — by XOR-combining corresponding components — the B garblings of $\mathcal{C}_0, \dots, \mathcal{C}_{B-1}$ into a single garbled artifact. The evaluator can then *unstack* locally once it learns (obliviously) the randomness used for all but the active (i.e., the α -th) branch. Building on this observation, [HK20a] achieves garbled size $\mathcal{O}(BI\lambda + C\lambda)$ at the cost of $\mathcal{O}(B^2)$ computation. The improved scheme LogStack [HK21] retains the same size/communication profile, while reducing party computation to $\mathcal{O}(B \log B)$.

SGC eliminates the *branch-duplication* blowup — stacking reduces the garbled size for B branches from $\mathcal{O}(BC\lambda)$ to $\mathcal{O}(C\lambda)$ — but incurs an additive $\mathcal{O}(BI\lambda)$ term, intuitively the cost of *privately routing* an I -bit input to the selected branch. Consequently, SGC is most beneficial when each branch performs substantial computation relative to its input length. In contrast, when I is comparable to (or larger than¹) C , the improvement over the naïve approach becomes marginal and may even disappear.

This regime is indeed common in practice: many systems implement a large *dispatch* in which an n -bit “mode” selector α chooses one of 2^n simple behaviors, and the selected branch then performs only light processing on a large input. Notable examples include network packet filtering, where α selects one of many filtering rules and the system outputs accept/drop based on a few header fields while the payload may be large. Similarly, in ML inference pipelines, a model index α selects one of many lightweight prediction modules (e.g., a small linear classifier) applied to a large embedding.

This leaves open a natural question:

Can we retain SGC’s $\mathcal{O}(C\lambda)$ benefit while decoupling the dependence on B from the input length I ?

1.1 Our Results

We answer this affirmatively. Within the standard OWF-based (Yao) garbling framework, we give a new SGC scheme with garbled size $\mathcal{O}(I \log B \cdot \lambda + B\lambda + C\lambda)$ and party computation $\mathcal{O}(B \log B)$. This asymptotically improves the prior state of the art LogStack [HK21], which has size $\mathcal{O}(BI\lambda + C\lambda)$ with the same computation.

¹This is possible because different branches may receive different subsets of the inputs.

We remark that this improvement is practically meaningful. The dominant concrete communication term in LogStack is $4BI\lambda$ bits, whereas ours is $8I \log B \cdot \lambda$ bits, up to lower-order terms. Thus, the two schemes are comparable for $B = 2, 4$, while ours is strictly smaller for every $B \geq 8$ and all I .

Theorem 1 (Improved SGC, informal). *Let $\mathcal{C}_0, \dots, \mathcal{C}_{B-1}$ be Boolean circuits of size C with a common interface $\{0, 1\}^I \rightarrow \{0, 1\}$. There exists an SGC scheme for computing $y \leftarrow \mathcal{C}_\alpha(x)$ from (black-box) OWFs whose garbled size is $\mathcal{O}(I \log B \cdot \lambda + B\lambda + C\lambda)$, and computation of each party is $\mathcal{O}(B \log B)$.*

Our improvement stems from a new perspective on the $\mathcal{O}(BI\lambda)$ bottleneck in SGC. We show that the relevant *demultiplexing* gadget (see next) can be cast as a one-point “programming” task closely related to *programmable PRFs* [BLW17, BKM17, CC17, PS18], and that the SGC setting only needs a weaker variant. This motivates a new primitive — *suffix-invariant programmable PRFs* (*sipPRFs*) — which we formalize, construct from black-box OWFs, and use to optimize SGC. We sketch this connection next, and defer a more detailed overview to Section 2.

Demultiplexing as one-point programming. The $\mathcal{O}(BI\lambda)$ overhead in prior SGC schemes comes from the *demultiplexer*: a garbled gadget that, on a hidden selector $\alpha \in \{0, 1\}^n$, must provide the evaluator with the I input-wire labels for branch α , while producing *dummy* labels for every $t \neq \alpha$. These dummy labels must (i) be independent of the real input x (so G can precompute the inactive-branch behavior) and (ii) be indistinguishable from real GC labels (so the active branch remains hidden). Existing constructions satisfy these constraints by essentially hard-coding B branch outcomes, incurring $\mathcal{O}(BI\lambda)$ size.

Suffix-invariant programmable PRFs (sipPRFs). We observe that demultiplexing is naturally a *one-point programming* task, reminiscent of (adaptive) *programmable PRFs* (apPRFs) [BLW17, BKM17, CC17, PS18]. An apPRF consists of a PRF family $F_{\text{msk}}(\cdot)$ together with a *programming* procedure that, given $(\text{msk}, \alpha, \beta)$, outputs a *programmed key* $\text{sk}_{\alpha \mapsto \beta}$ defining a related function $F_{\text{sk}_{\alpha \mapsto \beta}}(\cdot)$. This function satisfies $F_{\text{sk}_{\alpha \mapsto \beta}}(\alpha) = \beta$, while for every $t \neq \alpha$ we have $F_{\text{sk}_{\alpha \mapsto \beta}}(t) \equiv F_{\text{msk}}(t)$, i.e., it behaves like the underlying PRF off the programmed point. Moreover, when β is uniform, the programmed key should hide the programmed index α (*index-hiding*). In the SGC setting, we take the payload to be the active-branch input-label vector, and the off-point requirement corresponds to outputs that are independent of β and indistinguishable from real GC labels.

Unfortunately, known index-hiding apPRFs rely on comparatively strong assumptions (e.g., LWE [CC17, BTVW17, PS18] or $i\mathcal{O}$ [BLW17]), and their programming procedure is not obviously GC-friendly. For SGC, however, full adaptive programmability is unnecessary: it suffices that off-point behavior is independent of β (though it may depend on α), and that α is hidden when β is uniform. Moreover, to preserve the computational efficiency of LogStack, the required programmable PRF must satisfy a structural restriction that we call *suffix invariance*. For an unprogrammed query $t \neq \alpha$, suffix invariance requires the output at t to be independent of — i.e., invariant under changes to — $\alpha_{\ell+1..n}$, where ℓ is the first bit where t and α differ. Thus, for each fixed t , there are at most $\log B$ possible off-point outputs as α varies, rather than B ; see Section 2.1 for more details. This structure lets the garbler precompute inactive-branch behavior efficiently. This motivates sipPRFs, which admit efficient OWF-based realizations and GC-friendly offline-online forms, and which we use to construct SGC demultiplexers.

Our technical contribution. Motivated by the SGC application, we systematically study the sipPRF primitive. Our main technical contribution includes:

- **A new primitive: sipPRF.** We formalize suffix-invariant programmable PRFs (sipPRFs), a weakening of index-hiding apPRFs tailored to SGC. Concretely, sipPRFs capture single-point programming with β -independent off-point behavior and index hiding under uniform programming, while additionally satisfying suffix invariance, a correctness property that bounds the set of possible outputs on unprogrammed points and is needed to preserve LogStack’s computational efficiency.

We also define an offline-online variant that decomposes the programmed key into an (α, β) -independent offline key (which can be used as garbled material) and a short online key produced by the (garbled) programming procedure.

- **Short online keys from black-box OWFs.** For polynomial-size domains, we construct offline-online sipPRFs from black-box OWFs with short online keys. Interestingly, our constructions are inspired by SGC: the offline key is stackable, and the online key consists solely of standard Yao garbled labels encoding (α, β) , making the programming procedure inherently GC-friendly.

Theorem 2 (Short-key offline-online sipPRFs, informal). *Assuming one-way functions via black-box use, there exists a correct and index-hiding offline-online sipPRF over $\{0, 1\}^n \rightarrow \{0, 1\}^m$ for $2^n \leq \text{poly}(\lambda)$. Moreover, its offline programmed key has length $8nm\lambda + 4n^2\lambda - 2n\lambda + m$ bits, and its online programmed key has length $(n + m)\lambda$ bits.*

- **Improved stacked garbling.** We instantiate LogStack’s demultiplexer using our offline-online sipPRFs, obtaining garbled size $\mathcal{O}(I \log B \cdot \lambda + B\lambda + C\lambda)$ while preserving LogStack’s $\mathcal{O}(B \log B)$ computation (cf. Theorem 1).
- **apPRFs from black-box OWFs.** As an additional result of independent interest, we show that *programmable* DPFs [BGIK22b] imply *adaptive* programmable PRFs; this yields the first OWF-based apPRF construction (for polynomial-size domains). In contrast, prior apPRF constructions either target superpolynomial domains under LWE [CC17, BTVW17, PS18] or *iO* [BLW17], or are OWF-based for polynomial domains but incur a $1/\text{poly}(\lambda)$ security loss [GZSP25].

Theorem 3 (Programmable DPF $\Rightarrow$ apPRF, informal). *Assuming one-way functions (via black-box use), there exists a correct and index-hiding apPRF over $\{0, 1\}^n \rightarrow \{0, 1\}^m$ for $2^n \leq \text{poly}(\lambda)$.*

1.2 Related Work

Stacked garbling. The technique of *stacking* for oblivious branching was introduced in [HK20b] in the GC-ZK [JKO13] setting (i.e., privacy-free garbling), and later extended to standard Yao garbling in [HK20a, HK21]. It has since been applied and extended in several settings: [HKP20, GHAHJ22] extend it to MPC branching; [HHK⁺22, NK25] use it to build garbled PIR; [HKN24] uses it for garbled lookup tables; and [YPHK23] applies it to build a constant-round 2PC system for RAM programs. We advance stacked garbling by reducing the dominant garbled-size overhead that scales with the product of the number of branches and the per-branch input length.

Programmable PRFs. Programmable PRFs augment a standard PRF with a *one-point programming* capability: from a key, one can derive a programmed key whose evaluation at a designated index α is forced to a chosen value β , while evaluations on all other inputs remain unchanged [BW13, BGI14]. A central refinement is *index hiding* (also called *programming privacy*, or privately programmable PRFs, PPPRFs), which requires that the programmed key hide the programmed point α as long as the programmed

value β is uniform [BLW17, BKM17, CC17, PS18]. Without index hiding, programmable PRFs follow easily from one-way functions via the classical puncturable PRF, essentially by embedding (α, β) into the punctured key [BW13, BGI14, GGM86], yielding simple and concretely efficient schemes. In contrast, with programming privacy, known constructions rely on substantially stronger assumptions — notably lattice-based assumptions such as LWE [CC17, BTVW17, PS18] or indistinguishability obfuscation [BLW17] — and are considerably more involved, typically layering homomorphic-encryption machinery. In this work, we use “(adaptive) programmable PRF” to mean an index-hiding construction.

A recent attempt to obtain an OWF-based index-hiding construction for polynomial-size domains appears in [GZSP25], but incurs a $1/\text{poly}(\lambda)$ security loss. While our focus is on suffix-invariant and non-adaptive programmable PRFs, we also obtain the first *adaptive* OWF-based construction for polynomial-size domains *without* this inverse-polynomial loss.

Recently, Boudgoust and Simkin [BS24] studied a related notion of non-adaptive programmable PRFs, which they refer to as NAPs, and established their feasibility from black-box use of OWFs. Our notion of sipPRFs is also inherently non-adaptive, and thus weaker than adaptive programmable PRFs. A key difference between sipPRFs and NAPs is suffix invariance, namely, on unprogrammed points, the output is restricted by the first position where the queried point differs from the programmed point. This property is crucial to achieving our main result on stacked garbling. For a more detailed discussion on the differences between the two notions, see Section 2.1.

Distributed point functions. The first two-party DPF construction from one-way functions was given by Gilboa and Ishai [GI14], with subsequent works improving efficiency and concrete performance [BGI15, BGI16, BGH⁺25]. In this work, we use classical OWF-based two-party DPF constructions [GI14, BGI15] as black boxes.

Beyond the two-party setting, recent work [GWW25] gives the first OWF-based multiparty DPF construction. Earlier works from public-key assumptions (e.g., DCR and LPN) achieve succinct communication for a fixed number of parties [CK24], later extended to arbitrarily many parties [ARS24, CKY25]. Separately, [BGIK22a] studies information-theoretic DPFs for three and four parties tolerating one corruption.

2 Technical Overview

We present our techniques in sufficient detail to understand our contribution. Let $\mathcal{X} := \{0, 1\}^n$ denote a domain and let $\mathcal{Y} := \{0, 1\}^m$ denote a range. We often refer to elements $\alpha \in \mathcal{X}$ as *indexes* or *points*, and to elements $\beta \in \mathcal{Y}$ as *values*.

2.1 Suffix-Invariant Programmable PRFs (sipPRFs)

A sipPRF consists of three PPT procedures:

- KeyGen samples a master key msk ;
- Prog takes as input a master key msk , an index $\alpha \in \mathcal{X}$, and a value $\beta \in \mathcal{Y}$. It outputs a programmed key sk on input $(\text{msk}, \alpha, \beta)$, where intuitively sk ’s index α is programmed to output β .
- Eval takes as input a programmed key sk , a point $x \in \mathcal{X}$ and outputs a value.

It satisfies the following informal correctness and security guarantees:

- **On-point correctness:** $\text{Eval}(\text{sk}, \alpha) = \beta$, i.e., the evaluation at α outputs β .
- **Off-point value independence (correctness):** For every unprogrammed point $x \neq \alpha$, all β_1, β_2 , $\text{Eval}(\text{Prog}(\text{msk}, \alpha, \beta_1), x) = \text{Eval}(\text{Prog}(\text{msk}, \alpha, \beta_2), x)$, i.e., at $x \neq \alpha$, the evaluation output is fixed by α (and msk), but not by β .
- **Suffix invariance (correctness):** For any $x \neq \alpha$, let ℓ be the first position where x and α differ. Then $\text{Eval}(\text{Prog}(\text{msk}, \alpha, \beta), x)$ depends on α only through $\alpha_{1..\ell}$, and is invariant under changes to the suffix $\alpha_{\ell+1..n}$.
- **Index-hiding security:** For uniform β , the programmed key sk hides α .

We will later prove that these guarantees already imply that for any fixed α and uniform β , the induced function $x \mapsto \text{Eval}(\text{Prog}(\text{msk}, \alpha, \beta), x)$ is a PRF.

Suffix invariance is a structural correctness property needed for stacked garbling applications (cf. Section 2.3). For any fixed x , as α ranges over $\{0, 1\}^n \setminus \{x\}$, it guarantees only n possible off-point outputs, one per first-difference position ℓ , rather than $2^n - 1$. For instance, when $n = 3$ and $x = 000$, the three cases are $\alpha \in \{001\}, \{010, 011\}$, and $\{100, 101, 110, 111\}$.

Offline-online sipPRF. We consider an *offline-online* variant of sipPRFs, where the programmed key sk splits as $(\text{sk}_{\text{off}}, \text{sk}_{\text{on}})$. Here the offline key sk_{off} is sampled by KeyGen and is independent of (α, β) , while the online key sk_{on} is the (α, β) -dependent part produced by Prog . That is, $\text{KeyGen}(1^\lambda)$ outputs $(\text{msk}, \text{sk}_{\text{off}})$, $\text{Prog}(\text{msk}, \alpha, \beta)$ outputs sk_{on} , and Eval takes $\text{sk}_{\text{off}}, \text{sk}_{\text{on}}, x$ as inputs.

This variant's correctness and security notions carry over verbatim. This split is w.l.o.g.: any sipPRF is offline-online by setting $\text{sk}_{\text{off}} = \perp$, and any offline-online sipPRF is a standard sipPRF by folding sk_{off} into msk .

The main motivation for considering this variant is online efficiency: sk_{off} , which may be large, can be transmitted before (α, β) is known. Then in the online phase, once (α, β) is fixed, only the short key sk_{on} needs to be transmitted.

Key metrics. We measure efficiency mainly by (i) the total programmed-key size $|\text{sk}|$ and (ii) the online part $|\text{sk}_{\text{on}}|$. We require $|\text{sk}| \leq \text{poly}(n, m, \lambda)$, and call an offline-online sipPRF *short-key* if $|\text{sk}_{\text{on}}|$ is linear in $n+m$ (up to $\text{poly}(\lambda)$ factors).

Relation to napPRFs and NAPs. For modularity of presentation, we use *non-adaptive programmable PRF* (napPRF) to denote the baseline notion obtained from the sipPRF definition above by dropping suffix invariance. Thus, sipPRFs strengthen napPRFs with suffix invariance, while the offline-online and short-key refinements carry over directly to napPRFs.

This baseline napPRF notion follows the same non-adaptive programming intuition as the NAPs of Boudgoust and Simkin [BS24], but uses a slightly different syntax. In their formulation, the programming point is fixed during key generation: the master key is generated as $\text{KeyGen}(1^\lambda, \alpha)$, and the programming algorithm later takes only the programmed value β as input. In our formulation, KeyGen samples an α -independent master key, while Prog takes both α and β and outputs the programmed key. This syntax is more convenient for our garbling applications, where the programming procedure is itself garbled and therefore naturally receives (α, β) as input. Conceptually, however, the underlying non-adaptive guarantee is the same: the programmed key hides α when β is uniform, while evaluations away from α are unaffected by the choice of β .

Difference from adaptive programmable PRFs. An apPRF (see, e.g., [PS18, GZSP25]) has a *single* master function $F_{\text{msk}}(x)$, fixed by msk , that programming must preserve: for any programmed point (α, β) , evaluation agrees with F_{msk} on all $x \neq \alpha$. In contrast, napPRFs only guarantee β -*invariance* off the programmed point: for fixed α , changing β does not affect outputs on $x \neq \alpha$, but those off-point outputs may still depend on α . SipPRFs strengthen this guarantee with suffix invariance, but still do not require full cross- α consistency with a single master function. Thus, apPRF correctness enforces cross- α consistency, while generic napPRFs and sipPRFs need not. In particular, apPRFs satisfy the sipPRF requirements, while sipPRFs satisfy the napPRF requirements.

See Section 4 for formal definitions.

2.2 Stacked Garbling: A Concise Overview

Stacked garbling (SGC) [HK20a, HK21] optimizes *oblivious branching*: a computation selects one of $B = 2^n$ subcircuits $\mathcal{C}_0, \dots, \mathcal{C}_{B-1}$ (with $B \leq \text{poly}(\lambda)$), where each $\mathcal{C}_j : \{0, 1\}^I \rightarrow \{0, 1\}$ has C gates. We consider subcircuits with one-bit output for simplicity. The evaluator E holds labels encoding a selector $\alpha \in \{0, 1\}^n$ and input labels encoding $x \in \{0, 1\}^I$ (possibly arising as intermediate values), and must get the output label for $\mathcal{C}_\alpha(x)$. Naïvely garbling all branches costs $\mathcal{O}(BC\lambda)$.

Heath and Kolesnikov [HK20a] show that the garbler G can *stack* the B garblings into a single artifact by simply XORing them; later, E can *unstack* locally once it obviously learns seed randomness for all but the active branch $\mathcal{C}_\alpha$. This yields garbled size $\mathcal{O}(BI\lambda + C\lambda)$, with the computation improved to $\mathcal{O}(B \log B)$ in [HK21].

The $\mathcal{O}(BI\lambda)$ term comes from a garbling gadget called the **demultiplexer**. Unstacking produces B candidate branch GCs, only one of which (the α -th) is well-formed, and E cannot tell which. To evaluate candidate branch t , E must also obtain the corresponding I input-wire labels for that branch. The demultiplexer provides these labels: on the active branch it outputs the real labels encoding x , and on inactive branches it outputs *dummy* labels.

These dummy labels must satisfy two constraints: (i) they must be independent of x (since G does not know x but must precompute/merge the inactive-branch behavior), and (ii) they must be indistinguishable from fresh GC labels, so E learns nothing about which branch is active. Prior schemes implement this by naïvely hard-coding B branch outcomes, each consisting of $\mathcal{O}(I)$ λ -bit labels.

2.3 Demultiplexers from Programmable PRFs for Stacked Garbling

We outline how a programmable PRF yields the *demultiplexer* gadget at the heart of SGC. Conceptually, the demultiplexer enables E to derive a candidate input-wire label vector for *every* branch. On the *active* branch, this candidate must equal the *true* input-wire labels consistent with the garbled input; on every *inactive* branch, it should look like an unrelated dummy label vector, independent of the garbled input and indistinguishable from the active one.

- To start with, a direct approach — used by [HK20a, HK21] — would communicate input-wire labels separately for each branch (e.g., via a per-branch garbled selection table). With $B = 2^n$ branches and input length I , this incurs $\Theta(BI)$ overhead — precisely the cost we seek to remove.
- **A tempting but insufficient compression attempt.** One might hope to generate all dummy label vectors from a short PRF key, so E can derive them locally. However, this runs into an immediate obstacle: on the active branch, the correct labels necessarily depend on the garbled input, so they cannot be fixed

independently of the input. Still, this attempt is instructive: on every inactive branch, the PRF output looks pseudorandom (hence indistinguishable from the active branch) and is independent of the garbled input.

- **Programmability resolves the obstacle.** We instead use a *programmable* PRF: we generate a key that is “patched” at exactly one point so that its output equals the *correct* input-wire labels on the active branch, while its outputs on all other branches remain pseudorandom. Crucially, the programmed key hides the patched point and hence does not reveal the active branch identity.
- **Non-adaptive programmability is sufficient.** At this point, a full programmable PRF would impose more structure than SGC needs. In SGC, inactive-branch label vectors can depend on *which* branch is active; we only require that they be independent of the garbled input. That is, once the active branch is fixed, changing the programmed output should not affect the PRF output on any other branch — this is exactly the baseline non-adaptive guarantee underlying sipPRFs. We will explain shortly why suffix invariance is crucial for efficiency.
- **Garbled programmer gadget.** Programming requires (a) the identity of the active branch and (b) the input-wire labels for that branch. In our setting, neither party can simply write these values down in the clear; they are available only as garbled labels. Our solution is to realize the programming procedure itself as a small GC (a “programmer gadget”) that takes the selector labels together with the input-wire labels and outputs a programmed key.
- **Offline-online sipPRF.** In particular, we use an *offline-online* sipPRF so that the programmed key decomposes into an *offline* part that is independent of both the active branch and the input, and a short *online* part that depends on them. The garbled programmer gadget outputs only the online part, while the offline part can be sent in the clear. This is strictly beneficial: outputting a long offline key from a garbled circuit would bloat the gadget (one output wire per bit), yet hiding it buys no security.
- **Two remaining challenges.** We have two further major challenges.
 1. Garbling the sipPRF programming procedure would potentially lead to non-black-box use of the underlying cryptographic primitives (including OWFs): something we are trying to avoid.
 2. At first glance, the programmer gadget seems to need access to the input-wire labels of GC for *every* branch, since it is not known in advance which branch will be programmed. Storing these labels explicitly would make the gadget (and hence the garbling) too large, essentially reintroducing a $\Theta(BI)$ cost. A natural workaround is to generate all per-branch label vectors from a short PRG seed, and have the gadget store only this seed. However, this pushes the burden into the gadget: it must now *expand* the PRG (and thus rely on its internal structure) inside the garbled circuit, which is exactly the kind of non-black-box use we seek to avoid.

We next preview how to address the remaining challenges; but even before doing so, if Prog has small Boolean circuit complexity (polynomial in n, m, λ , where $n = \log B$ and $m = I\lambda$ here), this blueprint yields an SGC demultiplexer of garbled size $\text{poly}(\log B, I, \lambda)$, already improving over prior $\mathcal{O}(BI\lambda)$ -size demultiplexers.

Resolving the challenges via our offline-online sipPRFs. In this work, we construct a *short-key* offline-online sipPRF over a polynomial-size domain from OWFs in a black-box way. In our construction,

Prog essentially selects among $n + m$ standard Yao label pairs, so Challenge 1 can be handled using standard garbled truth tables (black-box in OWFs). Plugging this sipPRF into the blueprint yields an SGC demultiplexer of garbled size $\mathcal{O}(nI \cdot \text{poly}(\lambda) + n^2\lambda)$, and total SGC size $\mathcal{O}(nI \cdot \text{poly}(\lambda) + B\lambda + C\lambda)$, where the $\mathcal{O}(B\lambda)$ term is inherited from SGC’s oblivious seed-delivery gadget. We overview this sipPRF construction in Section 2.5.

We further tailor this offline-online sipPRF to SGC by *coupling* its internal GGM/PRG expansions with SGC’s label generation, so the programmed value need not be materialized as an explicit $I\lambda$ -bit string inside a garbled circuit. That is, the tailored sipPRF directly outputs GC labels (e.g., when $I = 1$, programming $(\alpha, 0)$ makes $\text{Eval}(\text{sk}, \alpha)$ output the *actual* SGC λ -bit wire label encoding 0 on branch α , and similarly for 1). This makes Challenge 2 fully black-box with respect to OWFs as well, yielding an SGC demultiplexer of garbled size $\mathcal{O}(nI\lambda + B\lambda)$ and total SGC size $\mathcal{O}(nI\lambda + B\lambda + C\lambda)$.

Toward preserving state-of-the-art SGC computation complexity. Beyond reducing garbled size, we also preserve LogStack’s $\mathcal{O}(B \log B)$ computation for both parties [HK21]. This relies on our offline-online sipPRFs as follows.

- **Garbler.** The garbler runs KeyGen, incurring $\mathcal{O}(B)$ computation.
- **Evaluator.** The evaluator requires outputs for all $t \in \{0, 1\}^n$; rather than $B = 2^n$ separate calls to Eval, our sipPRFs support a FullEval procedure that evaluates the sipPRF on the entire domain in $\mathcal{O}(B \log B)$ time.
- **Suffix invariance and garbage collection.** We remark that suffix invariance is crucial for preserving LogStack’s $\mathcal{O}(B \log B)$ computation. After E evaluates all B candidate branch garbled circuits, it obtains one candidate output label per branch: one real output label and $B - 1$ garbage output labels. The garbler G must then help merge these B candidate labels into a single consistent output label, without revealing which branch is active; this step is known as *garbage collection*.

The cost of garbage collection is governed by how many different garbage outputs a fixed branch $t \in \{0, 1\}^n$ may produce as the active branch α varies. Without suffix invariance, the dummy input labels for branch t may depend on the full identity of α , yielding up to $B - 1$ possible garbage inputs and hence up to $\mathcal{O}(B)$ possible garbage outputs for that branch. This would force G to account for $\mathcal{O}(B)$ possibilities per branch, leading to total work $\mathcal{O}(B^2)$. Suffix invariance prevents this blowup: for fixed t , the dummy labels depend only on the first position ℓ where t and α differ, and therefore there are only $n = \log B$ possible garbage inputs/outputs per branch. Thus, garbage collection costs only $\mathcal{O}(B \log B)$, matching LogStack’s computation complexity.

See Section 7 for full details. The remainder of this section focuses on constructing the offline-online sipPRFs needed for our SGC application, using napPRFs as a baseline technical layer.

2.4 Black-Box Connections Between napPRFs and Two-Party DPFs

Recall that a two-party DPF encodes a point function $P_{\alpha, \gamma} : \mathcal{X} \rightarrow \mathcal{Y}$ — which outputs γ on input α and 0^m elsewhere — into two short keys (k_1, k_2) . Given k_b and x , party $b \in \{1, 2\}$ locally outputs y_b such that $y_1 \oplus y_2 = P_{\alpha, \gamma}(x)$, while each individual key hides (α, γ) .

It is easy to see that a napPRF implies a two-party DPF in a black-box way.

Here, we briefly recall the converse direction, following [BS24]. Consider $\mathcal{Y} = \{0, 1\}$ and a DPF key pair $(k_1, k_2) \leftarrow \text{Gen}(1^\lambda, \alpha, 1)$. By DPF correctness, the two local DPF evaluations coincide at every

$x \neq \alpha$, while at $x = \alpha$ they differ by 1 (i.e., one evaluates to 0 and the other to 1). To obtain a correct napPRF (including off-point correctness), it suffices to output as the programmed key whichever of k_1 and k_2 evaluates to β at α . This DPF-based trick echoes the one-point equivocation underlying *somewhere equivocal encryption* [HJO⁺16].

This approach ensures index-hiding security for the induced napPRF. At a high level, when $\beta \xleftarrow{\$} \{0, 1\}$ is uniform, Prog outputs k_1 or k_2 with probability exactly $1/2$ each. By DPF security, each share distribution is simulatable without α , and hence so is their half-half mixture.

To support range $\{0, 1\}^m$ for any m , we simply repeat this procedure independently for each output bit and concatenate the resulting keys.

See Section 5 for formal details of this equivalence. Instantiating it with the well-known OWF-based DPF constructions [GI14, BGI15] yields a napPRF from black-box OWFs with total key size $nm\lambda + 2nm + m\lambda$ bits. However, this particular OWF-based instantiation has two drawbacks:

1. **No meaningful offline-online split.** The programmed key is generated entirely as a function of (α, β) , i.e., there is no nontrivial (α, β) -independent offline component (and the total key size scales with nm).
2. **Missing suffix invariance.** The induced napPRF does *not* satisfy suffix invariance: off-point behavior depends on the full identity of α . Thus, this construction does not yield a sipPRF.

These limitations persist for the constructions of [GI14, BGI15] absent substantial modifications. Intuitively, the so-called *correction words* underlying these constructions depend on α , so the resulting key structure varies with α .

Finally, we highlight an additional result of independent interest. In Section 5.4, we demonstrate that programmable DPFs [BGIK22b] imply *adaptive* programmable PRFs via the same black-box reduction, giving the first OWF-based construction (for polynomial-size domains) of apPRFs.²

2.5 Constructing Short-Key Offline-Online sipPRFs

We discuss how to construct short-key offline-online sipPRFs.

If we allow non-black-box techniques, there is a straightforward way to obtain short online keys for napPRFs: given any napPRF $\Pi = (\text{KeyGen}, \text{Prog}, \text{Eval})$ whose Prog has small Boolean circuit complexity, we garble Prog [Yao86] and place the garbling in sk_{off} , so that sk_{on} consists only of the input labels for (α, β) and has size $|\text{sk}_{\text{on}}| = (n + m)\lambda$.

However, this generic compilation does not meet our goals. First, it is inherently non-black-box: when instantiated from OWF-based DPF-induced napPRFs, it requires garbling large subcircuits, notably the $\mathcal{O}(nm)$ PRG expansions, leading to substantial overhead in sk_{off} . Second, it does not endow suffix invariance: if the underlying napPRF lacks suffix invariance, then the compilation does not yield a sipPRF.

To this end, we give a nontrivial construction of offline-online sipPRFs over polynomial-size domains from black-box OWFs. Interestingly, this construction is inspired by the stacked garbling technique.

At a high level, our construction proceeds in two steps. We first present a recursive blueprint for building short-key offline-online napPRFs by extending the domain one bit at a time while keeping both the offline and online keys short. This part is best understood at the napPRF level and does not yet rely on suffix invariance. We then show that the recursion preserves suffix invariance, so starting from a trivially suffix-invariant singleton-domain base case yields a short-key offline-online sipPRF.

²We note that [GZSP25] gives an attempt, but it incurs a $1/\text{poly}(\lambda)$ security loss.

A warm-up: extending the domain by one bit. Before presenting the full construction, it is helpful to isolate an instructive subproblem. Suppose we have an offline-online napPRF $\Pi^{(k)} : \{0, 1\}^k \rightarrow \{0, 1\}^m$ whose online key consists of only $k + m$ Yao labels (e.g., as obtained by the generic garbling-based compiler). Can we construct an offline-online napPRF $\Pi^{(k+1)} : \{0, 1\}^{k+1} \rightarrow \{0, 1\}^m$ — i.e., extend the domain by one bit — treating $\Pi^{(k)}$ as a black-box building block?

Here is a **natural first (but flawed) attempt**. The offline key simply contains *two independent copies* of the k -bit napPRF’s offline key, one meant for inputs whose first bit is 0 and the other for inputs whose first bit is 1.

To program a point $\alpha = \alpha_1 \| \alpha'$ with value β , we proceed as follows. If $\alpha_1 = 0$, we run the k -bit programming procedure on the “0-copy” at index α' with value β and take the resulting $k+m$ labels as the online key. If $\alpha_1 = 1$, we do the same but on the “1-copy.” In either case, the online key consists of exactly $k+m$ garbled labels; the only difference is which copy they belong to.

To evaluate at an input $x = x_1 \| x'$, we route to the corresponding copy: we run the k -bit evaluation procedure using the offline key of the x_1 -copy, together with the online key and the suffix x' .

It is easy to see that this attempt satisfies on-point correctness: when $x = \alpha$, the output of the (routed) k -bit evaluation procedure is exactly β .

This attempt fails as it only *partially* satisfies off-point value independence. To see this, w.l.o.g. assume $\alpha = 0^{k+1}$, so programming is performed on the 0-copy at point $\alpha' = 0^k$. When $x = 0 \| x'$, we do achieve off-point value independence for this α : evaluation always invokes the 0-copy, so it reduces to the k -bit napPRF’s off-point value independence. However, when $x = 1 \| x'$, we fail even for this α : evaluation invokes the 1-copy but is given an *ill-formed* online key for that copy. Indeed, the online key consists of labels produced for the 0-copy and varies with β , so we cannot appeal to the k -bit napPRF off-point correctness of the 1-copy.

This suggests a simple patch. During the $(k+1)$ -bit programming procedure, we additionally include a uniformly sampled *dummy* online key for the $\bar{\alpha}_1$ -copy (i.e., $k+m$ random garbled labels). The online key now contains two parts, one for each copy, and evaluation routes to the appropriate part. Crucially, when evaluation goes into the $\bar{\alpha}_1$ -copy, its online key is independent of β .

We briefly examine the index-hiding security of this $(k+1)$ -bit construction. Both components hide α' : the real one by the index-hiding of $\Pi^{(k)}$, and the dummy one by uniformity. Thus, the only remaining potential leakage is α_1 , via distinguishing which online key is real. Ensuring this indistinguishability depends on how we instantiate $\Pi^{(k)}$, so we defer it to the end.

Garble-then-recurse. Given a $(k+1)$ -bit offline-online napPRF, the natural next step is to extend the domain further to $(k+2)$ bits by applying the above idea once more. However, an immediate issue arises: the $(k+1)$ -bit online key already contains two components (one per copy), and thus has size proportional to $2(k+m)$. Repeating the same transformation would double the number of components again, causing the online key to grow exponentially with the recursion depth. (A similar issue arises for the offline key; we address it shortly.)

The fix is simple: we further apply the garbling-based compiler idea to the $(k+1)$ -bit construction, thereby *compressing* its online key. Concretely, we represent the $(k+1)$ -bit programming procedure itself as a small garbled circuit stored in the offline key, so that the online key consists only of the input labels for (α, β) and has size $((k+1) + m)\lambda$. More explicitly, we proceed as follows:

- In the master key, sample $(k+1) + m$ Yao label pairs corresponding to the bits of $\alpha \in \{0, 1\}^{k+1}$ and $\beta \in \{0, 1\}^m$. The online key is simply the selection of one label per bit, determined by (α, β) .

- In the offline key, include a garbling of the “programmer gadget” implementing the $(k+1)$ -bit programming procedure described above. Given the input labels for (α, β) , this garbled circuit outputs the two *internal* online keys (one real and one dummy) for the two k -bit copies.

Crucially, the $(k+1)$ -bit programming procedure performs only simple selection among garbled labels from the two k -bit copies, together with dummy padding. Hence, the programmer gadget can be implemented via simple garbled tables.

For example, consider the case $m = 1$, so $\beta \in \{0, 1\}$ is a single bit. $\text{KeyGen}^{(k+1)}$ samples Yao labels for the selector bit α_1 and the programmed bit β :

$$\mathcal{L}_{\alpha_1}^{(0)}, \mathcal{L}_{\alpha_1}^{(1)}, \mathcal{L}_{\beta}^{(0)}, \mathcal{L}_{\beta}^{(1)} \in \{0, 1\}^\lambda$$

The two k -bit copies come with their own β -labels:

$$\mathcal{L}_{\beta,0}^{(0)}, \mathcal{L}_{\beta,0}^{(1)} \in \{0, 1\}^\lambda \text{ (0-copy)} \quad \mathcal{L}_{\beta,1}^{(0)}, \mathcal{L}_{\beta,1}^{(1)} \in \{0, 1\}^\lambda \text{ (1-copy)}$$

To implement the programmer gadget for β , $\text{KeyGen}^{(k+1)}$ samples a dummy label $\mathcal{J}_\beta$ and garbles the 2×2 translation table as below:

$$\begin{array}{cc} \text{Enc}_{\mathcal{L}_{\alpha_1}^{(0)}, \mathcal{L}_{\beta}^{(0)}}(\mathcal{L}_{\beta,0}^{(0)} \parallel \mathcal{J}_\beta) & \text{Enc}_{\mathcal{L}_{\alpha_1}^{(0)}, \mathcal{L}_{\beta}^{(1)}}(\mathcal{L}_{\beta,0}^{(1)} \parallel \mathcal{J}_\beta) \\ \text{Enc}_{\mathcal{L}_{\alpha_1}^{(1)}, \mathcal{L}_{\beta}^{(0)}}(\mathcal{J}_\beta \parallel \mathcal{L}_{\beta,1}^{(0)}) & \text{Enc}_{\mathcal{L}_{\alpha_1}^{(1)}, \mathcal{L}_{\beta}^{(1)}}(\mathcal{J}_\beta \parallel \mathcal{L}_{\beta,1}^{(1)}) \end{array}$$

(with the usual random permutation / point-and-permute indexing). Given the online labels $(\mathcal{L}_{\alpha_1}^{(\alpha_1)}, \mathcal{L}_{\beta}^{(\beta)})$, one can decrypt one entry and obtain a pair consisting of the *real* label for copy α_1 and a dummy label for the other copy — exactly what we need. For instance, if $(\alpha_1, \beta) = (0, 0)$, decryption yields $\mathcal{L}_{\beta,0}^{(0)} \parallel \mathcal{J}_\beta$.

We can now obtain the $(k+2)$ -bit construction by applying the above transformation to the compressed-online-key $(k+1)$ -bit construction. More generally, repeating this step yields a $(k+n)$ -bit construction for any $n \geq 1$ whose online key has size $((k+n) + m)\lambda$, *once we can control the growth of the offline key*.

Stacking the child offline keys. We now show how to avoid exponential growth in the offline key. So far, the $(k+1)$ -bit construction stores two child offline keys (two k -bit copies); if repeated naïvely, the offline key would double at each level. Inspired by the stacked garbling technique, we instead store only a *stacked* representation of the two child offline keys (i.e., their XOR). During evaluation, the napPRF evaluator can “unstack” and recover the offline key for the α_1 -copy using randomness revealed for the other copy.

In more detail, $\text{KeyGen}^{(k+1)}$ runs $\text{KeyGen}^{(k)}$ twice with seeds seed_0 and seed_1 (PRG-expanded if needed), producing two child-copy offline keys $\text{sk}_{\text{off},0}$ and $\text{sk}_{\text{off},1}$. It then sets its offline key $\text{sk}_{\text{off}}^{(k+1)}$ to include:

- The stacked child offline key $\text{sk}_{\text{stack}} := \text{sk}_{\text{off},0} \oplus \text{sk}_{\text{off},1}$.
- A tiny garbled table that reveals the *other* seed (i.e., the unprogrammed half):

$$\text{Enc}_{\mathcal{L}_{\alpha_1}^{(0)}}(\text{seed}_1) \quad \text{Enc}_{\mathcal{L}_{\alpha_1}^{(1)}}(\text{seed}_0) \quad (\text{randomly permuted})$$

Given $\mathcal{L}_{\alpha_1}^{(\alpha_1)}$ in the online key, $\text{Eval}^{(k+1)}$ decrypts the table and learns $\text{seed}_{\bar{\alpha}_1}$. It then recomputes $\text{sk}_{\text{off}, \bar{\alpha}_1}$ from $\text{seed}_{\bar{\alpha}_1}$ and unstacks by XORing with sk_{stack} :

$$\text{sk}_{\text{off}, \alpha_1} = \text{sk}_{\text{stack}} \oplus \text{sk}_{\text{off}, \bar{\alpha}_1}$$

Finally, it invokes $\text{Eval}^{(k)}$ using the recovered child offline key.

We briefly revisit off-point value independence. Note that, regardless of x , the unstacked offline key is always $\text{sk}_{\text{off}, \alpha_1}$. In particular, when $x_1 \neq \alpha_1$, $\text{Eval}^{(k+1)}$ pairs $\text{sk}_{\text{off}, \alpha_1}$ with the *dummy* internal online key — both are independent of β .

Finally, with stacked offline keys in place, the garble-then-recurse blueprint extends to any depth. Concretely, for any $n \geq 1$ it yields an offline-online napPRF $\Pi^{(k+n)}$ whose online key has size $((k+n)+m)\lambda$ and whose offline key has size $\mathcal{O}(nm\lambda + n^2\lambda + |\text{sk}_{\text{off}}^{(k)}|)$. Crucially, the construction treats $\Pi^{(k)}$ as a black box and uses only black-box OWFs.

Our construction: starting from $\Pi_{\text{sipPRF}}^{(0)}$. Our short-key offline-online sipPRFs follow the above bit-by-bit extension blueprint, starting from a simple base case offline-online *and suffix-invariant* napPRF (i.e., an offline-online sipPRF) $\Pi_{\text{sipPRF}}^{(0)}$ for the singleton domain.

For the singleton domain $\{0, 1\}^0 = \{\epsilon\}$, programming outputs m garbled labels. The offline key $\text{sk}_{\text{off}}^{(0)}$ contains one decoding bit per wire (via point-and-permute [BMR90]), so that $\text{Eval}^{(0)}$ can recover $\beta \in \{0, 1\}^m$ from the m labels in $\text{sk}_{\text{on}}^{(0)}$. We define the base case in this form to match the interface of our blueprint.

Then $\text{sk}_{\text{off}}^{(0)}$ defines schemes $\{\Pi_{\text{sipPRF}}^{(k)}\}_{k=0}^n$, where $\Pi_{\text{sipPRF}}^{(k)}$ has domain $\{0, 1\}^k$.

See Section 6 for formal details; we conclude this section with several remarks:

- **Suffix invariance.** Our offline-online sipPRFs satisfy suffix invariance, and the proof is by a simple induction on k . Consider the $(k+1)$ -bit construction. If $x_1 \neq \alpha_1$, evaluation uses the α_1 -copy offline key (recovered by unstacking) together with the dummy labels; neither depends on the programmed suffix α' , so the output is invariant under changing α' . If $x_1 = \alpha_1$, evaluation uses the same α_1 -copy offline key together with the real labels, and suffix invariance reduces to that of the k -bit instance. Since $\Pi_{\text{sipPRF}}^{(0)}$ is trivially suffix-invariant, it follows that $\Pi_{\text{sipPRF}}^{(k)}$ is suffix-invariant for all k (including $k = n$).
- **Security.** We briefly discuss why our offline-online sipPRFs satisfy index-hiding. Recall that, in the $(k+1)$ -bit step, the only new leakage channel is the first programmed bit α_1 via distinguishing the real from the dummy.

In particular, fix the $(k+1)$ -bit construction. First, the *seed table* reveals only the $\bar{\alpha}_1$ -copy seed; the entry containing the α_1 -copy seed is undecryptable. Second, the *translation table* reveals only the entry decrypted by the online labels, and in that entry exactly one half contains the *real* child labels (for α_1 -copy) while the other half is dummy. Thus, the adversary's view is consistent with the recovered α_1 -copy offline key together with a candidate internal online key. Consequently, distinguishing whether this candidate internal online key is *well-formed* (real) or uniformly random reduces to a k -bit guarantee: given the corresponding offline key, the online key output by $\text{Prog}^{(k)}$ is indistinguishable from a uniformly random bundle of $k+m$ garbled labels.

The base case $k = 0$ is immediate when β is uniformly sampled: $\text{sk}_{\text{on}}^{(0)}$ is just a uniformly random choice among the two labels on each wire (up to public permutation bits), hence indistinguishable from uniform.

Assuming the claim for level k , it follows that the real internal online key at level $(k+1)$ can be replaced by a uniform dummy without changing the adversary's view; this hides which component is real and therefore hides α_1 .

- **Polynomial-size domains.** Finally, we remark that this construction targets polynomial-size domains (sufficient for our SGC application). Conceptually, the bit-by-bit extension unfolds into a full binary recursion tree of depth n , with 2^n leaves, and thus incurs $\Theta(2^n)$ work in $\text{KeyGen}^{(n)}$ and $\text{Eval}^{(n)}$.

3 Preliminaries

3.1 Notation

Throughout, λ denotes the security parameter and $\approx_c$ denotes computational indistinguishability (for ensembles of random variables or oracles). We write n and m for the input (domain) and output (range) lengths (often suppressing their dependence on λ), and let $B := 2^n$. Unless stated otherwise, $\mathcal{X} := \{0, 1\}^n$ and $\mathcal{Y} := \{0, 1\}^m$ are public parameters known to all algorithms (including simulators). We write $\xleftarrow{\$}$ for uniform sampling, use 1-based indexing for vectors unless noted, and let $[k] = \{1, 2, \dots, k\}$ with $[0] := \emptyset$. When we quantify the computation cost of a construction, we count the number of invocations on the dominating cryptographic primitive.

3.2 Distributed Point Functions

A (two-party) *distributed point function* (DPF) [BGI15, GI14] enables a dealer to secret share a point function as two compact keys.

Point functions. Let $\mathcal{X}$ denote a finite domain and $\mathcal{Y}$ an abelian group. For $\alpha \in \mathcal{X}$ and $\beta \in \mathcal{Y}$, define the point function $P_{\alpha, \beta} : \mathcal{X} \rightarrow \mathcal{Y}$ by $P_{\alpha, \beta}(x) = \beta$ if $x = \alpha$, and $P_{\alpha, \beta}(x) = 0$ otherwise.

Definition 1 (Two-party DPF). A two-party *distributed point function* (DPF) scheme consists of two PPT algorithms:

- $\text{Gen}(1^\lambda, \alpha, \beta)$ outputs a pair of keys (k_1, k_2) , and
- $\text{Eval}(k, x)$ outputs a value in $\mathcal{Y}$ for $k \in \{k_1, k_2\}$ and $x \in \mathcal{X}$.

The scheme is **correct** if for all $\lambda \in \mathbb{N}$, all $\alpha \in \mathcal{X}$, all $\beta \in \mathcal{Y}$, and all $x \in \mathcal{X}$, the following equality holds with probability 1:

$$\text{Eval}(k_1, x) + \text{Eval}(k_2, x) = P_{\alpha, \beta}(x) \mid (k_1, k_2) \leftarrow \text{Gen}(1^\lambda, \alpha, \beta)$$

Security. A two-party DPF scheme $\Pi = (\text{Gen}, \text{Eval})$ is *secure* if for each $b \in \{1, 2\}$ there exists a PPT simulator Sim_b such that for all $\alpha \in \mathcal{X}$ and $\beta \in \mathcal{Y}$, the following ensembles are computationally indistinguishable:

$$\left\{ k_b \mid (k_1, k_2) \leftarrow \text{Gen}(1^\lambda, \alpha, \beta) \right\}_\lambda \approx_c \left\{ \text{Sim}_b(1^\lambda) \right\}_\lambda$$

Lemma 1 (DPFs from OWFs [GI14, BGI15]). Assuming one-way functions, there exists a correct and secure two-party DPF scheme Π_{DPF} for functions $\{0, 1\}^n \rightarrow \{0, 1\}$, with only black-box use of OWFs, such that each share k_b (for $b \in [2]$) has length $n\lambda + 2n + \lambda$ bits. Moreover, both Gen and Eval run in time $\mathcal{O}(n)$ and make $\mathcal{O}(n)$ calls to a length-doubling PRG.

3.3 Pseudorandom Functions

A *pseudorandom function* (PRF) [GGM86] is an efficiently computable function family that is computationally indistinguishable from a truly random function.

Definition 2 (PRF). Let λ be the security parameter. A function family $F = \{F_k : \mathcal{X} \rightarrow \mathcal{Y}\}_{k \in \mathcal{K}}$ is a **pseudorandom function** (PRF) if there exists a PPT key generation algorithm $\text{KeyGen}(1^\lambda)$ that outputs a key $k \in \mathcal{K}$, and a deterministic poly-time evaluation algorithm s.t. the following oracles are indistinguishable:

$$\left\{ F_k(\cdot) \mid k \leftarrow \text{KeyGen}(1^\lambda) \right\}_\lambda \approx_c \left\{ R(\cdot) \mid R \leftarrow \text{Rand}(\mathcal{X}, \mathcal{Y}) \right\}_\lambda$$

Here, $\text{Rand}(\mathcal{X}, \mathcal{Y})$ denotes the uniform distribution over all functions mapping $\mathcal{X}$ to $\mathcal{Y}$, and $\mathcal{A}$ is given oracle access to either $F_k(\cdot)$ or a truly random function $R(\cdot)$.

3.4 Programmable PRFs

We recall the definition of (adaptive) programmable PRFs from [PS18, GZSP25].

Definition 3 (apPRF). Fix input length n and output length m , and let $\mathcal{X} := \{0, 1\}^n$ and $\mathcal{Y} := \{0, 1\}^m$. An **(adaptive) programmable PRF** (apPRF) for a single point is a tuple of PPT algorithms $\Pi = (\text{KeyGen}, \text{MEval}, \text{Prog}, \text{Eval})$ with the following interfaces (we omit 1^λ when clear):

$$\begin{aligned} \text{msk} &\leftarrow \text{KeyGen}(1^\lambda) \\ y &:= \text{MEval}(\text{msk}, x) \in \mathcal{Y} && (\text{deterministic}) \\ \text{sk} &:= \text{Prog}(\text{msk}, \alpha, \beta) && (\text{deterministic}) \\ y &:= \text{Eval}(\text{sk}, x) \in \mathcal{Y} && (\text{deterministic}) \end{aligned}$$

The scheme is **correct** if for all $\lambda \in \mathbb{N}$, all $\text{msk} \leftarrow \text{KeyGen}(1^\lambda)$, all indices $\alpha \in \mathcal{X}$, all values $\beta \in \mathcal{Y}$, let $\text{sk} := \text{Prog}(\text{msk}, \alpha, \beta)$, the following properties hold.

1. **Correct programming at α .** That is, $\text{Eval}(\text{sk}, \alpha) = \beta$.
2. **Off-point value unchanged.** $\forall x \in \mathcal{X} \setminus \{\alpha\}, \text{Eval}(\text{sk}, x) = \text{MEval}(\text{msk}, x)$.

Security (index hiding). An apPRF scheme $\Pi = (\text{KeyGen}, \text{MEval}, \text{Prog}, \text{Eval})$ is *index hiding* if there exists a PPT simulator Sim such that for every index $\alpha \in \mathcal{X}$ the following ensembles are computationally indistinguishable:

$$\left\{ \text{sk} \mid \begin{array}{l} \text{msk} \leftarrow \text{KeyGen}(1^\lambda), \\ \beta \xleftarrow{\$} \mathcal{Y}, \\ \text{sk} := \text{Prog}(\text{msk}, \alpha, \beta) \end{array} \right\}_\lambda \approx_c \left\{ \text{Sim}(1^\lambda) \right\}_\lambda$$

That is, when programmed value β is uniform, the distribution of sk can be efficiently simulated without the knowledge of programmed index α .

Security (PRF under uniform programming). An apPRF scheme $\Pi = (\text{KeyGen}, \text{MEval}, \text{Prog}, \text{Eval})$ is *PRF-secure under uniform programming* if for every $\alpha \in \mathcal{X}$, the following oracle ensembles are computationally indistinguishable:

$$\left\{ \text{Eval}(\text{sk}, \cdot) \left| \begin{array}{l} \text{msk} \leftarrow \text{KeyGen}(1^\lambda), \\ \beta \xleftarrow{\$} \mathcal{Y}, \\ \text{sk} := \text{Prog}(\text{msk}, \alpha, \beta) \end{array} \right. \right\}_\lambda \approx_c \left\{ R(\cdot) \xleftarrow{\$} \text{Rand}(\mathcal{X} \rightarrow \mathcal{Y}) \right\}_\lambda$$

where $R : \mathcal{X} \rightarrow \mathcal{Y}$ is a uniformly random function. That is, when the programmed value is uniform, the induced oracle $x \mapsto \text{Eval}(\text{sk}, x)$ is a PRF.

Remarks. We make several remarks on the above definitions.

- The index-hiding notion is also referred to as *private programmability* (or *privately programmable*) in prior work [PS18, GZSP25].
- Our results in Section 4 show that *correctness* together with *index hiding* already suffices to derive *PRF security under uniform programming*. To the best of our knowledge, this implication has not been explicitly observed before.
- When the domain size $|\mathcal{X}|$ is superpolynomial, PRF security under uniform programming further implies that $\text{MEval}(\text{msk}, \cdot)$ can be used as a standard PRF. In contrast, for polynomial-size domains, this implication fails in general. We provide a proof and an explicit counterexample in Appendix A.
- Existing constructions of apPRFs with index hiding rely on strong assumptions (e.g., LWE and/or indistinguishability obfuscation). Although our main focus is on sipPRFs, in Section 5.4, we show an apPRF scheme with index hiding can be built only from one-way functions for a polynomial-size domain.

4 Definitions of Suffix-Invariant Programmable PRFs

4.1 Baseline Non-Adaptive Programmability

Definition 4 (napPRF). Fix input length n and output length m , and let $\mathcal{X} := \{0, 1\}^n$ and $\mathcal{Y} := \{0, 1\}^m$. A **non-adaptive programmable PRF** (napPRF)³ for a single point is a tuple of PPT algorithms $\Pi = (\text{KeyGen}, \text{Prog}, \text{Eval})$ with the following interfaces (we omit 1^λ when clear):

$$\begin{aligned} \text{msk} &\leftarrow \text{KeyGen}(1^\lambda) \\ \text{sk} &:= \text{Prog}(\text{msk}, \alpha, \beta) && (\text{deterministic}) \\ y &:= \text{Eval}(\text{sk}, x) \in \mathcal{Y} && (\text{deterministic}) \end{aligned}$$

where msk is the master secret key, sk is the programmed key. We view Eval as defining a keyed function $F_{\text{sk}}(\cdot) : \mathcal{X} \rightarrow \mathcal{Y}$, $F_{\text{sk}}(x) := \text{Eval}(\text{sk}, x)$.

The scheme is **correct** if for all $\lambda \in \mathbb{N}$, all $\text{msk} \leftarrow \text{KeyGen}(1^\lambda)$, all indices $\alpha \in \mathcal{X}$, and all values $\beta, \beta_1, \beta_2 \in \mathcal{Y}$, the following properties hold.

1. **Correct programming at α .** Let $\text{sk} := \text{Prog}(\text{msk}, \alpha, \beta)$, $\text{Eval}(\text{sk}, \alpha) = \beta$.
2. **Off-point value independence.** Let $\text{sk}^{(i)} := \text{Prog}(\text{msk}, \alpha, \beta_i)$ for $i \in \{1, 2\}$. Then for every $x \in \mathcal{X} \setminus \{\alpha\}$, $\text{Eval}(\text{sk}^{(1)}, x) = \text{Eval}(\text{sk}^{(2)}, x)$.

³[BS24] study a closely related notion under the name NAP. Our formulation is slightly different; see Section 2.1.

Security (index hiding). A napPRF scheme $\Pi = (\text{KeyGen}, \text{Prog}, \text{Eval})$ is *index hiding* if there exists a PPT simulator Sim such that for every index $\alpha \in \mathcal{X}$ the following ensembles are computationally indistinguishable:

$$\left\{ \text{sk} \left| \begin{array}{l} \text{msk} \leftarrow \text{KeyGen}(1^\lambda), \\ \beta \xleftarrow{\$} \mathcal{Y}, \\ \text{sk} := \text{Prog}(\text{msk}, \alpha, \beta) \end{array} \right. \right\}_\lambda \approx_c \left\{ \text{Sim}(1^\lambda) \right\}_\lambda$$

That is, when programmed value β is uniform, the distribution of sk can be efficiently simulated without the knowledge of programmed index α .

Security (PRF under uniform programming). A napPRF $\Pi = (\text{KeyGen}, \text{Prog}, \text{Eval})$ is *PRF-secure under uniform programming* if for every $\alpha \in \mathcal{X}$, the following oracle ensembles are computationally indistinguishable:

$$\left\{ \text{Eval}(\text{sk}, \cdot) \left| \begin{array}{l} \text{msk} \leftarrow \text{KeyGen}(1^\lambda), \\ \beta \xleftarrow{\$} \mathcal{Y}, \\ \text{sk} := \text{Prog}(\text{msk}, \alpha, \beta) \end{array} \right. \right\}_\lambda \approx_c \left\{ R(\cdot) \xleftarrow{\$} \text{Rand}(\mathcal{X} \rightarrow \mathcal{Y}) \right\}_\lambda$$

where $R : \mathcal{X} \rightarrow \mathcal{Y}$ is a uniformly random function. That is, when the programmed value is uniform, the induced oracle $x \mapsto \text{Eval}(\text{sk}, x)$ is a PRF.

Perhaps surprisingly, in Section 5 we show that correctness and index hiding already imply PRF security under uniform programming, that is:

Theorem 4 (Index-hiding napPRF $\Rightarrow$ PRF under uniform programming). *Let $\Pi_{\text{napPRF}} = (\text{KeyGen}, \text{Prog}, \text{Eval})$ be a napPRF. If Π_{napPRF} is correct and index hiding, then it is PRF-secure under uniform programming.*

Definition 5 (Offline-online napPRF). *An offline-online napPRF is a tuple of PPT algorithms $\Pi = (\text{KeyGen}, \text{Prog}, \text{Eval})$ such that $\text{KeyGen}(1^\lambda)$ samples $(\text{msk}, \text{sk}_{\text{off}})$, $\text{Prog}(\text{msk}, \alpha, \beta)$ outputs sk_{on} , and $\text{Eval}(\text{sk}_{\text{off}}, \text{sk}_{\text{on}}, x)$ outputs $y \in \mathcal{Y}$. The programmed key is $\text{sk} := (\text{sk}_{\text{off}}, \text{sk}_{\text{on}})$, where sk_{off} is independent of (α, β) .*

All correctness/security notions extend by viewing the programmed key as $\text{sk} = (\text{sk}_{\text{off}}, \text{sk}_{\text{on}})$. This is w.l.o.g.: a standard napPRF is the special case $\text{sk}_{\text{off}} = \perp$, and conversely an offline-online napPRF can be viewed as a standard one by folding sk_{off} into msk and having Prog output it. Henceforth, we write napPRF for the standard syntax and *offline-online* napPRF when we use the split-key syntax.

Primary efficiency metrics and short-key offline-online napPRFs. Our primary efficiency measures are (i) the total programmed-key size $|\text{sk}|$ and (ii) the online-key size $|\text{sk}_{\text{on}}|$. We require $|\text{sk}| \leq \text{poly}(n, m, \lambda)$. We call an offline-online napPRF *short-key* if $|\text{sk}_{\text{on}}|$ is linear in $n + m$ (up to $\text{poly}(\lambda)$ factors).

Evaluation over the full domain. We also introduce a full-domain evaluation procedure. For any programmed key sk (possibly $\text{sk} = (\text{sk}_{\text{off}}, \text{sk}_{\text{on}})$), define

$$\text{FullEval}(\text{sk}) \in (\{0, 1\}^m)^{2^n}$$

as the vector indexed by $x \in \{0, 1\}^n$ whose x -th entry is $\text{Eval}(\text{sk}, x)$.

Note that an implementation of FullEval is not required to compute the vector by explicitly invoking Eval on every input; in particular, a construction may admit a more efficient procedure for generating the full-domain evaluation.

4.2 Non-Adaptive vs. Adaptive Programmable PRFs

We clarify the distinction between napPRFs and apPRFs (cf. Section 3.4).

Indeed, any apPRF $\Pi_{\text{apPRF}} = (\text{KeyGen}, \text{MEval}, \text{Prog}, \text{Eval})$ that is correct and index-hiding immediately yields a (standard) napPRF in a black-box way by simply dropping the master evaluation algorithm MEval . Correctness and index hiding of Π_{apPRF} immediately carry over.

Note that Theorem 4 also applies to the napPRF induced from an apPRF as above. In particular, under this embedding, we have $\Pi_{\text{napPRF}}.\text{Eval} \equiv \Pi_{\text{apPRF}}.\text{Eval}$. This implies that for an apPRF, *correctness* together with *index hiding* already suffices to derive *PRF security under uniform programming* (cf. Section 3.4).

We now consider the converse direction and explain why, in general, a correct and index-hiding napPRF does *not* yield a correct and index-hiding apPRF via a black-box transformation. Interestingly, the obstruction is not security but *correctness*: an apPRF requires the existence of a single “master” evaluation procedure (i.e., MEval) that is preserved under programming.

To see this, recall that in an apPRF we have a master function $F(x) := \Pi_{\text{apPRF}}.\text{MEval}(\text{msk}, x)$ such that programming at (α, β) produces a key sk whose induced oracle agrees with F everywhere except at α : $\forall x \neq \alpha, \Pi_{\text{apPRF}}.\text{Eval}(\text{sk}, x) = F(x)$. That is, the off-point value is independent of *both* α and β .

In contrast, napPRF correctness guarantees only *off-point invariance with respect to the programmed value*: For every fixed α and all β, β' , the evaluations on $x \neq \alpha$ are identical, but they may still depend arbitrarily on α . That is, while a napPRF can ensure that changing β does not affect off-point outputs, it does not enforce that programming different indices α ’s is consistent with a single underlying master function. This missing cross- α consistency is exactly what prevents a generic black-box construction of an apPRF from a napPRF.

Definition 6 (Cross- α consistency). A napPRF $\Pi = (\text{KeyGen}, \text{Prog}, \text{Eval})$ is cross- α consistent if for all $\lambda \in \mathbb{N}$, all $\text{msk} \leftarrow \text{KeyGen}(1^\lambda)$, all indices $\alpha_1, \alpha_2 \in \mathcal{X}$, and all $x \in \mathcal{X} \setminus \{\alpha_1, \alpha_2\}$ the following property holds:

$$\text{Eval}(\text{Prog}(\text{msk}, \alpha_1, 0), x) = \text{Eval}(\text{Prog}(\text{msk}, \alpha_2, 0), x)$$

Indeed, any napPRF that is correct, index-hiding, and cross- α consistent immediately yields a correct and index-hiding apPRF via a black-box construction. The transformation mirrors the trivial embedding of apPRFs into napPRFs, except for one caveat: we must define a master evaluation algorithm $\Pi_{\text{apPRF}}.\text{MEval}$. Cross- α consistency allows us to do so unambiguously: to evaluate at x , pick any $\alpha \in \mathcal{X} \setminus \{x\}$, and output $\Pi_{\text{napPRF}}.\text{Eval}(\Pi_{\text{napPRF}}.\text{Prog}(\text{msk}, \alpha, 0), x)$.

In Section 5.4, we show how to construct a correct, index-hiding, and cross- α consistent napPRF over a polynomial-size domain from one-way functions. By the discussion above, this immediately yields a correct and index-hiding apPRF from one-way functions in the same regime, which (to our knowledge) was not previously known. We emphasize, however, that since the domain is only polynomial-size, this does not by itself imply that the resulting apPRF’s master evaluation MEval is a standard PRF (cf. Sections 3.4 and 5.4).

4.3 Suffix-Invariant Programmable PRFs

We now formalize suffix invariance, the structural correctness property that upgrades the baseline napPRF notion into sipPRFs and is needed for our stacked garbling application. See Sections 2.1 and 2.3 for a concise explanation.

Definition 7 (Suffix invariance). Let $\Pi = (\text{KeyGen}, \text{Prog}, \text{Eval})$ be a napPRF over $\mathcal{X} = \{0, 1\}^n$ and $\mathcal{Y}$. For $x, \alpha \in \{0, 1\}^n$ with $x \neq \alpha$, define the divergence

$$\div(x, \alpha) := \min\{i \in [n] : x_i \neq \alpha_i\} \in [n]$$

and (optionally) set $\div(x, x) := n + 1$.

We say that Π is suffix-invariant if for all $\lambda \in \mathbb{N}$, all $\text{msk} \leftarrow \text{KeyGen}(1^\lambda)$, all $x \in \{0, 1\}^n$, all $\alpha, \alpha' \in \{0, 1\}^n \setminus \{x\}$, and all $\beta, \beta' \in \mathcal{Y}$,

$$\div(x, \alpha) = \div(x, \alpha') \implies \text{Eval}(\text{Prog}(\text{msk}, \alpha, \beta), x) = \text{Eval}(\text{Prog}(\text{msk}, \alpha', \beta'), x)$$

Definition 8 (sipPRF). A suffix-invariant programmable PRF (sipPRF) is a tuple of algorithms satisfying the napPRF correctness and index-hiding guarantees, together with suffix invariance. An offline-online sipPRF, respectively a short-key offline-online sipPRF, is defined by additionally imposing the corresponding offline-online, respectively short-key, requirement.

Remark 1. Suffix invariance strictly strengthens napPRF correctness while still remaining strictly weaker than apPRF correctness. Thus, apPRFs are a special case of sipPRFs, which are in turn a special case of napPRFs.

5 Relationship Between napPRFs and Two-Party DPFs

In this section, we formalize a *black-box* equivalence between napPRFs and two-party DPFs. Specifically, we provide black-box reductions in both directions.

We emphasize that the reduction from napPRFs to two-party DPFs is already known from prior work [BS24, HJO⁺16], explicitly or implicitly. We include the equivalence for completeness and to support our independent side result on OWF-based apPRFs over polynomial-size domains (see Section 5.4).

5.1 Individual Pseudorandomness for DPFs

We record a useful consequence of standard DPF security (cf. Section 3.2).

Individual pseudorandomness. A two-party DPF scheme $(\text{Gen}, \text{Eval})$ satisfies *individual pseudorandomness* if each individual key induces a function that is computationally indistinguishable from a uniformly random function. Formally, for each $b \in \{1, 2\}$ and for all $\alpha \in \mathcal{X}$ and $\beta \in \mathcal{Y}$, the following oracle distributions are computationally indistinguishable:

$$\left\{ \text{Eval}(k_b, \cdot) \mid (k_1, k_2) \leftarrow \text{Gen}(1^\lambda, \alpha, \beta) \right\}_\lambda \approx_c \left\{ R(\cdot) \mid R \leftarrow \text{Rand}(\mathcal{X} \rightarrow \mathcal{Y}) \right\}_\lambda$$

where $R : \mathcal{X} \rightarrow \mathcal{Y}$ is a uniformly random function. That is, each share k_b defines a PRF over domain $\mathcal{X}$ and range $\mathcal{Y}$. Interestingly, correctness and standard DPF security already imply individual pseudorandomness:

Lemma 2 (DPF security $\implies$ Individual pseudorandomness). Let $\Pi = (\text{Gen}, \text{Eval})$ be a two-party DPF scheme. Suppose Π is correct and secure in the sense of per-share simulation security for both shares (defined in Section 3.2). Then Π satisfies individual pseudorandomness.

We defer the proof to Appendix B.1. Jumping ahead, we use Lemma 2 to justify why index hiding implies PRF security under uniform programming in napPRFs.

5.2 napPRF $\implies$ DPF

Let $\Pi_{\text{napPRF}} = (\text{KeyGen}, \text{Prog}, \text{Eval})$ be a (standard) napPRF scheme over domain $\mathcal{X} = \{0, 1\}^{n(\lambda)}$ and range $\mathcal{Y} = \{0, 1\}^{m(\lambda)}$ (where $\mathcal{Y}$ is viewed as an abelian group under $\oplus$). We construct a two-party DPF scheme $\Pi_{\text{DPF}} = (\text{Gen}, \text{Eval})$ by using Π_{napPRF} as a black box, as depicted in Figure 1. We have:

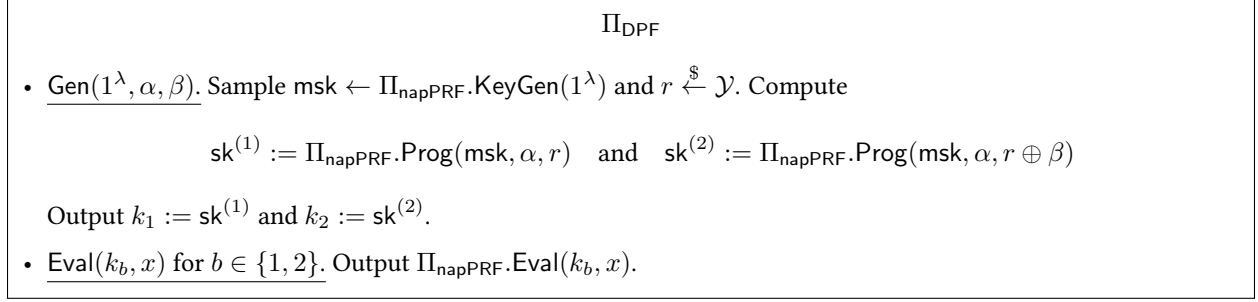


Figure 1: Constructing Π_{DPF} from $\Pi_{\text{napPRF}} = (\text{KeyGen}, \text{Prog}, \text{Eval})$ as a black box.

Theorem 5 (napPRF $\Rightarrow$ DPF). *If $\Pi_{\text{napPRF}} = (\text{KeyGen}, \text{Prog}, \text{Eval})$ is a correct and index-hiding napPRF for a single point over domain $\mathcal{X}$ and range $\mathcal{Y}$, then the construction in Figure 1 yields a two-party DPF $\Pi_{\text{DPF}} = (\text{Gen}, \text{Eval})$ over domain $\mathcal{X}$ and range $\mathcal{Y}$ that is correct and secure (defined in Section 3.2).*

The intuition behind Theorem 5 is natural. DPF correctness follows since at $x = \alpha$ we have outputs r and $r \oplus \beta$, whose XOR equals β , while for every $x \neq \alpha$, off-point value independence gives $\Pi_{\text{napPRF}}.\text{Eval}(\text{sk}^{(1)}, x) = \Pi_{\text{napPRF}}.\text{Eval}(\text{sk}^{(2)}, x)$, so the XOR is 0. DPF security follows because both r and $r \oplus \beta$ are uniform in $\mathcal{Y}$; hence each share key k_b is distributed as a napPRF key programmed with a uniform value, which is simulatable by napPRF’s index hiding and thus hides (α, β) . We defer the full proof to Appendix B.2.

We conclude this section with a corollary of Theorem 5 and Lemma 2, which we formalize in Theorem 4. By Theorem 5, our construction turns any correct and index-hiding napPRF into a two-party DPF that is correct and secure; by Lemma 2, this DPF moreover satisfies *individual pseudorandomness*.

Unwinding the construction, we obtain the following statement about the underlying napPRF. Fix any index $\alpha \in \mathcal{X}$ and let $\beta \xleftarrow{\$} \mathcal{Y}$ be uniform. Sample $\text{msk} \leftarrow \Pi_{\text{napPRF}}.\text{KeyGen}(1^\lambda)$ and set $\text{sk} := \Pi_{\text{napPRF}}.\text{Prog}(\text{msk}, \alpha, \beta)$. Then the induced oracle $\text{Eval}(\text{sk}, \cdot) : \mathcal{X} \rightarrow \mathcal{Y}$ is computationally indistinguishable from a uniformly random function. That is, when the programmed value is uniform, $\Pi_{\text{napPRF}}.\text{Eval}$ behaves like a PRF for any fixed α .

5.3 DPF $\Rightarrow$ napPRF

Let $\Pi_{\text{DPF}} = (\text{Gen}, \text{Eval})$ be a DPF scheme over domain $\mathcal{X} = \{0, 1\}^{n(\lambda)}$ and range $\{0, 1\}$. Assume that $\Pi_{\text{DPF}}.\text{Gen}$ uses at most $R(\lambda)$ random bits. We construct a napPRF scheme $\Pi_{\text{napPRF}} = (\text{KeyGen}, \text{Prog}, \text{Eval})$ (over domain $\mathcal{X}$ and range $\mathcal{Y} = \{0, 1\}^m$) by using Π_{DPF} as a black box, as depicted in Figure 2. We have:

Theorem 6 (DPF $\Rightarrow$ napPRF). *If $\Pi_{\text{DPF}} = (\text{Gen}, \text{Eval})$ is a correct and secure two-party DPF over domain $\mathcal{X}$ and range $\{0, 1\}$, then the construction in Figure 2 yields a napPRF $\Pi_{\text{napPRF}} = (\text{KeyGen}, \text{Prog}, \text{Eval})$ over domain $\mathcal{X}$ and range $\mathcal{Y}$ that is correct and index-hiding (defined in Section 4).*

We note that the intuition behind Theorem 6 already appears in Section 2.4; we defer the full proof to Appendix B.3.

Instantiating with OWF-based DPF. Plugging the standard OWF-based DPF constructions [GI14, BGI15] (cf. Lemma 1) into Theorem 6 yields an OWF-based napPRF whose programmed key has size $nm\lambda + 2nm + m\lambda$ bits. In this instantiation, the programmed key is generated entirely as a function of

Π_{napPRF}
• KeyGen(1^λ). Sample $\text{msk} \xleftarrow{\\$} \{0, 1\}^{mR}$. Output msk. • Prog($\text{msk}, \alpha, \beta$). Parse $\beta = \beta_1 \parallel \dots \parallel \beta_m$ where each $\beta_i \in \{0, 1\}$. Parse $\text{msk} = \text{msk}_1 \parallel \dots \parallel \text{msk}_m$ where each $\text{msk}_i \in \{0, 1\}^R$. For each $i \in [m]$: 1. Compute $(k_1^{(i)}, k_2^{(i)}) := \Pi_{\text{DPF}}.\text{Gen}(1^\lambda, \alpha, 1; \text{msk}_i)$, i.e., generate DPF shares of the point function $P_{\alpha,1}$ using random tape fixed as msk_i. 2. Let $b_i := \Pi_{\text{DPF}}.\text{Eval}(k_1^{(i)}, \alpha)$. If $b_i = \beta_i$, set $k^{(i)} := k_1^{(i)}$; else, set $k^{(i)} := k_2^{(i)}$. Output $\text{sk} := k^{(1)} \parallel \dots \parallel k^{(m)}$. • Eval(sk, x). Parse $\text{sk} = k^{(1)} \parallel \dots \parallel k^{(m)}$. Compute $y_i := \Pi_{\text{DPF}}.\text{Eval}(k^{(i)}, x)$ for each $i \in [m]$. Output $y := y_1 \parallel \dots \parallel y_m$.

Figure 2: Constructing Π_{napPRF} from $\Pi_{\text{DPF}} = (\text{Gen}, \text{Eval})$ as a black box.

(α, β) (i.e., there is no offline part), and it also fails suffix invariance (cf. Definition 7), even with non-black-box access to the constructions of [GI14, BGI15] (absent substantial modifications). Intuitively, in these DPFs the so-called *correction words* are α -dependent, so changing α changes the resulting key structure.

5.4 Programmable DPF $\Rightarrow$ apPRF

Instantiating the transformation in Section 5.3 with a correct and secure *programmable* DPF [BGIK22b] yields a napPRF that is not only correct and index hiding, but also *cross- α consistent*. Consequently, by Section 4.2, this napPRF further implies a correct and index-hiding apPRF via a black-box construction.

Programmable DPFs. At a high level, a programmable DPF is a DPF scheme whose key-generation algorithm Gen can be decomposed into two stages, Gen_1 and Gen_2 . In Gen_1 , the first key k_1 is sampled without knowledge of (α, β) ; then Gen_2 produces the second key k_2 as a (randomized) function of k_1 and (α, β) . In particular, once the randomness of Gen is fixed, the induced function $x \mapsto \text{DPF}.\text{Eval}(k_1, x)$ is fully determined for all $x \in \mathcal{X}$.

Cross- α consistency of the induced napPRF. Recall that cross- α consistency requires that for all (distinct) $\alpha_1, \alpha_2 \in \mathcal{X}$ and all $x \in \mathcal{X} \setminus \{\alpha_1, \alpha_2\}$,

$$\text{Eval}(\text{Prog}(\text{msk}, \alpha_1, 0), x) = \text{Eval}(\text{Prog}(\text{msk}, \alpha_2, 0), x)$$

Consider our napPRF in Section 5.3 and fix any coordinate $i \in [m]$. For each α , Prog computes $(k_1^{(i)}, k_2^{(i)}) \leftarrow \Pi_{\text{DPF}}.\text{Gen}(1^\lambda, \alpha, 1; \text{msk}_i)$ (using fixed randomness msk_i), and then outputs either $k_1^{(i)}$ or $k_2^{(i)}$. For any $x \neq \alpha$, DPF correctness implies $\Pi_{\text{DPF}}.\text{Eval}(k_1^{(i)}, x) = \Pi_{\text{DPF}}.\text{Eval}(k_2^{(i)}, x)$, and hence the i -th output bit of $\text{Eval}(\text{sk}, x)$ equals $\Pi_{\text{DPF}}.\text{Eval}(k_1^{(i)}, x)$, regardless of if Prog chose $k_1^{(i)}$ or $k_2^{(i)}$.

Now take two indices α_1, α_2 and a point $x \notin \{\alpha_1, \alpha_2\}$. In the programmable DPF, the first key $k_1^{(i)}$ produced with randomness msk_i is independent of α , so the same $k_1^{(i)}$ is used in both executions with α_1 and with α_2 . Therefore, the i -th output bit at x is identical in both cases, and concatenating over $i \in [m]$ gives cross- α consistency for the induced napPRF.

Building apPRF from [BGIK22b]. Boyle *et al.* [BGIK22b] construct programmable DPFs for polynomial-size domains from one-way functions. Combining their OWF-based programmable DPF with the implications above, we obtain (for polynomial-size domains) a correct and index-hiding apPRF from one-way functions via a black-box reduction; to our knowledge, such an OWF-based construction was not previously available even in this restricted regime.⁴

We emphasize that, since the domain is only polynomial-size, the resulting apPRF guarantee pertains to *PRF security under uniform programming*. That is, it implies pseudorandomness of the programmed evaluation oracle $\text{Eval}(\text{sk}, \cdot)$ when the programmed value is uniform, but does not by itself imply that the master evaluation $\text{MEval}(\text{msk}, \cdot)$ is a standard PRF (cf. Section 3.4). It remains to identify additional conditions or instantiations under which the induced master evaluation MEval also enjoys standard PRF security. It is plausible that the specific OWF-based programmable DPF construction of [BGIK22b] already yields such a guarantee; we leave a careful investigation to future work.

6 Toward Short-Key Offline-Online sipPRFs

In this section, we construct *short-key offline-online* sipPRFs from one-way functions. Recall that sipPRFs satisfy the baseline non-adaptive programming guarantees together with suffix invariance. In the offline-online syntax, the programmed key splits as $\text{sk} = (\text{sk}_{\text{off}}, \text{sk}_{\text{on}})$, where $(\text{msk}, \text{sk}_{\text{off}}) \leftarrow \text{KeyGen}(1^\lambda)$ is independent of (α, β) and $\text{sk}_{\text{on}} := \text{Prog}(\text{msk}, \alpha, \beta)$ is the only *online* material depending on the programmed point. Our goal is to make sk_{on} short, namely linear in $n + m$ up to $\text{poly}(\lambda)$ factors, while keeping the total key size $|\text{sk}| \leq \text{poly}(n, m, \lambda)$.

First, we describe a generic garbling-based compiler that gives short online keys for any napPRF whose programming algorithm has small Boolean circuit complexity. This route is conceptually simple, but it is non-black-box and does not by itself provide suffix invariance. Second, for polynomial-size domains, we give our main construction: a black-box OWF-based short-key offline-online napPRF whose recursion preserves suffix invariance, and hence yields a sipPRF.

6.1 Garbling-Based Short-Key Compiler

We begin with a simple approach for building a short-key offline-online napPRF from a standard napPRF whose Prog has small Boolean circuit complexity. We use Yao garbling to *compile* the programming algorithm Prog into offline key material sk_{off} . That is, we garble a circuit for Prog and include the resulting GC in sk_{off} . The online key sk_{on} then consists only of the input labels encoding α and β . Since this requires $(n + m)$ labels, we obtain $|\text{sk}_{\text{on}}| = (n + m)\lambda$.

This compiler is conceptually simple; we defer the formalism to Appendix C.

Instantiation via the DPF-based napPRF. The napPRF from Section 5.3 (instantiated with OWF-based DPFs [GI14, BGI15]) indeed admits a Prog procedure with small circuit complexity: Prog is dominated by $\mathcal{O}(nm)$ calls to a length-doubling PRG. Thus, via non-black-box use of cryptography (i.e., garbling $\mathcal{O}(nm)$ PRG expansions), the above compiler yields a short-key OWF-based offline-online napPRF with total key size $|\text{sk}| = \mathcal{O}(nm \cdot \text{poly}(\lambda))$.

This instantiation supports superpolynomial-size domains, but is non-black-box and does not provide the suffix-invariance needed for our SGC application.

⁴We note that [GZSP25] gives an attempt, but it incurs a $1/\text{poly}(\lambda)$ security loss.

6.2 Short-Key sipPRFs via Recursively Stacked Offline Keys

We construct short-key offline-online sipPRFs for polynomial-size domains from OWFs, using all underlying primitives in a black-box manner. As in the intuitive overview (cf. Section 2.5), we present the construction in two steps. First, we define a recursive transformation that builds short-key offline-online napPRFs by extending the domain one bit at a time. Second, we show that this transformation preserves suffix invariance. Thus, starting from a trivially suffix-invariant singleton-domain base case, the final scheme is a short-key offline-online sipPRF.

Specifically, we define a sequence $\Pi_{\text{napPRF}}^{(0)}, \dots, \Pi_{\text{napPRF}}^{(n)}$, where $\Pi_{\text{napPRF}}^{(0)}$ is a correct, index-hiding offline-online napPRF over the singleton domain $\{0, 1\}^0 = \{\epsilon\}$ and range $\{0, 1\}^m$, and for each $k \in \{1, \dots, n\}$, the scheme $\Pi_{\text{napPRF}}^{(k)}$ treats $\Pi_{\text{napPRF}}^{(k-1)}$ purely as a black box and forms a correct, index-hiding offline-online napPRF over domain $\{0, 1\}^k$ and range $\{0, 1\}^m$. We later show that every $\Pi_{\text{napPRF}}^{(k)}$ is suffix-invariant, and hence $\Pi_{\text{napPRF}}^{(n)}$ is a sipPRF, denoted $\Pi_{\text{sipPRF}}^{(n)}$.

PRG (used by all levels). Let $G : \{0, 1\}^\lambda \rightarrow \{0, 1\}^*$ be a PRG (from OWFs).

Encryption schemes (used by all levels). Let $F : \{0, 1\}^\lambda \times \{0, 1\}^* \rightarrow \{0, 1\}^{2\lambda}$ be a PRF (from OWFs). All ciphertexts below have length λ or 2λ , and every garbled-table entry is associated with a public *unique* tweak $\tau \in \{0, 1\}^*$.

- **Single-key encryption.** For messages $M \in \{0, 1\}^\lambda$ define

$$\text{Enc}_K^{(1)}(\tau, M) := M \oplus F_K(\tau).\text{first} \quad \text{Dec}_K^{(1)}(\tau, C) := C \oplus F_K(\tau).\text{first}$$

where $F_K(\tau).\text{first}$ denotes the first λ bits of $F_K(\tau) \in \{0, 1\}^{2\lambda}$. By PRF security, if K is uniform and hidden, then $\text{Enc}_K^{(1)}(\tau, M)$ is computationally indistinguishable from uniform over $\{0, 1\}^\lambda$ (even given τ and M).

- **Two-key encryption.** For messages $M \in \{0, 1\}^{2\lambda}$ define

$$\begin{aligned} \text{Enc}_{K_1, K_2}^{(2)}(\tau, M) &:= M \oplus F_{K_1}(\tau \| 0) \oplus F_{K_2}(\tau \| 1) \\ \text{Dec}_{K_1, K_2}^{(2)}(\tau, C) &:= C \oplus F_{K_1}(\tau \| 0) \oplus F_{K_2}(\tau \| 1) \end{aligned}$$

By PRF security, if at least one of K_1, K_2 is uniform and hidden, then $\text{Enc}_{K_1, K_2}^{(2)}(\tau, M)$ is computationally indistinguishable from uniform over $\{0, 1\}^{2\lambda}$ (even given τ, M , and the other key).

Least significant bit. For $x \in \{0, 1\}^\lambda$, $\text{Lsb}(x)$ denotes its least significant bit.

Base case $\Pi_{\text{napPRF}}^{(0)}$. For a singleton domain, an (offline-online) napPRF is immediate (one may take $\text{sk} = \beta$). We nevertheless give a slightly more verbose formulation — decoding garbled labels — to match the interface of our recursive construction. The scheme $\Pi_{\text{napPRF}}^{(0)}$ is depicted in Figure 3, and satisfies:

Theorem 7 (Base case). Let G be a PRG. $\Pi_{\text{napPRF}}^{(0)}$ (cf. Figure 3) is a correct and index-hiding offline-online napPRF over domain $\{0, 1\}^0$ and range $\{0, 1\}^m$.

The proof is straightforward; we defer the details to Appendix B.4.

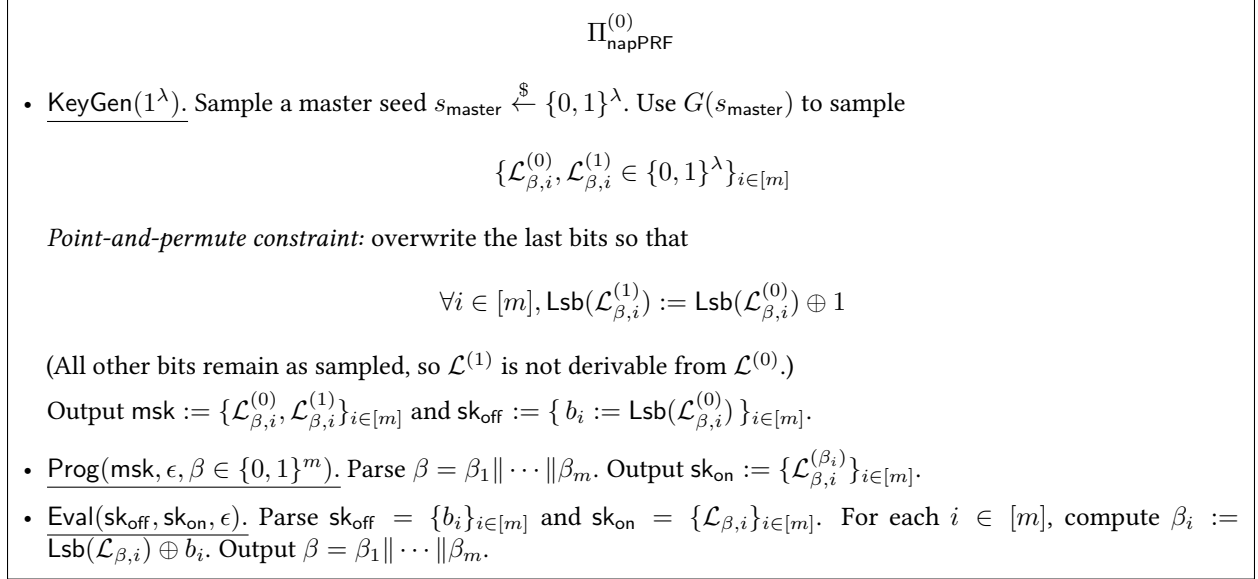


Figure 3: $\Pi_{\text{napPRF}}^{(0)}$ for domain $\{0, 1\}^0 = \{\epsilon\}$ and range $\{0, 1\}^m$.

Recursive definitions. Fix $k \geq 1$. We define $\Pi_{\text{napPRF}}^{(k)} = (\text{KeyGen}, \text{Prog}, \text{Eval})$ over domain $\mathcal{X} = \{0, 1\}^k$ and range $\mathcal{Y} = \{0, 1\}^m$. The construction of $\Pi_{\text{napPRF}}^{(k)}$ makes *black-box* use of $\Pi_{\text{napPRF}}^{(k-1)}$.

We write $\tau_{\text{seed},r}^{(k)}$, $\tau_{\beta,i,r,c}^{(k)}$, and $\tau_{\alpha,i,r,c}^{(k)}$ for public, *unique* tweaks for the corresponding (garbled) table entries. For notational convenience, we write

$$(\text{msk}, \text{sk}_{\text{off}}) := \Pi_{\text{napPRF}}. \text{KeyGen}(1^\lambda; s)$$

to denote running $\Pi_{\text{napPRF}}. \text{KeyGen}$ with randomness set as $s \in \{0, 1\}^\lambda$. In particular, we will ensure that each KeyGen uses only a λ -bit random tape. The scheme $\Pi_{\text{napPRF}}^{(k)}$ is depicted in Figures 4 and 5, and satisfies:

Theorem 8 (Main). *Let G be a PRG and let F be a PRF. For every $k \geq 1$, the scheme $\Pi_{\text{napPRF}}^{(k)}$ (cf. Figures 4 and 5) is a correct and index-hiding offline-online napPRF over domain $\{0, 1\}^k$ and range $\{0, 1\}^m$.*

We note that the intuition behind Theorem 8 already appears in Section 2.5; we defer the full proof to Appendix B.5.

Suffix invariance. It remains to show that the recursive transformation preserves suffix invariance; see Section 2.5 for an intuitive justification.

Lemma 3. *If $\Pi_{\text{napPRF}}^{(k-1)}$ is suffix-invariant, then the recursively constructed $\Pi_{\text{napPRF}}^{(k)}$ is suffix-invariant. Consequently, since $\Pi_{\text{napPRF}}^{(0)}$ is trivially suffix-invariant, $\Pi_{\text{napPRF}}^{(n)}$ is suffix-invariant for every n .*

Together with Theorem 8, this implies that the final scheme $\Pi_{\text{napPRF}}^{(n)}$ is a short-key offline-online sipPRF. When emphasizing this suffix-invariant structure, we denote the same scheme by $\Pi_{\text{sipPRF}}^{(n)}$.

$$\Pi_{\text{napPRF}}^{(k)}.\text{KeyGen}(1^\lambda)$$

1. Sample a master seed $s_{\text{master}} \xleftarrow{\$} \{0, 1\}^\lambda$. Use $G(s_{\text{master}})$ to sample

$$s_{\text{left}}, s_{\text{right}} \in \{0, 1\}^\lambda \quad \{\mathcal{L}_{\alpha,i}^{(0)}, \mathcal{L}_{\alpha,i}^{(1)} \in \{0, 1\}^\lambda\}_{i \in [k]} \quad \{\mathcal{L}_{\beta,i}^{(0)}, \mathcal{L}_{\beta,i}^{(1)} \in \{0, 1\}^\lambda\}_{i \in [m]}$$

$$\{\mathcal{J}_{\alpha,i} \in \{0, 1\}^\lambda\}_{i \in [k-1]} \quad \{\mathcal{J}_{\beta,i} \in \{0, 1\}^\lambda\}_{i \in [m]}$$

Point-and-permute constraint: overwrite the last bits so that

$$\forall i \in [k], \text{Lsb}(\mathcal{L}_{\alpha,i}^{(1)}) := \text{Lsb}(\mathcal{L}_{\alpha,i}^{(0)}) \oplus 1 \quad \forall i \in [m], \text{Lsb}(\mathcal{L}_{\beta,i}^{(1)}) := \text{Lsb}(\mathcal{L}_{\beta,i}^{(0)}) \oplus 1$$

(All other bits remain as sampled, so $\mathcal{L}^{(1)}$ is not derivable from $\mathcal{L}^{(0)}$.)

2. **Generate two child instances.** For $j \in \{\text{left}, \text{right}\}$, compute $(\text{msk}_j, \text{sk}_{j,\text{off}}) := \Pi_{\text{napPRF}}^{(k-1)}.\text{KeyGen}(1^\lambda; s_j)$. By construction, msk_j includes the child wire labels $\{\mathcal{L}_{\alpha,j,i}^{(0)}, \mathcal{L}_{\alpha,j,i}^{(1)}\}_{i \in [k-1]}$ and $\{\mathcal{L}_{\beta,j,i}^{(0)}, \mathcal{L}_{\beta,j,i}^{(1)}\}_{i \in [m]}$, whereas $\text{sk}_{j,\text{off}}$ is the level- $(k-1)$ offline key output by the recursive call (in a fixed canonical serialization).

3. Output $\text{msk} := \{\mathcal{L}_{\alpha,i}^{(0)}, \mathcal{L}_{\alpha,i}^{(1)}\}_{i \in [k]}$, $\{\mathcal{L}_{\beta,i}^{(0)}, \mathcal{L}_{\beta,i}^{(1)}\}_{i \in [m]}$ and sk_{off} that contains:
 - **Stacked child offline keys:** $\text{sk}_{\text{stack}} := \text{sk}_{\text{left},\text{off}} \oplus \text{sk}_{\text{right},\text{off}}$.
 - **Seed table $\mathcal{T}_{\text{seed}}$.** Let $r := \text{Lsb}(\mathcal{L}_{\alpha,1}^{(0)})$. Define

$$\mathcal{T}_{\text{seed}}[r] := \text{Enc}_{\mathcal{L}_{\alpha,1}^{(0)}}^{(1)}(\tau_{\text{seed},r}^{(k)}, s_{\text{right}}) \quad \mathcal{T}_{\text{seed}}[r \oplus 1] := \text{Enc}_{\mathcal{L}_{\alpha,1}^{(1)}}^{(1)}(\tau_{\text{seed},r \oplus 1}^{(k)}, s_{\text{left}})$$

- **α -translation tables $\{\mathcal{T}_{\alpha,i}\}_{i \in [k-1]}$.** Let $r := \text{Lsb}(\mathcal{L}_{\alpha,1}^{(0)})$.

For each $i \in [k-1]$: Let $j := i+1$, $c := \text{Lsb}(\mathcal{L}_{\alpha,j}^{(0)})$ and define

$$\begin{aligned} \mathcal{T}_{\alpha,i}[r][c] &:= \text{Enc}_{\mathcal{L}_{\alpha,1}^{(0)}, \mathcal{L}_{\alpha,j}^{(0)}}^{(2)}(\tau_{\alpha,i,r,c}^{(k)}, \mathcal{L}_{\alpha,\text{left},i}^{(0)} \parallel \mathcal{J}_{\alpha,i}) \\ \mathcal{T}_{\alpha,i}[r][c \oplus 1] &:= \text{Enc}_{\mathcal{L}_{\alpha,1}^{(0)}, \mathcal{L}_{\alpha,j}^{(1)}}^{(2)}(\tau_{\alpha,i,r,c \oplus 1}^{(k)}, \mathcal{L}_{\alpha,\text{left},i}^{(1)} \parallel \mathcal{J}_{\alpha,i}) \\ \mathcal{T}_{\alpha,i}[r \oplus 1][c] &:= \text{Enc}_{\mathcal{L}_{\alpha,1}^{(1)}, \mathcal{L}_{\alpha,j}^{(0)}}^{(2)}(\tau_{\alpha,i,r \oplus 1,c}^{(k)}, \mathcal{J}_{\alpha,i} \parallel \mathcal{L}_{\alpha,\text{right},i}^{(0)}) \\ \mathcal{T}_{\alpha,i}[r \oplus 1][c \oplus 1] &:= \text{Enc}_{\mathcal{L}_{\alpha,1}^{(1)}, \mathcal{L}_{\alpha,j}^{(1)}}^{(2)}(\tau_{\alpha,i,r \oplus 1,c \oplus 1}^{(k)}, \mathcal{J}_{\alpha,i} \parallel \mathcal{L}_{\alpha,\text{right},i}^{(1)}) \end{aligned}$$

- **β -translation tables $\{\mathcal{T}_{\beta,i}\}_{i \in [m]}$.** Let $r := \text{Lsb}(\mathcal{L}_{\alpha,1}^{(0)})$.

For each $i \in [m]$: Let $c := \text{Lsb}(\mathcal{L}_{\beta,i}^{(0)})$ and define

$$\begin{aligned} \mathcal{T}_{\beta,i}[r][c] &:= \text{Enc}_{\mathcal{L}_{\alpha,1}^{(0)}, \mathcal{L}_{\beta,i}^{(0)}}^{(2)}(\tau_{\beta,i,r,c}^{(k)}, \mathcal{L}_{\beta,\text{left},i}^{(0)} \parallel \mathcal{J}_{\beta,i}) \\ \mathcal{T}_{\beta,i}[r][c \oplus 1] &:= \text{Enc}_{\mathcal{L}_{\alpha,1}^{(0)}, \mathcal{L}_{\beta,i}^{(1)}}^{(2)}(\tau_{\beta,i,r,c \oplus 1}^{(k)}, \mathcal{L}_{\beta,\text{left},i}^{(1)} \parallel \mathcal{J}_{\beta,i}) \\ \mathcal{T}_{\beta,i}[r \oplus 1][c] &:= \text{Enc}_{\mathcal{L}_{\alpha,1}^{(1)}, \mathcal{L}_{\beta,i}^{(0)}}^{(2)}(\tau_{\beta,i,r \oplus 1,c}^{(k)}, \mathcal{J}_{\beta,i} \parallel \mathcal{L}_{\beta,\text{right},i}^{(0)}) \\ \mathcal{T}_{\beta,i}[r \oplus 1][c \oplus 1] &:= \text{Enc}_{\mathcal{L}_{\alpha,1}^{(1)}, \mathcal{L}_{\beta,i}^{(1)}}^{(2)}(\tau_{\beta,i,r \oplus 1,c \oplus 1}^{(k)}, \mathcal{J}_{\beta,i} \parallel \mathcal{L}_{\beta,\text{right},i}^{(1)}) \end{aligned}$$

Note that entries in each table are stored in the canonical order.

Figure 4: $\Pi_{\text{napPRF}}^{(k)}.\text{KeyGen}$ for domain $\{0, 1\}^k$ and range $\{0, 1\}^m$.

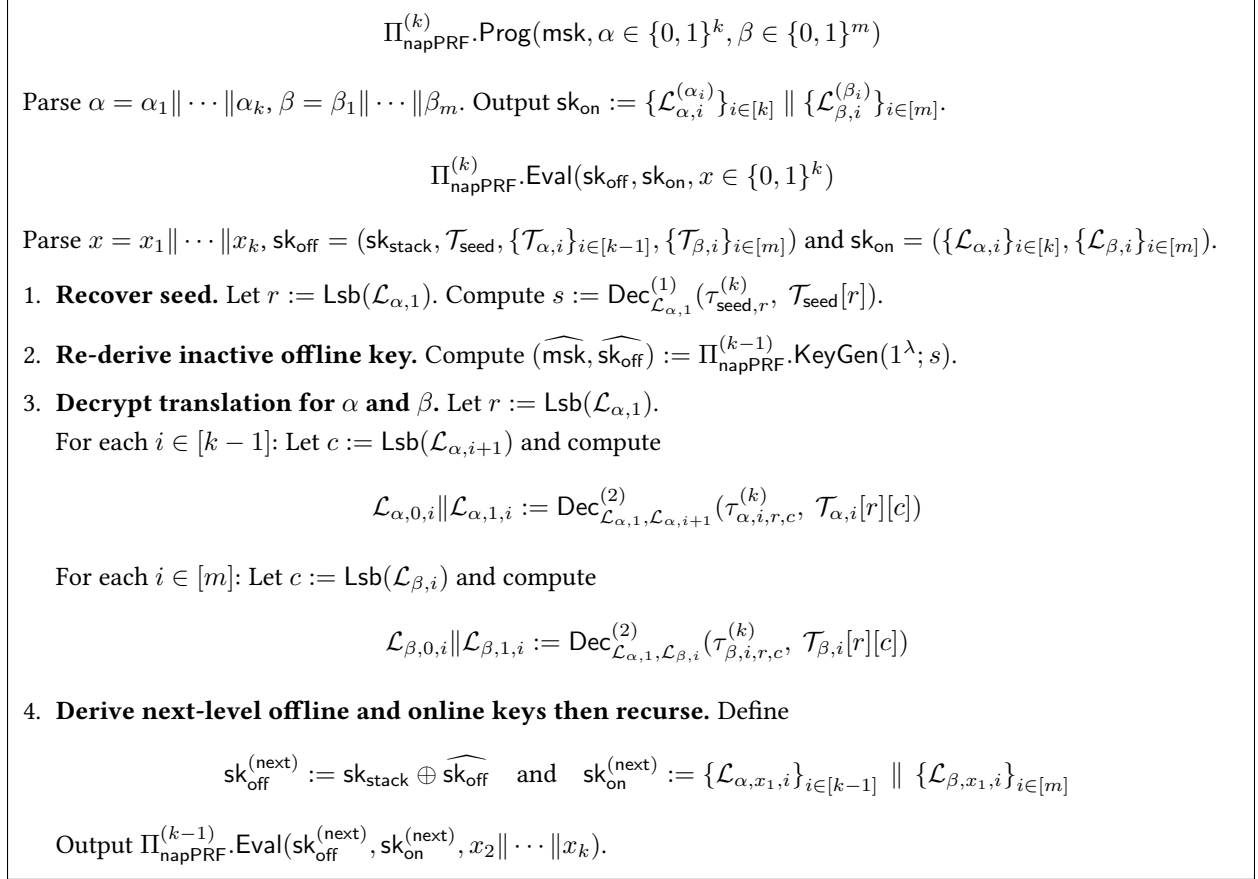


Figure 5: $\Pi_{\text{napPRF}}^{(k)}.\text{Prog}$ and $\Pi_{\text{napPRF}}^{(k)}.\text{Eval}$ for domain $\{0, 1\}^k$ and range $\{0, 1\}^m$.

Efficiency. We briefly discuss the efficiency of our final scheme $\Pi_{\text{napPRF}/\text{sipPRF}}^{(n)}$.

At each recursive step, sk_{off} grows only by the garbled tables introduced at that level: the two child offline keys are stacked (via XOR), so lower-level offline material is not duplicated across the recursion. Therefore, $|\text{sk}_{\text{off}}| = 8nm\lambda + 4n^2\lambda - 2n\lambda + m$. Moreover, sk_{on} consists of n λ -bit α -related labels together with m λ -bit β -related labels, giving $|\text{sk}_{\text{on}}| = (n + m)\lambda$.

We next analyze the computational complexity. Our metric is the number of calls to $\text{Enc}^{(1/2)}$ and $\text{Dec}^{(1/2)}$, which dominates the running time. Specifically:

- $\text{KeyGen}^{(n)}$ makes $(2^n - 1)(4m + 6) - 4n$ calls;
- $\text{Prog}^{(n)}$ makes 0 calls; and
- $\text{Eval}^{(n)}$ makes $(2^n - n - 1)(4m + 6) + n(m + 1) - \frac{3}{2}n(n - 1)$ calls.

The appearance of 2^n terms explains why the construction is intended for polynomial-size domains. Finally, we remark that $\Pi_{\text{napPRF}/\text{sipPRF}}^{(n)}$ admits a more efficient FullEval by reusing the inactive-branch offline key $\widehat{\text{sk}}_{\text{off}}$ (and decrypted components):

- $\text{FullEval}^{(n)}$ makes $2^{n+1}nm - 3(2^n - 1)m + 2^n(3n - 8) + 3n + 8$ calls.

7 Improving SGC via Short-Key Offline-Online sipPRFs

In this section, we show how our short-key offline-online sipPRF from Section 6.2 can be used — both in an off-the-shelf and a tailored manner — to improve the state-of-the-art stacked garbling (SGC) scheme LogStack [HK21], which optimizes (oblivious) branching in Yao garbled circuits.

Concretely, consider oblivious branching among $B = 2^n$ Boolean subcircuits $\mathcal{C}_0, \dots, \mathcal{C}_{B-1}$, where each $\mathcal{C}_j : \{0, 1\}^I \rightarrow \{0, 1\}$ has size C .⁵ Given selector-wire labels for $\alpha \in \{0, 1\}^n$ and input-wire labels for $x \in \{0, 1\}^I$, the evaluator must obtain the output label encoding $\mathcal{C}_\alpha(x)$. While LogStack achieves garbled size $\mathcal{O}(BI\lambda + C\lambda)$ with $\mathcal{O}(B \log B)$ computation for both parties, we reduce the $\mathcal{O}(BI\lambda)$ term without increasing the $\mathcal{O}(B \log B)$ computation.

A concise overview of stacked garbling appears in Section 2.2, and an overview of our sipPRF-based improvement appears in Section 2.3.

7.1 Off-the-Shelf Use of Our Offline-Online sipPRF

The additive $\mathcal{O}(BI\lambda)$ term in LogStack’s garbled-circuit size arises from a garbling gadget called the *demultiplexer*. At a high level, the demultiplexer enables E, given wire labels for the selector bits $\alpha_1, \dots, \alpha_n$ and the inputs $x_1, \dots, x_I$, to obtain *candidate* input-wire labels for every branch $t \in \{0, 1\}^n$. Crucially, on the *active* branch $t = \alpha := \alpha_1 \parallel \dots \parallel \alpha_n$, the candidate labels must encode the underlying input values $x_1, \dots, x_I$, while on every inactive branch $t \neq \alpha$ the candidate labels must be indistinguishable from the active ones. Moreover (for technical reasons in SGC), the candidate label vector on each inactive branch must be a fixed dummy value that is independent of x .

This is exactly the functionality provided by an offline-online sipPRF.⁶ Let $\beta \in \{0, 1\}^{I\lambda}$ denote the concatenation of the I input-wire labels for the active branch (on x). In principle, if E were given a programmed key $\text{sk} = (\text{sk}_{\text{off}}, \text{sk}_{\text{on}})$ with $\text{sk}_{\text{on}} := \text{Prog}(\text{msk}, \alpha, \beta)$, then for every $t \in \{0, 1\}^n$ it could compute a candidate label vector $\beta_t := \text{Eval}(\text{sk}, t)$. On-point correctness gives $\beta_\alpha = \beta$, while off-point correctness guarantees that for all $t \neq \alpha$ we have $\beta_t = \beta_\perp$ for a fixed dummy value $\beta_\perp$ that is independent of β (and hence of x) and distributed like valid labels. Since sk_{off} depends only on KeyGen, G can simply include sk_{off} as part of the garbled material. The remaining issue is how to deliver the online key sk_{on} to E (via garbling).

We exploit the internal structure of SGC (including LogStack): the active-branch input-wire labels are obtained by evaluating an n -level GGM tree along the selector path α and then expanding the resulting leaf seed via a PRG. Consequently, G can garble a circuit that (i) derives the leaf seed on the path $\alpha_1, \dots, \alpha_n$, (ii) expands it into the I input-wire label pairs for that branch, and (iii) for each $i \in [I]$, selects the label consistent with the evaluator’s input label x_i . Equivalently, from the selector-wire labels (encoding α) and the input-wire labels (encoding x), the circuit computes — in GC label form — the sipPRF programmed value β for the active branch.

Finally, recall that in our construction $\Pi_{\text{sipPRF}}^{(n)}$ (cf. Section 6.2), the procedure Prog outputs precisely the GC labels that constitute sk_{on} (i.e., the labels corresponding to the programmed index and the programmed value). Thus, once the circuit above computes the appropriate β , we can use standard garbled truth tables to let E get the corresponding sk_{on} from its labels for $\alpha_1, \dots, \alpha_n$ and β .

Garbled size. The dominant garbling components of this approach are:

⁵SGC supports multiple outputs; we focus on a single output.

⁶A standard sipPRF can also offer similar functionality; see Section 2.3.

1. Computing the GGM path seed costs $\mathcal{O}(n \cdot \text{poly}(\lambda))$.
2. Expanding the leaf seed via PRG to get $\mathcal{O}(I)$ input labels costs $\mathcal{O}(I \cdot \text{poly}(\lambda))$.
3. A garbled truth table that delivers the online key sk_{on} given GC labels for (α, β) . Here we use GC labels to encode (the bits of) $\beta \in \{0, 1\}^{I\lambda}$, so this contributes $\mathcal{O}((n + I\lambda)\lambda)$.
4. The offline key sk_{off} . In our instantiation (cf. Section 6.2) with range length $m = I\lambda$, we have $|\text{sk}_{\text{off}}| = \mathcal{O}(nI\lambda^2 + n^2\lambda)$.

Overall, this yields a demultiplexer of size $\mathcal{O}(nI \cdot \text{poly}(\lambda) + n^2\lambda)$, improving over the original $\mathcal{O}(BI\lambda)$ -size construction. With this new demultiplexer in place, the overall garbled-circuit size for branching in LogStack becomes $\mathcal{O}(nI \cdot \text{poly}(\lambda) + B\lambda + C\lambda)$, where the additive $\mathcal{O}(B\lambda)$ term is inherited from LogStack's oblivious seed-delivery gadget (called SortingHat).

Correctness and security. We use the sipPRF guarantees as follows:

- **On-point correctness.** Correct programming ensures that E obtains the intended input-wire labels on the active branch $t = \alpha$.
- **Fixed off-point behavior.** For every $t \neq \alpha$, sipPRF evaluation yields a fixed dummy value $\beta_{\perp}$ that is independent of β (and hence of x).
- **Index hiding.** Since β consists of GC labels (and thus is pseudorandom-looking), index hiding implies that sk does not reveal which branch α is active.

Computation complexity. We discuss how the (offline-online) sipPRF-based demultiplexer affects the computation complexity of both G and E. On the garbler side, generating the sipPRF keys and the additional garbled material (needed to let E recover sk_{on}) incurs work proportional to B (dominated by KeyGen). On the evaluator side, by our FullEval procedure, evaluating the sipPRF over the entire domain $\{0, 1\}^n$ incurs work proportional to $B \log B$.

Maintaining LogStack's overall computation complexity, however, requires more than matching these asymptotic bounds. In a subsequent *garbage-collection* step, G must predict, for each branch $t \in \{0, 1\}^n$, the dummy output that E obtains on branch t (when it is inactive) in order to combine B candidate outputs into a single garbled label. Without additional structure, this dummy output for a fixed t could vary with the full identity of the active index α , yielding up to B possibilities per branch and total work $\mathcal{O}(B^2)$ (indeed, this is the bottleneck in the original SGC scheme [HK20a]). LogStack avoids this blow-up by requiring *suffix invariance*: for each inactive branch $t \neq \alpha$, the dummy output on t depends on α only through the first divergence position ℓ between t and α (i.e., $t_{1..\ell} = \alpha_{1..\ell}$ but $t_{\ell+1} \neq \alpha_{\ell+1}$), which limits a fixed t to only $\mathcal{O}(n)$ possible dummy outcomes and keeps the total work at $\mathcal{O}(B \log B)$.

Our construction $\Pi_{\text{sipPRF}}^{(n)}$ from Section 6.2 satisfies this requirement when instantiated as the demultiplexer, because it is suffix-invariant.

Summary. We conclude with the following theorem.

Theorem 9. *Let $\mathcal{C}_0, \dots, \mathcal{C}_{B-1}$ be Boolean circuits, each of size C , sharing a common interface $\{0, 1\}^I \rightarrow \{0, 1\}$, and suppose $B = 2^n = \text{poly}(\lambda)$. Then there exists a (non-black-box) OWF-based garbling scheme —*

achieving the same correctness and security guarantees as LogStack [HK21] — such that, given garbled inputs for (α, x) , an evaluator can compute $y = \mathcal{C}_\alpha(x)$ with total garbled size

$$\mathcal{O}(nI \cdot \text{poly}(\lambda) + B\lambda + C\lambda)$$

Moreover, both the garbler and evaluator run in time $\mathcal{O}(B \log B)$.

7.2 Tailoring Our Offline-Online sipPRF for Better Integration

The sipPRF-based demultiplexer from Section 7.1 incurs a $\text{poly}(\lambda)$ overhead, and it must garble the GGM/PRG label-generation procedure inside a circuit. In this section, we tailor our sipPRF construction from Section 6.2 to LogStack, eliminating the need to garble GGM/PRG for a more efficient SGC integration.

Intuition. The $\text{poly}(\lambda)$ overhead in Section 7.1 is an artifact of composition. To produce the online key sk_{on} , the demultiplexer must first recover the active-branch input-label vector β , represented as GC wires (i.e., labels encoding labels). In SGC, these labels are generated by evaluating a GGM tree on the selector α and then expanding the resulting leaf seed via PRG. Thus, the off-the-shelf approach ends up garbling this entire GGM/PRG label-generation process.

We can avoid this by exploiting that the same “GGM tree $\rightarrow$ PRG expansion $\rightarrow$ wire labels” structure already appears in our sipPRF construction from Section 6.2. In $\Pi_{\text{sipPRF}}^{(n)}$, the master seed s_{master} deterministically induces a full binary tree of child seeds (e.g., $s_{\text{left}}, s_{\text{right}}$ at the top level), and each leaf seed is expanded via a PRG to generate the corresponding Yao wire labels. We therefore *couple* the randomness so that the sipPRF’s internal GC labels at the leaf corresponding to α are *exactly* the active-branch labels that LogStack would generate.

A strawman attempt. Suppose we instantiate $\Pi_{\text{sipPRF}}^{(n)}$ using the same top-level master seed and PRG as LogStack’s label-generation procedure. If we then program the sipPRF at index α with a *bit* $b \in \{0, 1\}$ (as opposed to the GC label for b), the PRG expansion at the leaf corresponding to α already determines the GC wire label for b (prior to the final LSB-based decoding). In particular, evaluating the sipPRF at α yields the desired active-branch wire label “for free,” without any additional garbled PRG computation.

However, this naïve coupling raises two issues:

1. In $\Pi_{\text{sipPRF}}^{(0)}$, sk_{off} includes output-decoding information that would allow E to decode a wire label (e.g., the GC label encoding the input of active branch) into the underlying bit. We can avoid this by simply omitting the decoding information from sk_{off} and treating sipPRF outputs purely as GC labels.
2. A second, more subtle issue arises from the SortingHat gadget in LogStack. To achieve the claimed $\mathcal{O}(B \log B)$ computation cost, SortingHat provides E with a *candidate seed* for every node of the balanced binary tree whose B leaves correspond to the branches. A node receives its *real* seed (from GGM) if and only if it is a sibling root of the active leaf α ; every other node receives a fixed dummy seed that is indistinguishable from a real one. At a high level, this ensures that for any fixed inactive branch, as α varies, unstacking yields only $\mathcal{O}(n)$ (i.e., $\mathcal{O}(\log B)$) possible dummy GCs rather than $\mathcal{O}(B)$.

These evaluator-visible seeds rule out the naive idea of reusing LogStack’s master seed as the top-level seed of $\Pi_{\text{sipPRF}}^{(n)}$. Indeed, $\Pi_{\text{sipPRF}}^{(n)}.\text{Eval}$ explicitly reveals a seed for the sibling subtree opposite the programmed path at the first divergence. If this seed were derived from the same master seed underlying SortingHat, then E could simply compare the seed leaked by Eval to the candidate seeds output by SortingHat to identify the corresponding sibling subtree (and, by iterating this argument down the tree, recover α). This would violate oblivious branching and break the security of SGC.

Absorbing SortingHat into the sipPRF seed delivery. To address the second issue, we “pad” the seed information exposed by $\Pi_{\text{sipPRF}}^{(n)}$ with the SortingHat view already present in LogStack. In other words, we tailor our offline-online sipPRF constructions in Section 6.2 by absorbing the SortingHat.

Concretely, rather than leak a single sibling seed via $\mathcal{T}_{\text{seed}}$, we drop $\mathcal{T}_{\text{seed}}$ altogether and let msk contain the seed material needed to emulate SortingHat: a full binary tree of GGM-expanded seeds together with an independent full binary tree of i.i.d. dummy seeds (sampled via the PRG). In Prog, we output the online key sk_{on} consisting of (i) the usual garbled labels encoding the programmed pair (α, β) and (ii) a *candidate-seed tree*, populated so that precisely the sibling roots on the path to α carry the *real* GGM seeds, while every other node carries the corresponding dummy seed. As a result, the seeds that Eval uses to unstack sk_{off} are delivered exactly as in SortingHat. Hence, the tailored sipPRF-based demultiplexer subsumes SortingHat’s candidate-seed view, and the new SGC no longer requires invoking it as a separate gadget.

We include formal descriptions of this LogStack-tailored sipPRF in Appendix D. Here we briefly discuss its Eval, correctness, security, and efficiency.

Eval. On input $(\text{sk}_{\text{off}}, \text{sk}_{\text{on}})$ and $x \in \{0, 1\}^n$, the Eval parses the online key sk_{on} into (i) the programmed labels for (α, β) and (ii) the candidate-seed tree. It then follows the Eval procedure of $\Pi_{\text{sipPRF}}^{(n)}$, except that whenever Eval would recover an (inactive) seed from $\mathcal{T}_{\text{seed}}$, it instead *reads* the corresponding sibling seed from the candidate-seed tree. Using this seed, Eval locally runs the next-level KeyGen routine to reconstruct an offline key sk_{off} for the sibling subtree, and recurses into the other subtree as dictated by the first bit of x . Note, Eval never needs to determine which candidate seeds are “real” versus “dummy.”

Correctness and suffix invariance. If $x = \alpha$, then at every level the sibling seed read from the candidate-seed tree is a *real* GGM seed, and hence the locally unstacked sk_{off} coincides with the offline key that KeyGen would generate along the programmed path. The recursion therefore reaches the leaf corresponding to α , which yields exactly the active-branch Yao wire-label vector encoding β .

If $x \neq \alpha$, let ℓ be the first position where x and α differ, and consider the recursive call made after this divergence. At level ℓ , Eval reads a *dummy* sibling seed from the candidate-seed tree, sampled independently of β ; consequently, the re-derived child offline key sk'_{off} is independent of β . Moreover, the child online key passed down is also independent of β : at this level Eval decrypts the translation ciphertexts to obtain the next-level labels, but on the diverged side these decryptions yield only fixed junk labels, and in our LogStack-tailored variant, *all* candidate seeds below this node are dummy. Thus, the recursive invocation receives a programmed key $(\text{sk}'_{\text{off}}, \text{sk}'_{\text{on}})$ whose distribution is independent of β .

Note that the same reasoning above also establishes suffix invariance.

Security. Security follows by essentially the same hybrid argument (cf. Appendix B.5) as for $\Pi_{\text{sipPRF}}^{(n)}$ in Section 6.2, and by the security of SortingHat [HK21].

Efficiency. This LogStack-tailored sipPRF incurs two additional overheads:

1. The online key sk_{on} additionally carries a full SortingHat-style tree of candidate seeds of total size $\mathcal{O}(B)$; this is acceptable in our application since SGC (including LogStack) already accounts for $\mathcal{O}(B)$ seed material.
2. In FullEval, each level invokes next-level KeyGen twice (once per candidate seed) rather than once, but this does not affect the asymptotic running time; the total computation complexity remains proportional to $B \log B$.

Summary. We conclude with the following theorem.

Theorem 10. *Let $\mathcal{C}_0, \dots, \mathcal{C}_{B-1}$ be Boolean circuits, each of size C , sharing a common interface $\{0, 1\}^I \rightarrow \{0, 1\}$, and suppose $B = 2^n = \text{poly}(\lambda)$. Then there exists a (black-box) OWF-based garbling scheme — achieving the same correctness and security guarantees as LogStack [HK21] — such that, given garbled inputs for (α, x) , an evaluator can compute $y = \mathcal{C}_\alpha(x)$ with total garbled size*

$$\mathcal{O}(nI\lambda + B\lambda + C\lambda)$$

Moreover, both the garbler and evaluator run in time $\mathcal{O}(B \log B)$.

Formal improved LogStack. With our tailored offline-online sipPRFs, the improved LogStack is largely plug-and-play: it simply replaces LogStack’s demultiplexer with our sipPRF-based one. For completeness, following [HK20a, HK21], we formalize the resulting SGC as a *garbling scheme* [BHR12] in Appendix E.

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A Pseudorandomness of Master Evaluation Procedures

In this section, we study adaptive programmable PRFs (cf. Section 3.4) and ask whether PRF-under-uniform-programming security implies that the corresponding master evaluation procedure MEval is itself a PRF.

Security (PRF, no programming). An apPRF scheme $\Pi = (\text{KeyGen}, \text{MEval}, \text{Prog}, \text{Eval})$ is *PRF-secure without programming* if the following oracle ensembles are computationally indistinguishable:

$$\left\{ \text{MEval}(\text{msk}, \cdot) \mid \text{msk} \leftarrow \text{KeyGen}(1^\lambda) \right\}_\lambda \approx_c \left\{ R(\cdot) \stackrel{\$}{\leftarrow} \text{Rand}(\mathcal{X} \rightarrow \mathcal{Y}) \right\}_\lambda$$

where $R : \mathcal{X} \rightarrow \mathcal{Y}$ is a uniformly random function. That is, when the programmed value is uniform, the induced oracle $x \mapsto \text{MEval}(\text{msk}, x)$ is a PRF.

A.1 Superpolynomial-Size Domains

When the domain size $|\mathcal{X}|$ is superpolynomial, PRF security under uniform programming further implies PRF security without programming.

Theorem 11 (PRF under uniform programming $\Rightarrow$ PRF without programming). *Let $\Pi_{\text{apPRF}} = (\text{KeyGen}, \text{MEval}, \text{Prog}, \text{Eval})$ be a correct apPRF over a domain $\mathcal{X}$ of superpolynomial size in λ . If Π_{apPRF} is PRF-secure under uniform programming, it is PRF-secure without programming.*

Proof. Fix any PPT distinguisher $\mathcal{A}$ making at most $q(\lambda)$ oracle queries. Consider the standard PRF experiment for $F_{\text{msk}}(\cdot) := \text{MEval}(\text{msk}, \cdot)$, where the oracle is either $\text{MEval}(\text{msk}, \cdot)$ for $\text{msk} \leftarrow \text{KeyGen}(1^\lambda)$ or a uniform random function $R \stackrel{\$}{\leftarrow} \text{Rand}(\mathcal{X} \rightarrow \mathcal{Y})$. We build a PPT distinguisher $\mathcal{B}$ for the *PRF-under-uniform-programming* experiment as follows.

Distinguisher $\mathcal{B}^{O(\cdot)}(1^\lambda)$. Sample $\alpha \stackrel{\$}{\leftarrow} \mathcal{X}$.⁷ Run $\mathcal{A}$ and answer each oracle query x of $\mathcal{A}$ as follows: if $x = \alpha$, then $\mathcal{B}$ aborts and outputs a uniform bit; otherwise, $\mathcal{B}$ forwards x to its own oracle $O(\cdot)$ and returns the reply to $\mathcal{A}$. If $\mathcal{A}$ halts and outputs b , then $\mathcal{B}$ outputs b .

In the *real* uniform-programming experiment, $\mathcal{B}$'s oracle is $O(\cdot) = \text{Eval}(\text{sk}, \cdot)$ for $\text{msk} \leftarrow \text{KeyGen}(1^\lambda)$, $\beta \stackrel{\$}{\leftarrow} \mathcal{Y}$, and $\text{sk} := \text{Prog}(\text{msk}, \alpha, \beta)$. In the *ideal* experiment, $O(\cdot) = R(\cdot)$ for $R \stackrel{\$}{\leftarrow} \text{Rand}(\mathcal{X} \rightarrow \mathcal{Y})$.

Let Bad be the event that $\mathcal{A}$ ever queries α (i.e., that $\mathcal{B}$ aborts). We first bound $\Pr[\text{Bad}]$ in either experiment. Conditioned on $\neg \text{Bad}$ up to any point, all oracle answers returned to $\mathcal{A}$ are taken at inputs $x \neq \alpha$. By apPRF correctness (off-point unchanged), for every $x \neq \alpha$ we have $\text{Eval}(\text{sk}, x) = \text{MEval}(\text{msk}, x)$, so the answers seen by $\mathcal{A}$ up to that point are distributed independently of α . Therefore, for each query index i ,

$$\Pr[x_i = \alpha \mid \neg \text{Bad on the first } i-1 \text{ queries}] = \frac{1}{|\mathcal{X}|}$$

and by a union bound,

$$\Pr[\text{Bad}] \leq \frac{q(\lambda)}{|\mathcal{X}|} \tag{1}$$

Next, we relate $\mathcal{A}$'s view in the standard PRF experiment for MEval to $\mathcal{B}$'s view in the uniform-programming PRF experiment. In the *real* case, conditioned on $\neg \text{Bad}$, every query made by $\mathcal{A}$ is some

⁷We use that PRF-security under uniform programming holds for every fixed $\alpha \in \mathcal{X}$; in particular, it holds for the value of α sampled above.

$x \neq \alpha$, hence by correctness $\text{Eval}(\text{sk}, x) = \text{MEval}(\text{msk}, x)$ for all answered queries; thus $\mathcal{A}$'s entire transcript is identically distributed to its transcript when given oracle access to $\text{MEval}(\text{msk}, \cdot)$. In the *ideal* case, conditioned on $\neg \text{Bad}$, $\mathcal{A}$ receives answers from a uniformly random function in both experiments. Note, $\mathcal{B}$ outputs a random bit upon Bad , so

$$\text{Adv}_{\text{MEval}}^{\text{prf}}(\mathcal{A}) \leq \text{Adv}_{\Pi_{\text{apPRF}}}^{\text{uProg-prf}}(\mathcal{B}) + \Pr[\text{Bad}] \leq \text{Adv}_{\Pi_{\text{apPRF}}}^{\text{uProg-prf}}(\mathcal{B}) + \frac{q(\lambda)}{|\mathcal{X}|}$$

by (1). Finally, since Π_{apPRF} is PRF-secure under uniform programming, the first term is negligible. Since $q(\lambda)$ is polynomial and $|\mathcal{X}|$ is superpolynomial, $q(\lambda)/|\mathcal{X}|$ is negligible. Hence $\text{Adv}_{\text{MEval}}^{\text{prf}}(\mathcal{A})$ is negligible for every PPT $\mathcal{A}$, so $\text{MEval}(\text{msk}, \cdot)$ is a standard PRF (i.e., Π is PRF-secure without programming). $\square$

A.2 Polynomial-Size Domains

In contrast, for polynomial-size domains, this implication can fail. We give a simple counterexample.

Let $\mathcal{X} = \{0, 1\}^n$ with $2^n = \text{poly}(\lambda)$ and $\mathcal{Y} = \{0, 1\}$. Define KeyGen to sample a master key msk that encodes the full truth table of $\text{MEval}(\text{msk}, \cdot)$ as follows: choose the first $2^n - 1$ outputs uniformly at random, and set the last output so that the overall parity is zero, that is,

$$\bigoplus_{x \in \{0, 1\}^n} \text{MEval}(\text{msk}, x) = 0$$

Then MEval is clearly *not* a PRF: a distinguisher can query all $x \in \{0, 1\}^n$ and check whether the XOR of the replies is zero.

Now define the programming procedure $\text{Prog}(\text{msk}, \alpha, \beta)$ to output a programmed key sk that is identical to msk except that the entry at position α is overwritten with β (and $\text{Eval}(\text{sk}, \cdot)$ simply reads the resulting table). If $\beta \xleftarrow{\$} \{0, 1\}$ is uniform, then $\text{Eval}(\text{sk}, \cdot)$ is distributed exactly as a uniformly random truth table over $\{0, 1\}^n \rightarrow \{0, 1\}$: the parity constraint is destroyed by re-randomizing a single entry. Hence, Π can satisfy PRF-under-uniform-programming security even though MEval is not a PRF (i.e., not PRF-secure without programming).

B Deferred Proofs

B.1 Proof of Lemma 2

Lemma 2. Let $\Pi = (\text{Gen}, \text{Eval})$ be a two-party DPF scheme. Suppose Π is *correct* and *secure* in the sense of per-share simulation security for both shares (defined in Section 3.2). Then Π satisfies *individual pseudorandomness*.

Proof. Fix $b = 1$; the argument for $b = 2$ is symmetric. For a key k , write $F_k : \mathcal{X} \rightarrow \mathcal{Y}$ for the induced function $F_k(x) := \text{Eval}(k, x)$.

Step 1a: Single-point shift invariance. Fix any $\alpha \in \mathcal{X}$ and $\beta \in \mathcal{Y}$. For each λ , sample $(k_{1,\lambda}^{\alpha,\beta}, k_{2,\lambda}^{\alpha,\beta}) \leftarrow \text{Gen}(1^\lambda, \alpha, \beta)$, and define the induced (random) functions $F_{1,\lambda}^{\alpha,\beta}(x) := \text{Eval}(k_{1,\lambda}^{\alpha,\beta}, x)$ and $F_{2,\lambda}^{\alpha,\beta}(x) := \text{Eval}(k_{2,\lambda}^{\alpha,\beta}, x)$.

By correctness, for every λ , we have for all $x \in \mathcal{X}$:

$$F_{1,\lambda}^{\alpha,\beta}(x) + F_{2,\lambda}^{\alpha,\beta}(x) = P_{\alpha,\beta}(x)$$

and hence, as functions,

$$F_{2,\lambda}^{\alpha,\beta} = P_{\alpha,\beta} - F_{1,\lambda}^{\alpha,\beta} \quad (*)$$

By security of share 2, for all β ,

$$\{k_{2,\lambda}^{\alpha,\beta}\}_\lambda \approx_c \{\text{Sim}_2(1^\lambda)\}_\lambda$$

Since Eval is PPT, this lifts to oracle indistinguishability, and in particular,

$$\{F_{2,\lambda}^{\alpha,\beta}\}_\lambda \approx_c \{F_{2,\lambda}^{\alpha,0}\}_\lambda \quad (\text{as oracles})$$

Combining with $(*)$ for β and for $\beta = 0$, we obtain

$$P_{\alpha,\beta} - \{F_{1,\lambda}^{\alpha,\beta}\}_\lambda \approx_c -\{F_{1,\lambda}^{\alpha,0}\}_\lambda \quad (\text{as oracles})$$

or equivalently,

$$\{F_{1,\lambda}^{\alpha,\beta}\}_\lambda \approx_c \{F_{1,\lambda}^{\alpha,0}\}_\lambda + P_{\alpha,\beta} \quad (\text{as oracles})$$

By security of share 1, $\{F_{1,\lambda}^{\alpha,\beta}\}_\lambda \approx_c \{F_{1,\lambda}^{\alpha,0}\}_\lambda$ as oracles, yielding

$$\{F_{1,\lambda}^{\alpha,0}\}_\lambda \approx_c \{F_{1,\lambda}^{\alpha,0}\}_\lambda + P_{\alpha,\beta} \quad (\text{as oracles}) \quad (1)$$

We remark that since α was arbitrary, (1) holds for every fixed $\alpha \in \mathcal{X}$ and $\beta \in \mathcal{Y}$.

Step 1b: Global shift invariance. Fix arbitrary $\alpha^* \in \mathcal{X}$ and $\beta^* \in \mathcal{Y}$, and let $F_\lambda^* := F_{1,\lambda}^{\alpha^*,\beta^*}$. Now, for any $\alpha \in \mathcal{X}$ and $\beta \in \mathcal{Y}$, by per-share simulation security for share 1, we have

$$\{F_\lambda^*\}_\lambda \approx_c \{F_{1,\lambda}^{\alpha,0}\}_\lambda \quad (\text{as oracles}) \quad (2)$$

Post-processing (2) by adding $P_{\alpha,\beta}$ yields

$$\{F_\lambda^*\}_\lambda + P_{\alpha,\beta} \approx_c \{F_{1,\lambda}^{\alpha,0}\}_\lambda + P_{\alpha,\beta} \quad (\text{as oracles}) \quad (3)$$

Applying (1) and combining with (2) and (3) yields:

$$\{F_\lambda^*\}_\lambda \approx_c \{F_{1,\lambda}^{\alpha,0}\}_\lambda \approx_c \{F_{1,\lambda}^{\alpha,0}\}_\lambda + P_{\alpha,\beta} \approx_c \{F_\lambda^*\}_\lambda + P_{\alpha,\beta} \quad (\text{as oracles}) \quad (4)$$

I.e., the oracle ensemble induced by a single DPF share is computationally indistinguishable from the same ensemble after adding an arbitrary point function.

Step 2: Hybrid argument (adaptive queries). Fix a stateful PPT adversary $\mathcal{A}$ with oracle access, and let $Q(\lambda)$ be a polynomial upper bound on its total number of oracle queries. We define a sequence of hybrid oracles $\{H_{i,\lambda}\}_{i=0}^{Q(\lambda)}$. Each hybrid maintains a table $T : \mathcal{X} \rightarrow \mathcal{Y}$ (initially empty) and a counter d (initially 0) that counts the number of *distinct* query points seen so far. On a query $x \in \mathcal{X}$, the oracle responds as follows:

1. If $T[x]$ is defined, return $T[x]$.
2. Otherwise increment $d \leftarrow d + 1$ and:
 - if $d \leq i$, sample $T[x] \xleftarrow{\$} \mathcal{Y}$ and return $T[x]$;

- if $d > i$, set $T[x] := F_\lambda^*(x)$ and return $T[x]$.

Equivalently, $H_{i,\lambda}$ answers the first i *distinct* queries according to a lazy-sampled truly random function table, and answers all later (new) points by F_λ^* .

Let $\mathcal{V}_{i,\lambda}$ denote $\mathcal{A}$'s output bit in an interaction with $H_{i,\lambda}$ (including $\mathcal{A}$'s internal randomness).

Claim 2.1. For every $i \in \{1, \dots, Q(\lambda)\}$, $\{\mathcal{V}_{i-1,\lambda}\}_\lambda \approx_c \{\mathcal{V}_{i,\lambda}\}_\lambda$.

Proof of Claim 2.1. Suppose for contradiction there exists an i and a distinguisher $\mathcal{A}$ such that $\{\mathcal{V}_{i-1,\lambda}\}_\lambda$ and $\{\mathcal{V}_{i,\lambda}\}_\lambda$ are distinguishable with non-negligible advantage. We construct a PPT distinguisher $\mathcal{D}$ that breaks the shift-invariance property (4). Distinguisher $\mathcal{D}$ is given oracle access to $O(\cdot)$, where O is either $F_\lambda^*(\cdot)$ or $F_\lambda^*(\cdot) + P_{\alpha,\beta}(\cdot)$ for some (α, β) . $\mathcal{D}$ runs $\mathcal{A}$ and answers $\mathcal{A}$'s oracle queries by simulating an oracle $\tilde{H}(\cdot)$ defined as follows. $\tilde{H}$ maintains a table T and a counter d exactly as above. On a query x :

1. If $T[x]$ is defined, return $T[x]$.
2. Otherwise increment $d \leftarrow d + 1$.
 - If $d < i$, sample $T[x] \xleftarrow{\$} \mathcal{Y}$ and return $T[x]$.
 - If $d = i$, set $\alpha := x$ and sample $\beta \xleftarrow{\$} \mathcal{Y}$. Then query $y \leftarrow O(x)$, define $T[x] := y - \beta$, and return $T[x]$.
 - If $d > i$, query $y \leftarrow O(x)$, set $T[x] := y$, and return $T[x]$.

Finally, $\mathcal{D}$ outputs whatever bit $\mathcal{A}$ outputs.

We emphasize that (α, β) are fixed by $\mathcal{D}$ *prior to learning any information about the challenger's oracle* O . Concretely, until the moment $d = i$, $\mathcal{D}$ answers $\mathcal{A}$ using only its own fresh randomness (the values sampled into $T[\cdot]$) and does not receive any oracle reply from O . Thus the random variables (α, β) depend only on $\mathcal{A}$'s internal coins and on $\mathcal{D}$'s fresh uniform samples, and are therefore independent of whether the challenger sampled O as F_λ^* or as $F_\lambda^* + P_{\alpha,\beta}$. This is crucial: it ensures that $\mathcal{D}$ indeed contradicts (4), which guarantees indistinguishability for every *fixed* (α, β) (and hence also for any (α, β) sampled independently of the challenge bit).

We now analyze the simulation under the two cases for O .

Case 1: $O = F_\lambda^*$. For the first $i - 1$ distinct points, $\tilde{H}$ returns independent uniform values, matching $H_{i,\lambda}$. On the i -th distinct point $x = \alpha$, $\tilde{H}$ returns $F_\lambda^*(\alpha) - \beta$, which is uniform over $\mathcal{Y}$ (since β is uniform and independent), and is stored for repeat-consistency. All later new points are answered by F_λ^* . Thus $\tilde{H}$ is distributed identically to $H_{i,\lambda}$, and hence $\mathcal{D}^{F_\lambda^*}$ perfectly simulates $\mathcal{A}^{H_{i,\lambda}}$.

Case 2: $O = F_\lambda^* + P_{\alpha,\beta}$. For the first $i - 1$ distinct points, the answers are again independent uniform values. On the i -th distinct point $x = \alpha$, we have $O(\alpha) = F_\lambda^*(\alpha) + \beta$, and so $\tilde{H}$ returns $O(\alpha) - \beta = F_\lambda^*(\alpha)$. All later new points are answered by $O(x) = F_\lambda^*(x)$ since $P_{\alpha,\beta}(x) = 0$ for $x \neq \alpha$. Thus $\tilde{H}$ is distributed identically to $H_{i-1,\lambda}$, and hence $\mathcal{D}^{F_\lambda^* + P_{\alpha,\beta}}$ perfectly simulates $\mathcal{A}^{H_{i-1,\lambda}}$.

Therefore, $\mathcal{D}$ distinguishes the oracle ensembles F_λ^* and $F_\lambda^* + P_{\alpha,\beta}$ with the same non-negligible advantage as $\mathcal{A}$ distinguishes $H_{i,\lambda}$ from $H_{i-1,\lambda}$, contradicting (4). This proves the claim. $\square$

Step 3: Conclude individual pseudorandomness. By a hybrid argument over $i = 0, 1, \dots, Q(\lambda)$ and Claim 2.1, we obtain

$$\{H_{0,\lambda}\}_\lambda \approx_c \{H_{Q(\lambda),\lambda}\}_\lambda \quad (\text{as oracles})$$

Observe that $H_{0,\lambda}$ answers every query as $F_\lambda^*(\cdot)$, and therefore $H_{0,\lambda} \equiv F_\lambda^*$. Moreover, $H_{Q(\lambda),\lambda}$ answers all (distinct) queries using the lazy-sampled table T , and hence is distributed identically to a truly random function oracle $R \leftarrow \text{Rand}(\mathcal{X} \rightarrow \mathcal{Y})$. Therefore, we have for any $\alpha^* \in \mathcal{X}$ and $\beta^* \in \mathcal{Y}$:

$$\{F_\lambda^*\}_\lambda \approx_c \{R \leftarrow \text{Rand}(\mathcal{X} \rightarrow \mathcal{Y})\}_\lambda \quad (\text{as oracles})$$

proving that the share-induced oracle ensemble is PRF-like. $\square$

B.2 Proof of Theorem 5

Theorem 5. If $\Pi_{\text{napPRF}} = (\text{KeyGen}, \text{Prog}, \text{Eval})$ is a *correct* and *index-hiding* napPRF for a single point over domain $\mathcal{X}$ and range $\mathcal{Y}$, then the construction in Figure 1 yields a two-party DPF $\Pi_{\text{DPF}} = (\text{Gen}, \text{Eval})$ over domain $\mathcal{X}$ and range $\mathcal{Y}$ that is *correct* and *secure* (defined in Section 3.2).

Proof. We prove correctness and security for our induced Π_{DPF} .

Correctness. Let $(k_1, k_2) \leftarrow \Pi_{\text{DPF}}.\text{Gen}(1^\lambda, \alpha, \beta)$, where $k_1 = \text{sk}^{(1)}$ and $k_2 = \text{sk}^{(2)}$ with $\text{msk} \leftarrow \Pi_{\text{napPRF}}.\text{KeyGen}(1^\lambda)$, $r \xleftarrow{\$} \mathcal{Y}$, and

$$\text{sk}^{(1)} := \Pi_{\text{napPRF}}.\text{Prog}(\text{msk}, \alpha, r) \quad \text{and} \quad \text{sk}^{(2)} := \Pi_{\text{napPRF}}.\text{Prog}(\text{msk}, \alpha, r \oplus \beta)$$

At $x = \alpha$, by correct programming of the napPRF,

$$\begin{aligned} & \Pi_{\text{DPF}}.\text{Eval}(k_1, \alpha) \oplus \Pi_{\text{DPF}}.\text{Eval}(k_2, \alpha) \\ &= \Pi_{\text{napPRF}}.\text{Eval}(\text{sk}^{(1)}, \alpha) \oplus \Pi_{\text{napPRF}}.\text{Eval}(\text{sk}^{(2)}, \alpha) = r \oplus (r \oplus \beta) = \beta \end{aligned}$$

For any $x \neq \alpha$, by off-point value independence of the napPRF applied to $\text{sk}^{(1)}$ and $\text{sk}^{(2)}$ (which share the same index α),

$$\begin{aligned} & \Pi_{\text{DPF}}.\text{Eval}(k_1, x) \oplus \Pi_{\text{DPF}}.\text{Eval}(k_2, x) \\ &= \Pi_{\text{napPRF}}.\text{Eval}(\text{sk}^{(1)}, x) \oplus \Pi_{\text{napPRF}}.\text{Eval}(\text{sk}^{(2)}, x) = 0 \end{aligned}$$

Therefore, $\Pi_{\text{DPF}}.\text{Eval}(k_1, \cdot) \oplus \Pi_{\text{DPF}}.\text{Eval}(k_2, \cdot) = P_{\alpha, \beta}(\cdot)$.

Security. Fix $b \in \{1, 2\}$. We show that there exists a PPT simulator Sim_b such that for all $\alpha \in \mathcal{X}$ and $\beta \in \mathcal{Y}$,

$$\left\{ k_b \mid (k_1, k_2) \leftarrow \Pi_{\text{DPF}}.\text{Gen}(1^\lambda, \alpha, \beta) \right\}_\lambda \approx_c \left\{ \text{Sim}_b(1^\lambda) \right\}_\lambda$$

Define $\text{Sim}_1 := \text{Sim}_2 := \text{Sim}$, where Sim is the simulator guaranteed by the napPRF index-hiding property. It remains to show that, for each b , the real distribution of k_b is exactly an instance covered by napPRF index hiding.

Case $b = 1$. By construction of $\Pi_{\text{DPF}}.\text{Gen}$, we have $k_1 = \text{sk}^{(1)}$ where

$$\text{msk} \leftarrow \Pi_{\text{napPRF}}.\text{KeyGen}(1^\lambda), \quad r \xleftarrow{\$} \mathcal{Y}, \quad \text{sk}^{(1)} := \Pi_{\text{napPRF}}.\text{Prog}(\text{msk}, \alpha, r)$$

Thus $\{k_1\}_\lambda$ is exactly the LHS ensemble in the napPRF index-hiding definition (with uniform programmed value), and hence $\{k_1\}_\lambda \approx_c \{\text{Sim}(1^\lambda)\}_\lambda$.

Case $b = 2$. By construction, $k_2 = \text{Prog}(\text{msk}, \alpha, r \oplus \beta)$ for uniform $r \xleftarrow{\$} \mathcal{Y}$. Since $r \oplus \beta$ is uniform over $\mathcal{Y}$ for every fixed β , $\{k_2\}_\lambda$ is identically distributed to the LHS ensemble in the napPRF index-hiding definition (i.e., with a uniform programmed value), and hence $\{k_2\}_\lambda \approx_c \{\text{Sim}(1^\lambda)\}_\lambda$.

Combining the two cases, we conclude that for each $b \in \{1, 2\}$ there exists a PPT simulator Sim_b satisfying the DPF security definition. $\square$

B.3 Proof of Theorem 6

Theorem 6. If $\Pi_{\text{DPF}} = (\text{Gen}, \text{Eval})$ is a *correct* and *secure* two-party DPF over domain $\mathcal{X}$ and range $\{0, 1\}$, then the construction in Figure 2 yields a $\text{napPRF } \Pi_{\text{napPRF}} = (\text{KeyGen}, \text{Prog}, \text{Eval})$ over domain $\mathcal{X}$ and range $\mathcal{Y}$ that is *correct* and *index-hiding* (defined in Section 4).

Proof. We prove correctness and security (index-hiding) for our induced Π_{napPRF} .

Correctness. Fix any $\lambda \in \mathbb{N}$ and any $\text{msk} \leftarrow \Pi_{\text{napPRF}}.\text{KeyGen}(1^\lambda)$. Write $\text{msk} = \text{msk}_1 \parallel \dots \parallel \text{msk}_m$ with each $\text{msk}_i \in \{0, 1\}^R$.

(1) *Correct programming at α .* Fix any $\alpha \in \mathcal{X}$ and any $\beta \in \mathcal{Y} = \{0, 1\}^m$, and write $\beta = \beta_1 \parallel \dots \parallel \beta_m$. Let $\text{sk} := \Pi_{\text{napPRF}}.\text{Prog}(\text{msk}, \alpha, \beta)$ and parse $\text{sk} = k^{(1)} \parallel \dots \parallel k^{(m)}$. For each $i \in [m]$, Prog computes

$$(k_1^{(i)}, k_2^{(i)}) := \Pi_{\text{DPF}}.\text{Gen}(1^\lambda, \alpha, 1; \text{msk}_i)$$

and then sets $k^{(i)} \in \{k_1^{(i)}, k_2^{(i)}\}$ so that $\Pi_{\text{DPF}}.\text{Eval}(k^{(i)}, \alpha) = \beta_i$. This choice is well-defined because DPF correctness for the point function $P_{\alpha, 1}$ guarantees

$$\Pi_{\text{DPF}}.\text{Eval}(k_1^{(i)}, \alpha) \oplus \Pi_{\text{DPF}}.\text{Eval}(k_2^{(i)}, \alpha) = P_{\alpha, 1}(\alpha) = 1,$$

hence exactly one of the two bits $\Pi_{\text{DPF}}.\text{Eval}(k_1^{(i)}, \alpha)$ and $\Pi_{\text{DPF}}.\text{Eval}(k_2^{(i)}, \alpha)$ equals β_i . Therefore,

$$\Pi_{\text{napPRF}}.\text{Eval}(\text{sk}, \alpha) = \Pi_{\text{DPF}}.\text{Eval}(k^{(1)}, \alpha) \parallel \dots \parallel \Pi_{\text{DPF}}.\text{Eval}(k^{(m)}, \alpha) = \beta_1 \parallel \dots \parallel \beta_m = \beta$$

(2) *Off-point value independence.* Fix any $\alpha \in \mathcal{X}$, any $\beta^{(1)}, \beta^{(2)} \in \mathcal{Y}$, and any $x \in \mathcal{X} \setminus \{\alpha\}$. Let $\text{sk}^{(j)} := \Pi_{\text{napPRF}}.\text{Prog}(\text{msk}, \alpha, \beta^{(j)})$ for $j \in \{1, 2\}$, and parse $\text{sk}^{(j)} = k^{(1,j)} \parallel \dots \parallel k^{(m,j)}$.

Fix any coordinate $i \in [m]$. Both executions of Prog (with $\beta^{(1)}$ and with $\beta^{(2)}$) invoke DPF key generation with the *same* inputs and *the same* fixed randomness:

$$(k_1^{(i)}, k_2^{(i)}) := \Pi_{\text{DPF}}.\text{Gen}(1^\lambda, \alpha, 1; \text{msk}_i)$$

They may differ only in whether they output $k_1^{(i)}$ or $k_2^{(i)}$ as the programmed key for coordinate i . However, since $x \neq \alpha$, DPF correctness implies

$$\Pi_{\text{DPF}}.\text{Eval}(k_1^{(i)}, x) \oplus \Pi_{\text{DPF}}.\text{Eval}(k_2^{(i)}, x) = P_{\alpha, 1}(x) = 0$$

and hence $\Pi_{\text{DPF}}.\text{Eval}(k_1^{(i)}, x) = \Pi_{\text{DPF}}.\text{Eval}(k_2^{(i)}, x)$. Therefore, regardless of whether $k^{(i,j)}$ equals $k_1^{(i)}$ or $k_2^{(i)}$, we have

$$\Pi_{\text{DPF}}.\text{Eval}(k^{(i,1)}, x) = \Pi_{\text{DPF}}.\text{Eval}(k_1^{(i)}, x) = \Pi_{\text{DPF}}.\text{Eval}(k_2^{(i)}, x) = \Pi_{\text{DPF}}.\text{Eval}(k^{(i,2)}, x)$$

This holds for every $i \in [m]$, so concatenating over i yields

$$\Pi_{\text{napPRF}}.\text{Eval}(\text{sk}^{(1)}, x) = \Pi_{\text{napPRF}}.\text{Eval}(\text{sk}^{(2)}, x)$$

establishing off-point value independence.

Security (index-hiding). We show that when $\beta \xleftarrow{\$} \mathcal{Y}$ is uniform, the distribution of sk can be simulated without α .

Fix any $\alpha \in \mathcal{X}$. For each $i \in [m]$ and each λ , let $K_{1,\lambda}^{(i)}, K_{2,\lambda}^{(i)}$ denote the random variables distributed as the two DPF keys output by $\Pi_{\text{DPF}}.\text{Gen}(1^\lambda, \alpha, 1; \text{msk}_i)$ for $\text{msk}_i \xleftarrow{\$} \{0, 1\}^R$, and let $K_\lambda^{(i)}$ denote the random variable output by Prog as $k^{(i)}$.

In the real experiment, for each $i \in [m]$, the napPRF scheme samples msk_i and computes $(k_1^{(i)}, k_2^{(i)}) := \Pi_{\text{DPF}}.\text{Gen}(1^\lambda, \alpha, 1; \text{msk}_i)$. Let $s_i := \Pi_{\text{DPF}}.\text{Eval}(k_1^{(i)}, \alpha) \in \{0, 1\}$. By DPF correctness, $\Pi_{\text{DPF}}.\text{Eval}(k_2^{(i)}, \alpha) = s_i \oplus 1$. Since $\beta_i \xleftarrow{\$} \{0, 1\}$ is uniform and independent of $(k_1^{(i)}, k_2^{(i)})$, the output key $k^{(i)}$ satisfies

$$\Pr[k^{(i)} = k_1^{(i)} \mid k_1^{(i)}, k_2^{(i)}] = \Pr[\beta_i = s_i] = 1/2 \quad \Pr[k^{(i)} = k_2^{(i)} \mid k_1^{(i)}, k_2^{(i)}] = 1/2$$

and hence for every fixed λ ,

$$\text{Dist}(K_\lambda^{(i)}) = \frac{1}{2} \text{Dist}(K_{1,\lambda}^{(i)}) + \frac{1}{2} \text{Dist}(K_{2,\lambda}^{(i)}) \quad (2)$$

where $(K_{1,\lambda}^{(i)}, K_{2,\lambda}^{(i)}) \leftarrow \Pi_{\text{DPF}}.\text{Gen}(1^\lambda, \alpha, 1)$.

By DPF security, for each $b \in \{1, 2\}$ there exists a PPT simulator Sim_b such that for all α , $\{K_{b,\lambda}^{(i)}\}_\lambda \approx_c \{\text{Sim}_b(1^\lambda)\}_\lambda$. Combining with (2), we obtain

$$\{K_\lambda^{(i)}\}_\lambda \approx_c \left\{ \frac{1}{2} \text{Sim}_1(1^\lambda) + \frac{1}{2} \text{Sim}_2(1^\lambda) \right\}_\lambda$$

That is, $K_\lambda^{(i)}$ is indistinguishable from the output of the PPT procedure that samples $c \xleftarrow{\$} \{1, 2\}$ and returns $\text{Sim}_c(1^\lambda)$, which is independent of α .

Define a simulator Sim for the napPRF experiment as follows: On input 1^λ , for each $i \in [m]$ sample $c_i \xleftarrow{\$} \{1, 2\}$ and set $k^{(i)} \leftarrow \text{Sim}_{c_i}(1^\lambda)$; finally output $\text{sk} := k^{(1)} \parallel \dots \parallel k^{(m)}$. A standard hybrid argument over $i \in [m]$ (replacing one coordinate at a time) shows that the distribution of sk in the real experiment is computationally indistinguishable from $\text{Sim}(1^\lambda)$. This proves index hiding. $\square$

B.4 Proof of Theorem 7

Theorem 7. Let G be a PRG. $\Pi_{\text{napPRF}}^{(0)}$ (cf. Figure 3) is a correct and index-hiding offline-online napPRF over domain $\{0, 1\}^0$ and range $\{0, 1\}^m$.

Proof. Correctness is trivial. For security (index-hiding), since $\mathcal{X} = \{0, 1\}^0$ has a single element (i.e., ϵ), a perfect simulator is: sample $(\text{msk}, \text{sk}_{\text{off}}) \leftarrow \text{KeyGen}^{(0)}(1^\lambda)$, sample $\beta \xleftarrow{\$} \{0, 1\}^m$, set $\text{sk}_{\text{on}} := \text{Prog}^{(0)}(\text{msk}, \epsilon, \beta)$, and output $(\text{sk}_{\text{off}}, \text{sk}_{\text{on}})$.

We also note a simpler simulator that will be useful later: $\text{Sim}^{(0)}(1^\lambda)$ runs $(\text{msk}, \text{sk}_{\text{off}}) \leftarrow \text{KeyGen}^{(0)}(1^\lambda)$ except it replaces $G(s_{\text{master}})$ with $2m$ i.i.d. uniform λ -bit strings, samples $U^{(0)} \xleftarrow{\$} \{0, 1\}^{m\lambda}$, and outputs $(\text{sk}_{\text{off}}, U^{(0)})$. This works because of the PRG security and the following simple fact. Fix any bit $p \in \{0, 1\}$. Let $(\mathcal{L}^{(0)}, \mathcal{L}^{(1)})$ be sampled as two i.i.d. uniform λ -bit strings, except that we overwrite the last bits so that $\text{Lsb}(\mathcal{L}^{(0)}) = p$ and $\text{Lsb}(\mathcal{L}^{(1)}) = p \oplus 1$. Let $\zeta \xleftarrow{\$} \{0, 1\}$ and define $\mathcal{L} := \mathcal{L}^{(\zeta)}$. Then $\mathcal{L}$ is uniform over $\{0, 1\}^\lambda$ and independent of p . $\square$

B.5 Proof of Theorem 8

Theorem 8. Let G be a PRG and let F be a PRF. For every $k \geq 1$, the scheme $\Pi_{\text{napPRF}}^{(k)}$ (cf. Figures 4 and 5) is a correct and index-hiding offline-online napPRF over domain $\{0, 1\}^k$ and range $\{0, 1\}^m$.

Proof. We prove correctness and security (index-hiding) for $\{\Pi_{\text{napPRF}}^{(k)}\}_{k \geq 0}$.

Correctness. We prove correctness by induction on k .

The base case $k = 0$ is proved in Theorem 7. Fix $k \geq 1$ and assume, inductively, that $\Pi_{\text{napPRF}}^{(k-1)}$ is correct, i.e., for every $(\text{msk}', \text{sk}'_{\text{off}}) \leftarrow \text{KeyGen}^{(k-1)}(1^\lambda)$, every $\alpha' \in \{0, 1\}^{k-1}$ and $\beta \in \{0, 1\}^m$, letting $\text{sk}'_{\text{on}} := \text{Prog}^{(k-1)}(\text{msk}', \alpha', \beta)$, we have: (i) $\text{Eval}^{(k-1)}(\text{sk}'_{\text{off}}, \text{sk}'_{\text{on}}, \alpha') = \beta$, and (ii) for every $x' \neq \alpha'$, the value $\text{Eval}^{(k-1)}(\text{sk}'_{\text{off}}, \text{sk}'_{\text{on}}, x')$ is independent of β .

Now fix λ , sample $(\text{msk}, \text{sk}_{\text{off}}) \leftarrow \text{KeyGen}^{(k)}(1^\lambda)$, fix $\alpha^* \in \{0, 1\}^k$ and $\beta^* \in \{0, 1\}^m$, and let $\text{sk}_{\text{on}} := \text{Prog}^{(k)}(\text{msk}, \alpha^*, \beta^*)$. Write $\alpha^* = \alpha_1 \| \dots \| \alpha_k$, $\beta^* = \beta_1 \| \dots \| \beta_m$ and $x = x_1 \| x'$ where $\alpha_1, \dots, \alpha_k, \beta_1, \dots, \beta_m, x_1 \in \{0, 1\}$.

(i) *On-point* ($x = \alpha^*$). Since sk_{on} contains $\mathcal{L}_{\alpha,1} = \mathcal{L}_{\alpha_1,1}^{(\alpha_1)}$, the $\text{Eval}^{(k)}$ in Item 1 decrypts the seed-table row $r := \text{Lsb}(\mathcal{L}_{\alpha,1}) = \text{Lsb}(\mathcal{L}_{\alpha_1,1}^{(\alpha_1)})$ and obtains $s := \text{Dec}_{\mathcal{L}_{\alpha,1}}^{(1)}(\tau_{\text{seed},r}^{(k)}, \mathcal{T}_{\text{seed}}[r])$. By the construction of $\mathcal{T}_{\text{seed}}$, this seed is precisely the *real* child seed used by $\text{KeyGen}^{(k)}$ to generate the child instance along the *inactive* branch $1 - \alpha_1$ (here branch 0 is “left” and branch 1 is “right”).

Next, in Item 2, the $\text{Eval}^{(k)}$ computes

$$(\widehat{\text{msk}}, \widehat{\text{sk}}_{\text{off}}) := \Pi_{\text{napPRF}}^{(k-1)}.\text{KeyGen}(1^\lambda; s) = \Pi_{\text{napPRF}}^{(k-1)}.\text{KeyGen}(1^\lambda; s_{1-\alpha_1}).$$

Hence $\widehat{\text{sk}}_{\text{off}}$ equals the offline key $\text{sk}_{1-\alpha_1, \text{off}}$ of the *opposite/inactive* child instance generated in $\text{KeyGen}^{(k)}$. Since $\text{sk}_{\text{stack}} = \text{sk}_{\text{left,off}} \oplus \text{sk}_{\text{right,off}}$, we have

$$\text{sk}_{\text{off}}^{(\text{next})} := \text{sk}_{\text{stack}} \oplus \widehat{\text{sk}}_{\text{off}} = (\text{sk}_{\text{left,off}} \oplus \text{sk}_{\text{right,off}}) \oplus \text{sk}_{1-\alpha_1, \text{off}} = \text{sk}_{\alpha_1, \text{off}}$$

namely the offline key of the *active* child instance (branch α_1).

Moreover, in Item 3, for each $i \in [k-1]$ the $\text{Eval}^{(k)}$ decrypts the unique entry of $\mathcal{T}_{\alpha,i}$ indexed by (r, c) where $c := \text{Lsb}(\mathcal{L}_{\alpha,i+1}) = \text{Lsb}(\mathcal{L}_{\alpha,i+1}^{(\alpha_{i+1})})$. By construction of $\mathcal{T}_{\alpha,i}$, the component selected by $x_1 = \alpha_1$ equals the child label $\mathcal{L}_{\alpha, \alpha_1, i}^{(\alpha_{i+1})}$, while the other component is junk. Similarly, for each $i \in [m]$, decrypting $\mathcal{T}_{\beta,i}[r][c]$ (with $c := \text{Lsb}(\mathcal{L}_{\beta,i})$) yields a pair whose component selected by $x_1 = \alpha_1$ equals the child label $\mathcal{L}_{\beta, \alpha_1, i}^{(\beta_i)}$.

Therefore,

$$\text{sk}_{\text{on}}^{(\text{next})} = \{\mathcal{L}_{\alpha, x_1, i}\}_{i \in [k-1]} \parallel \{\mathcal{L}_{\beta, x_1, i}\}_{i \in [m]} = \{\mathcal{L}_{\alpha, \alpha_1, i}^{(\alpha_{i+1})}\}_{i \in [k-1]} \parallel \{\mathcal{L}_{\beta, \alpha_1, i}^{(\beta_i)}\}_{i \in [m]}$$

which is exactly the online key that $\Pi_{\text{napPRF}}^{(k-1)}$ would output for index $\alpha' = \alpha_2 \| \dots \| \alpha_k$ and value β^* under the active child master key msk_{α_1} . Applying the inductive correctness of $\Pi_{\text{napPRF}}^{(k-1)}$ gives

$$\text{Eval}^{(k)}(\text{sk}_{\text{off}}, \text{sk}_{\text{on}}, \alpha^*) = \text{Eval}^{(k-1)}(\text{sk}_{\text{off}}^{(\text{next})}, \text{sk}_{\text{on}}^{(\text{next})}, \alpha') = \beta^*$$

(ii) *Off-point* ($x \neq \alpha^*$). If $x_1 \neq \alpha_1$, then $x_1 = 1 - \alpha_1$ and the decrypted seed is $s = s_{1-\alpha_1} = s_{x_1}$. Consequently, $(\widehat{\text{msk}}, \widehat{\text{sk}}_{\text{off}})$ and thus $\text{sk}_{\text{off}}^{(\text{next})} = \text{sk}_{\text{stack}} \oplus \widehat{\text{sk}}_{\text{off}}$ are independent of β^* . Crucially, in Item 3 the component selected by x_1 from each decrypted translation-table entry is the *junk* label by construction, so $\text{sk}_{\text{on}}^{(\text{next})}$ is independent of β^* as well. Thus, the final output is independent of β^* .

Otherwise $x_1 = \alpha_1$ but $x' \neq \alpha' = \alpha_2 \| \dots \| \alpha_k$. In this case, the argument in (i) shows that $\text{sk}_{\text{off}}^{(\text{next})} = \text{sk}_{\alpha_1, \text{off}}$ and $\text{sk}_{\text{on}}^{(\text{next})}$ is the correctly formed (offline and online) programmed key for index α' and value β^* in the active child instance. Since the recursive call evaluates at $x' \neq \alpha'$, the inductive hypothesis (off-point correctness for $\Pi_{\text{napPRF}}^{(k-1)}$) implies that $\text{Eval}^{(k-1)}(\text{sk}_{\text{off}}^{(\text{next})}, \text{sk}_{\text{on}}^{(\text{next})}, x')$ is independent of β^* . Therefore $\text{Eval}^{(k)}(\text{sk}_{\text{off}}, \text{sk}_{\text{on}}, x)$ is independent of β^* as well.

This completes the inductive proof of correctness for $\Pi_{\text{napPRF}}^{(k)}$.

Security (index-hiding). We prove index-hiding by induction on k .

The base case $k = 0$ is proved in Theorem 7. Fix $k \geq 1$ and assume inductively that $\Pi_{\text{napPRF}}^{(k-1)}$ is index-hiding.

Experiments. For $\alpha \in \{0, 1\}^k$, let $\text{REAL}_\alpha^{(k)}$ be the distribution of

$$(\text{sk}_{\text{off}}, \text{sk}_{\text{on}}) \mid (\text{msk}, \text{sk}_{\text{off}}) \leftarrow \text{KeyGen}^{(k)}(1^\lambda), \text{sk}_{\text{on}} := \text{Prog}^{(k)}(\text{msk}, \alpha, \beta \xleftarrow{\$} \{0, 1\}^m)$$

For later use, we also define $\text{REAL}_\epsilon^{(0)}$ in the same way.

Index-hiding asks for a PPT simulator $\text{Sim}^{(k)}$ such that for each α , $\text{REAL}_\alpha^{(k)} \approx_c \text{Sim}^{(k)}(1^\lambda)$. Clearly, it suffices to show $\text{REAL}_\alpha^{(k)} \approx_c \text{REAL}_{0^k}^{(k)}$ for all α , and then take the simulator $\text{Sim}^{(k)}(1^\lambda)$ as:

$$(\text{msk}, \text{sk}_{\text{off}}) \leftarrow \text{KeyGen}^{(k)}(1^\lambda), \beta \xleftarrow{\$} \{0, 1\}^m, \text{output}(\text{sk}_{\text{off}}, \text{Prog}^{(k)}(\text{msk}, 0^k, \beta))$$

Step 0: PRG-to-uniform for the master expansion. In $\text{KeyGen}^{(k)}$, all top-level randomness (the two seeds $s_{\text{left}}, s_{\text{right}}$, the k α -wire label pairs and the m β -wire label pairs in msk , and the top-level junk strings) is sampled via $G(s_{\text{master}})$ for uniform s_{master} . By PRG security, we may replace $G(s_{\text{master}})$ by uniform randomness of the same length, obtaining an experiment $\widetilde{\text{REAL}}_\alpha^{(k)}$ such that

$$\text{REAL}_\alpha^{(k)} \approx_c \widetilde{\text{REAL}}_\alpha^{(k)} \quad \text{for all } \alpha$$

Henceforth, we work in $\widetilde{\text{REAL}}_\alpha^{(k)}$ where all these top-level values are jointly uniform subject only to the point-and-permute LSB constraints.

For later use, we also define $\widetilde{\text{REAL}}_\epsilon^{(0)} \approx_c \text{REAL}_\epsilon^{(0)}$ in the same way.

Step 1: Identify decryptable vs. non-decryptable entries. Let $\alpha = \alpha_1 \parallel \alpha'$ with $\alpha' = \alpha_2 \parallel \dots \parallel \alpha_k \in \{0, 1\}^{k-1}$ and write $\beta = \beta_1 \parallel \dots \parallel \beta_m$. In sk_{on} , the adversary sees exactly one α -label per coordinate, $\mathcal{L}_{\alpha,i} = \mathcal{L}_{\alpha,i}^{(\alpha_i)}$ for $i \in [k]$, and one β -label per output bit, $\mathcal{L}_{\beta,i} = \mathcal{L}_{\beta,i}^{(\beta_i)}$ for $i \in [m]$. Consequently, it can decrypt:

- Exactly one entry of $\mathcal{T}_{\text{seed}}$ – the row indexed by $r := \text{Lsb}(\mathcal{L}_{\alpha,1}) = \text{Lsb}(\mathcal{L}_{\alpha,1}^{(\alpha_1)})$;
- For each $i \in [k-1]$, exactly one entry of $\mathcal{T}_{\alpha,i}$ – the cell indexed by (r, c_i) where $c_i := \text{Lsb}(\mathcal{L}_{\alpha,i+1})$;
- For each $i \in [m]$, exactly one entry of $\mathcal{T}_{\beta,i}$ – the cell indexed by (r, c'_i) where $c'_i := \text{Lsb}(\mathcal{L}_{\beta,i})$.

Every other entry in $\mathcal{T}_{\text{seed}}$, $\{\mathcal{T}_{\alpha,i}\}$, and $\{\mathcal{T}_{\beta,i}\}$ is encrypted under at least one key that is *not* contained in sk_{on} .

Step 2: PRF hybrids. Define $\text{Hyb}_\alpha^{(k)}$ from $\widetilde{\text{REAL}}_\alpha^{(k)}$ by replacing *every* table entry in $\mathcal{T}_{\text{seed}}$ (and $\{\mathcal{T}_{\alpha,i}\}$, $\{\mathcal{T}_{\beta,i}\}$, resp.) that is encrypted under at least one key not in sk_{on} by an independent uniform string in $\{0, 1\}^\lambda$ (and $\{0, 1\}^{2\lambda}$, resp.). (All tweaks are unique by construction.) Then for each fixed α ,

$$\widetilde{\text{REAL}}_\alpha^{(k)} \approx_c \text{Hyb}_\alpha^{(k)}$$

by a standard entry-by-entry hybrid argument using PRF security of F .

Thus, to prove index-hiding, it remains to show that for all α , $\text{Hyb}_\alpha^{(k)}$ is computationally indistinguishable from $\text{Hyb}_{0^k}^{(k)}$.

For later use, we define $\text{Hyb}_\epsilon^{(0)} \equiv \widetilde{\text{REAL}}_\epsilon^{(0)}$; note, there is no PRF in $\widetilde{\text{REAL}}_\epsilon^{(0)}$.

Step 3a: Removing suffix α' . Fix $\alpha_1 \in \{0, 1\}$. We show that $\forall \alpha' \in \{0, 1\}^{k-1}$,

$$\text{Hyb}_{\alpha_1 \parallel \alpha'}^{(k)} \approx_c \text{Hyb}_{\alpha_1 \parallel 0^{k-1}}^{(k)}$$

for $k \geq 1$. The case $k = 1$ is trivial, so we assume $k > 1$ below.

In $\text{Hyb}_{\alpha_1 \parallel \alpha'}^{(k)}$, all non-decryptable table entries have been replaced by independent uniform strings, hence they are independent of α' . Therefore, any dependence on α' can only arise from: (i) the revealed labels in sk_{on} and (ii) the decryptable cells (one row of $\mathcal{T}_{\text{seed}}$ and one cell per table in $\{\mathcal{T}_{\alpha,i}\}$ and $\{\mathcal{T}_{\beta,i}\}$).

First, the distribution of sk_{on} is independent of α' : for each $j \in \{2, \dots, k\}$ the revealed label $\mathcal{L}_{\alpha,j} = \mathcal{L}_{\alpha,j}^{(\alpha_j)}$ is uniform in $\{0, 1\}^\lambda$ subject only to the point-and-permute constraint, regardless of α_j . (The same holds for the revealed β -labels, and $\beta \xleftarrow{\$} \{0, 1\}^m$ is uniform.)

Next, we define an efficient extraction algorithm Ext_{α_1} that, given $(\text{sk}_{\text{off}}, \text{sk}_{\text{on}})$, outputs a level- $(k-1)$ pair $(\text{sk}_{\text{act,off}}, \text{sk}_{\text{act,on}})$ as follows:

1. Decrypt the unique seed-row $\mathcal{T}_{\text{seed}}[r]$ using $\mathcal{L}_{\alpha,1} \in \text{sk}_{\text{on}}$ and obtain s .
2. Compute $(\widehat{\text{msk}}, \widehat{\text{sk}}) := \Pi_{\text{napPRF}}^{(k-1)}.\text{KeyGen}(1^\lambda; s)$ and $\text{sk}_{\text{act,off}} := \text{sk}_{\text{stack}} \oplus \widehat{\text{sk}}$.
3. For each $i \in [k-1]$ (resp. $i \in [m]$), decrypt the unique opened cell of $\mathcal{T}_{\alpha,i}$ (resp. $\mathcal{T}_{\beta,i}$) using the corresponding two keys in sk_{on} . From each decrypted 2λ -bit plaintext, take the *active half*: the left half if $\alpha_1 = 0$ and the right half if $\alpha_1 = 1$. Concatenating these active-half labels yields $\text{sk}_{\text{act,on}}$.

By construction of $\Pi_{\text{napPRF}}^{(k)}$, extracted $(\text{sk}_{\text{act,off}}, \text{sk}_{\text{act,on}})$ is distributed *exactly* as

$$(\text{sk}'_{\text{off}}, \text{sk}'_{\text{on}}) \mid (\text{msk}', \text{sk}'_{\text{off}}) \leftarrow \text{KeyGen}^{(k-1)}(1^\lambda), \text{sk}'_{\text{on}} := \text{Prog}^{(k-1)}(\text{msk}', \alpha', \beta \xleftarrow{\$} \{0, 1\}^m)$$

namely $\text{REAL}_{\alpha'}^{(k-1)}$. Moreover, the remaining information in $(\text{sk}_{\text{off}}, \text{sk}_{\text{on}})$ outside of Ext_{α_1} is independent of α' in $\text{Hyb}_{\alpha_1 \parallel \alpha'}^{(k)}$ (it is either uniform filler or top-level values whose distribution does not depend on α').

Therefore, by the inductive index-hiding hypothesis for $\Pi_{\text{napPRF}}^{(k-1)}$ and closure under efficient post-processing / taking products with α' -independent auxiliary information, we may replace $\text{REAL}_{\alpha'}^{(k-1)}$ by $\text{REAL}_{0^{k-1}}^{(k-1)}$ inside the extracted component, obtaining $\text{Hyb}_{\alpha_1 \parallel \alpha'}^{(k)} \approx_c \text{Hyb}_{\alpha_1 \parallel 0^{k-1}}^{(k)}$.

Step 3b: Hiding α_1 . It remains to show

$$\text{Hyb}_{0^k}^{(k)} \approx_c \text{Hyb}_{1 \parallel 0^{k-1}}^{(k)}$$

for $k \geq 1$. Now, fix $k \geq 1$ and consider the distribution $\text{Hyb}_{0^k}^{(k)}$. We define $\text{FCHyb}_{0^k}^{(k)}$ by further modifying $\text{Hyb}_{0^k}^{(k)}$ *recursively down the leftmost path*:

Definition 9 ($\text{FCHyb}_{0^k}^{(k)}$). We define a distribution over $(\text{sk}_{\text{off}}, \text{sk}_{\text{on}})$ for $\alpha = 0^k$.

1. (**Generate sk_{off} with cleaning on the 0^k spine**). Run $\text{KeyGen}^{(k)}$ to obtain sk_{off} , except that for the left child instance seeded by s_{left} (and recursively its left child, etc., for k levels down to $\Pi_{\text{napPRF}}^{(0)}$), replace the master expansion $G(s_{\text{left}})$ used in that child $\text{KeyGen}^{(\cdot)}$ by uniform randomness of the same length (followed by the usual point-and-permute LSB overwrite).

2. **(Sample β and form top-level sk_{on}).** Sample $\beta \xleftarrow{\$} \{0, 1\}^m$ and form top-level sk_{on} exactly as in $\text{Hyb}_{0^k}^{(k)}$.
3. **(Hybridize unopened entries along the 0^k spine).** For each of the k instances on the left spine, apply Step 2 at that instance: within its sk_{off} , replace every table entry encrypted under at least one key that is not revealed by the corresponding (derived) programmed key at that level by an independent uniform λ -bit or 2λ -bit string.

All other parts of the transcript are unchanged.

For later use, we define $\text{FCHyb}_{\epsilon}^{(0)} \equiv \text{Hyb}_{\epsilon}^{(0)}$.

Let $U^{(k)} \xleftarrow{\$} \{0, 1\}^{(k+m)\lambda}$ be an independent uniform string with the same length/serialization as sk_{on} . We prove:

Lemma 4. For every $k \geq 0$,

$$(\text{sk}_{\text{off}}, \text{sk}_{\text{on}}) \mid (\text{sk}_{\text{off}}, \text{sk}_{\text{on}}) \leftarrow \text{FCHyb}_{0^k}^{(k)} \approx_c (\text{sk}_{\text{off}}, U^{(k)})$$

Proof. We prove by induction on k .

Base case $k = 0$. This follows directly from the proof of Theorem 7.

Inductive step. Assume the lemma holds for $k - 1$. Consider $\text{FCHyb}_{0^k}^{(k)}$. As in $\text{Hyb}_{0^k}^{(k)}$, all non-decryptable top-level entries are uniform fillers already, so any dependence between sk_{off} and sk_{on} can only arise through the decryptable top-level cells: the opened seed-row and the opened cells of $\{\mathcal{T}_{\alpha,i}\}_{i \in [k-1]}$ and $\{\mathcal{T}_{\beta,i}\}_{i \in [m]}$.

- *Decryptable seed-row.* Its plaintext is s_{right} , an independent uniform λ -bit string (this is ensured already by the k -th level Step 0), hence its ciphertext is uniform even given the seed-row key $\mathcal{L}_{\alpha,1} \in \text{sk}_{\text{on}}$.
- *Decryptable translation cells.* Fix any opened translation cell. In $\alpha = 0^k$, its plaintext is of the form $(\mathcal{L} \parallel \mathcal{J})$, where $\mathcal{J}$ is a top-level junk string (uniform), and $\mathcal{L}$ is the corresponding left-child label (an entry of the next-level online programmed key that evaluation would recurse on). By definition of $\text{FCHyb}_{0^k}^{(k)}$, the left child instance is distributed exactly as $\text{FCHyb}_{0^{k-1}}^{(k-1)}$ (using the same random β), including the recursive Step 2 hybridization within that child. Therefore, by the inductive hypothesis, the left-child online key (hence each such label $\mathcal{L}$) is indistinguishable from an independent uniform λ -bit string even given the entire left-child offline key. Consequently, each plaintext $(\mathcal{L} \parallel \mathcal{J})$ is indistinguishable from two independent uniform λ -bit strings, and its ciphertext is uniform even given its decryption keys.

Thus, conditioned on sk_{off} , all decryptable ciphertexts that involve keys from sk_{on} are (computationally) uniform and do not correlate with those keys; all other entries are uniform fillers. It follows that sk_{on} is computationally independent of sk_{off} , and since it is uniform, we may replace it by an independent $U^{(k)}$, completing the induction. $\square$

Similarly, define $\text{FCHyb}_{1 \parallel 0^{k-1}}^{(k)}$ starting from $\text{Hyb}_{1 \parallel 0^{k-1}}^{(k)}$ and recursively cleaning/hybridizing down the active path for $\alpha = 1 \parallel 0^{k-1}$, i.e., take the right child at the root and then the left child thereafter (always using the same β).

Claim 3b.1 (Cleaning both sides).

$$\text{Hyb}_{0^k}^{(k)} \approx_c \text{FCHyb}_{0^k}^{(k)} \quad \text{and} \quad \text{Hyb}_{1 \parallel 0^{k-1}}^{(k)} \approx_c \text{FCHyb}_{1 \parallel 0^{k-1}}^{(k)}$$

This holds naturally by PRG and PRF security.

Claim 3b.2 (Uniform sk_{on} given sk_{off} on both sides). Let $U^{(k)}$ be sampled as $(k + m)$ independent λ -bit strings. Then Lemma 4 gives

$$(\text{sk}_{\text{off}}, \text{sk}_{\text{on}}) \mid (\text{sk}_{\text{off}}, \text{sk}_{\text{on}}) \leftarrow \text{FCHyb}_{0^k}^{(k)} \approx_c (\text{sk}_{\text{off}}, U^{(k)})$$

and the symmetric argument yields

$$(\text{sk}_{\text{off}}, \text{sk}_{\text{on}}) \mid (\text{sk}_{\text{off}}, \text{sk}_{\text{on}}) \leftarrow \text{FCHyb}_{1\|0^{k-1}}^{(k)} \approx_c (\text{sk}_{\text{off}}, U^{(k)})$$

Claim 3b.3 (Once sk_{on} is uniform, α_1 disappears). Consider the two distributions obtained by replacing sk_{on} with $U^{(k)}$ in $\text{FCHyb}_{0^k}^{(k)}$ and in $\text{FCHyb}_{1\|0^{k-1}}^{(k)}$. We claim these are *identically distributed*. Indeed, conditioned on sk_{off} and $U^{(k)}$:

- All *non-decryptable* top-level table entries are uniform fillers by definition of $\text{Hyb}^{(k)}$ (and remain so in FCHyb).
- The unique decryptable seed-row plaintext is an independent uniform λ -bit string (by top-level Step 0), so its ciphertext is uniform even given the revealed seed-row key; hence it carries no α_1 -dependence.
- Each opened translation-cell plaintext at the top level is of the form

$$(\text{child label}) \parallel (\text{junk}) \quad \text{if } \alpha_1 = 0 \quad (\text{junk}) \parallel (\text{child label}) \quad \text{if } \alpha_1 = 1$$

After replacing sk_{on} by $U^{(k)}$, the “child label” component is just an independent uniform λ -bit string, and the junk half is uniform as well. Therefore each opened plaintext is a concatenation of two independent uniform λ -bit strings, whose distribution is invariant under swapping left/right halves. Thus the fact that the “real” half appears on the left for $\alpha_1 = 0$ and on the right for $\alpha_1 = 1$ becomes distributionally irrelevant.

Since every component of sk_{off} that could reflect the choice of α_1 is either (i) a uniform filler, or (ii) an encryption of a uniform plaintext, the overall joint distribution does not depend on α_1 . Hence, after replacing sk_{on} by $U^{(k)}$, the two FCHyb experiments are identically distributed.

Conclusion of Step 3b. Chaining the above,

$$\text{Hyb}_{0^k}^{(k)} \approx_c \text{FCHyb}_{0^k}^{(k)} \approx_c (\text{sk}_{\text{off}}, U^{(k)}) \equiv (\text{sk}_{\text{off}}, U^{(k)}) \approx_c \text{FCHyb}_{1\|0^{k-1}}^{(k)} \approx_c \text{Hyb}_{1\|0^{k-1}}^{(k)}$$

which completes Step 3b. $\square$

C Formalism of Garbling-Based Short-Key Compiler

C.1 Garbling Schemes

Definition 10 (Garbling). A *garbling scheme* is a pair of PPT algorithms:

- $\text{Gb}(1^\lambda, \mathcal{C})$ takes as input a standard fan-in-2 Boolean circuit $\mathcal{C} : \{0, 1\}^n \rightarrow \{0, 1\}^m$ and outputs a garbled circuit $\widehat{\mathcal{C}}$ with input labels $\{\mathcal{L}_i^{(0)}, \mathcal{L}_i^{(1)}\}_{i \in [n]}$.
- $\text{Ev}(1^\lambda, \mathcal{C}, \widehat{\mathcal{C}}, \mathcal{L}_1, \dots, \mathcal{L}_n)$ takes $\mathcal{C}$, $\widehat{\mathcal{C}}$, and one label $\mathcal{L}_i$ for each input wire $i \in [n]$, and outputs a string $y \in \{0, 1\}^m$.

The scheme is **correct** if for all $\lambda \in \mathbb{N}$, all Boolean circuits $\mathcal{C} : \{0, 1\}^n \rightarrow \{0, 1\}^m$, and all $\mathbf{x} \in \{0, 1\}^n$

$$\Pr \left[\text{Ev}(1^\lambda, \mathcal{C}, \widehat{\mathcal{C}}, \mathcal{L}_1^{(\mathbf{x}_1)}, \dots, \mathcal{L}_n^{(\mathbf{x}_n)}) = \mathcal{C}(\mathbf{x}) \mid (\widehat{\mathcal{C}}, \{\mathcal{L}_i^{(0)}, \mathcal{L}_i^{(1)}\}_{i \in [n]}) \leftarrow \text{Gb}(1^\lambda, \mathcal{C}) \right] = 1$$

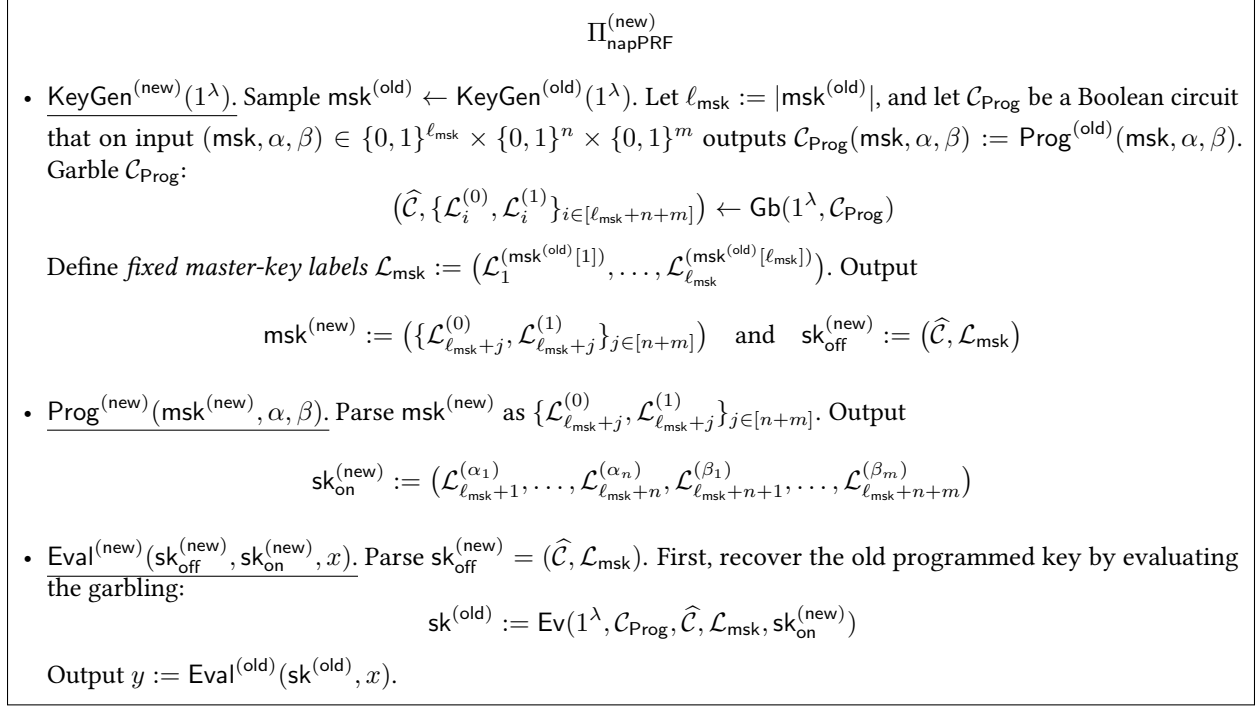


Figure 6: Garbling-based short-key compiler.

Security. A garbling scheme $\Pi = (\text{Gb}, \text{Ev})$ is *secure* if there exists a PPT simulator Sim such that for all Boolean circuits $\mathcal{C} : \{0, 1\}^n \rightarrow \{0, 1\}^m$, and all $x \in \{0, 1\}^n$, the following ensembles are computationally indistinguishable:

$$\left\{ (\hat{\mathcal{C}}, \mathcal{L}_1^{(x_1)}, \dots, \mathcal{L}_n^{(x_n)}) \mid (\hat{\mathcal{C}}, \{\mathcal{L}_i^{(0)}, \mathcal{L}_i^{(1)}\}_{i \in [n]}) \leftarrow \text{Gb}(1^\lambda, \mathcal{C}) \right\}_\lambda \approx_c \left\{ \text{Sim}(1^\lambda, \mathcal{C}, \mathcal{C}(x)) \right\}_\lambda$$

C.2 Formalism

Let $\Pi_{\text{napPRF}}^{(\text{old})} = (\text{KeyGen}^{(\text{old})}, \text{Prog}^{(\text{old})}, \text{Eval}^{(\text{old})})$ be a (standard) napPRF scheme with domain $\{0, 1\}^n$ and range $\{0, 1\}^m$, and let $\Pi_{\text{GC}} = (\text{Gb}, \text{Ev})$ be the (Yao) garbling scheme. We construct an offline-online napPRF scheme $\Pi_{\text{napPRF}}^{(\text{new})} = (\text{KeyGen}^{(\text{new})}, \text{Prog}^{(\text{new})}, \text{Eval}^{(\text{new})})$ as depicted in Figure 6.

Theorem 12. *Let $\Pi_{\text{napPRF}}^{(\text{old})} = (\text{KeyGen}^{(\text{old})}, \text{Prog}^{(\text{old})}, \text{Eval}^{(\text{old})})$ be a correct and index-hiding napPRF over domain $\mathcal{X} = \{0, 1\}^n$ and range $\mathcal{Y} = \{0, 1\}^m$. Let $\Pi_{\text{GC}} = (\text{Gb}, \text{Ev})$ be a correct and secure garbling scheme (cf. Appendix C.1). Then the construction in Figure 6 yields an offline-online napPRF $\Pi_{\text{napPRF}}^{(\text{new})} = (\text{KeyGen}^{(\text{new})}, \text{Prog}^{(\text{new})}, \text{Eval}^{(\text{new})})$ over the same domain and range that is correct and index-hiding.*

Moreover, in the resulting scheme the online key $\text{sk}_{\text{on}}^{(\text{new})}$ consists of exactly $(n + m)$ garbled input labels; in particular, under standard Yao garbling (with λ -bit labels) it has length $(n + m)\lambda$ bits.

Proof sketch. Correctness and index hiding are both straightforward to verify. We therefore only briefly describe the simulator for the index-hiding experiment.

The goal is to simulate the joint distribution of $(\text{sk}_{\text{off}}^{(\text{new})}, \text{sk}_{\text{on}}^{(\text{new})})$. Recall that $\text{sk}_{\text{off}}^{(\text{new})}$ consists of a garbled circuit $\tilde{C}$ for the public circuit C_{Prog} and a bundle of “fixed” garbled labels (corresponding to the $\text{msk}^{(\text{old})}$ input wires). The online key $\text{sk}_{\text{on}}^{(\text{new})}$ consists of the remaining $(n + m)$ garbled labels (corresponding to the (α, β) input wires). Evaluating the garbled circuit on these labels yields the old programmed key $\text{sk}^{(\text{old})}$.

The simulator proceeds as follows. First, it invokes the index-hiding simulator for $\Pi_{\text{napPRF}}^{(\text{old})}$ to obtain a simulated programmed key $\tilde{\text{sk}}^{(\text{old})}$. Next, since C_{Prog} is public, it invokes the garbling simulator on input $(1^\lambda, C_{\text{Prog}}, \tilde{\text{sk}}^{(\text{old})})$ to obtain a simulated garbled circuit $\tilde{C}$ together with a full set of active input labels consistent with output $\tilde{\text{sk}}^{(\text{old})}$. Finally, it partitions these active labels into (i) the labels for the $\text{msk}^{(\text{old})}$ input wires, which it places into $\tilde{\text{sk}}_{\text{off}}^{(\text{new})}$ alongside $\tilde{C}$, and (ii) the labels for the (α, β) input wires, which it outputs as $\tilde{\text{sk}}_{\text{on}}^{(\text{new})}$. $\square$

D Tailored Offline-Online sipPRF for LogStack

We formally present our offline-online sipPRF constructions tailored to the LogStack application; Section 7.2 includes overview. We emphasize that we still define the schemes – and their security – at the level of a napPRF, rather than as outputting GC labels, and then show that the resulting schemes satisfy suffix invariance. Adapting the construction to the variant used inside SGC’s demultiplexer is straightforward (it suffices to drop the decoding bits).

The constructions follow the paradigm from Section 6.2. The key difference is that we eliminate the seed table $\mathcal{T}_{\text{seed}}$ entirely. Instead, KeyGen samples a full binary tree of i.i.d. dummy seeds; this tree is stored into msk and, in Prog, is modified (as SortingHat) so that the sibling roots along the active path carry the real GGM seeds while all remaining nodes carry dummy seeds.

As Section 6.2, we (re)define a family of offline-online napPRF schemes $\{\Pi_{\text{napPRF}}^{(i)}\}_{i \in [n]}$, where each $\Pi_{\text{napPRF}}^{(i)} : \{0, 1\}^i \rightarrow \{0, 1\}^m$. $\Pi_{\text{napPRF}}^{(0)}$ is unchanged.

Recursive definitions. Fix $k \geq 1$. We (re)define $\Pi_{\text{napPRF}}^{(k)} = (\text{KeyGen}, \text{Prog}, \text{Eval})$ over domain $\mathcal{X} = \{0, 1\}^k$ and range $\mathcal{Y} = \{0, 1\}^m$. The construction of $\Pi_{\text{napPRF}}^{(k)}$ makes *black-box* use of $\Pi_{\text{napPRF}}^{(k-1)}$.

We write $\tau_{\text{seed}, r}^{(k)}$, $\tau_{\beta, i, r, c}^{(k)}$, and $\tau_{\alpha, i, r, c}^{(k)}$ for public, *unique* tweaks for the corresponding (garbled) table entries. For notational convenience, we write

$$(\text{msk}, \text{sk}_{\text{off}}) := \Pi_{\text{napPRF}}.\text{KeyGen}(1^\lambda; s)$$

to denote running $\Pi_{\text{napPRF}}.\text{KeyGen}$ with randomness set as $s \in \{0, 1\}^\lambda$. In particular, we will ensure that each KeyGen uses only a λ -bit random tape. The redefined scheme $\Pi_{\text{napPRF}}^{(k)}$ is depicted in Figures 7 and 8, and satisfies:

Theorem 13 (Main (tailored)). *Let G be a PRG and let F be a PRF. For every $k \geq 1$, the tailored scheme $\Pi_{\text{napPRF}}^{(k)}$ (cf. Figures 7 and 8) is a correct and index-hiding offline-online napPRF over domain $\{0, 1\}^k$ and range $\{0, 1\}^m$.*

Proof sketch. The proof follows the structure of Theorem 8; the only changes are that we drop $\mathcal{T}_{\text{seed}}$ and sk_{on} additionally contains a seed tree s . The correctness is trivial, so we only outline the index-hiding argument.

Redefined $\Pi_{\text{napPRF}}^{(k)}.\text{KeyGen}(1^\lambda)$

1. Sample a master seed $s_{\text{master}} \xleftarrow{\$} \{0, 1\}^\lambda$. Use $G(s_{\text{master}})$ to sample

$$s_{\text{left}}, s_{\text{right}} \in \{0, 1\}^\lambda \quad \{\mathcal{L}_{\alpha,i}^{(0)}, \mathcal{L}_{\alpha,i}^{(1)} \in \{0, 1\}^\lambda\}_{i \in [k]} \quad \{\mathcal{L}_{\beta,i}^{(0)}, \mathcal{L}_{\beta,i}^{(1)} \in \{0, 1\}^\lambda\}_{i \in [m]}$$

$$\{\mathcal{J}_{\alpha,i} \in \{0, 1\}^\lambda\}_{i \in [k-1]} \quad \{\mathcal{J}_{\beta,i} \in \{0, 1\}^\lambda\}_{i \in [m]}$$

$$\text{seed}' := \{s'_{i,j} \in \{0, 1\}^\lambda\}_{i \in [k], j \in \{0, 1, \dots, 2^i - 1\}}$$

Point-and-permute constraint: overwrite the last bits so that

$$\forall i \in [k], \text{Lsb}(\mathcal{L}_{\alpha,i}^{(1)}) := \text{Lsb}(\mathcal{L}_{\alpha,i}^{(0)}) \oplus 1 \quad \forall i \in [m], \text{Lsb}(\mathcal{L}_{\beta,i}^{(1)}) := \text{Lsb}(\mathcal{L}_{\beta,i}^{(0)}) \oplus 1$$

(All other bits remain as sampled, so $\mathcal{L}^{(1)}$ is not derivable from $\mathcal{L}^{(0)}$.)

2. **Generate two child instances.** For each $j \in \{\text{left}, \text{right}\}$, compute $(\text{msk}_j, \text{sk}_{j,\text{off}}) := \Pi_{\text{napPRF}}^{(k-1)}.\text{KeyGen}(1^\lambda; s_j)$.

By construction, msk_j includes the child wire labels $\{\mathcal{L}_{\alpha,j,i}^{(0)}, \mathcal{L}_{\alpha,j,i}^{(1)}\}_{i \in [k-1]}$ and $\{\mathcal{L}_{\beta,j,i}^{(0)}, \mathcal{L}_{\beta,j,i}^{(1)}\}_{i \in [m]}$ along with auxiliary seeds (i.e., the GGM-derived material from s_j and the level- $(k-1)$ dummy-seed tree). Note that this dummy-seed tree is irrelevant. $\text{sk}_{j,\text{off}}$ is the level- $(k-1)$ offline key output by the recursive call (in a fixed canonical serialization).

3. Let $\text{seed} := \{s_{i,j} \in \{0, 1\}^\lambda\}_{i \in [k], j \in \{0, 1, \dots, 2^i - 1\}}$ be the GGM seeds derived from s_{master} , i.e., the recursive expansion trace obtained by iterating from s_{left} and s_{right} .

Output $\text{msk} := \{\mathcal{L}_{\alpha,i}^{(0)}, \mathcal{L}_{\alpha,i}^{(1)}\}_{i \in [k]}, \{\mathcal{L}_{\beta,i}^{(0)}, \mathcal{L}_{\beta,i}^{(1)}\}_{i \in [m]}$, seed , seed' and sk_{off} containing:

- **Stacked child offline keys:** $\text{sk}_{\text{stack}} := \text{sk}_{\text{left},\text{off}} \oplus \text{sk}_{\text{right},\text{off}}$.

- **α -translation tables** $\{\mathcal{T}_{\alpha,i}\}_{i \in [k-1]}$. Let $r := \text{Lsb}(\mathcal{L}_{\alpha,1}^{(0)})$.

For each $i \in [k-1]$: Let $j := i+1$, $c := \text{Lsb}(\mathcal{L}_{\alpha,j}^{(0)})$ and define

$$\mathcal{T}_{\alpha,i}[r][c] := \text{Enc}_{\mathcal{L}_{\alpha,1}^{(0)}, \mathcal{L}_{\alpha,j}^{(0)}}^{(2)}(\tau_{\alpha,i,r,c}^{(k)}, \mathcal{L}_{\alpha,\text{left},i}^{(0)} \parallel \mathcal{J}_{\alpha,i})$$

$$\mathcal{T}_{\alpha,i}[r][c \oplus 1] := \text{Enc}_{\mathcal{L}_{\alpha,1}^{(0)}, \mathcal{L}_{\alpha,j}^{(1)}}^{(2)}(\tau_{\alpha,i,r,c \oplus 1}^{(k)}, \mathcal{L}_{\alpha,\text{left},i}^{(1)} \parallel \mathcal{J}_{\alpha,i})$$

$$\mathcal{T}_{\alpha,i}[r \oplus 1][c] := \text{Enc}_{\mathcal{L}_{\alpha,1}^{(1)}, \mathcal{L}_{\alpha,j}^{(0)}}^{(2)}(\tau_{\alpha,i,r \oplus 1,c}^{(k)}, \mathcal{J}_{\alpha,i} \parallel \mathcal{L}_{\alpha,\text{right},i}^{(0)})$$

$$\mathcal{T}_{\alpha,i}[r \oplus 1][c \oplus 1] := \text{Enc}_{\mathcal{L}_{\alpha,1}^{(1)}, \mathcal{L}_{\alpha,j}^{(1)}}^{(2)}(\tau_{\alpha,i,r \oplus 1,c \oplus 1}^{(k)}, \mathcal{J}_{\alpha,i} \parallel \mathcal{L}_{\alpha,\text{right},i}^{(1)})$$

- **β -translation tables** $\{\mathcal{T}_{\beta,i}\}_{i \in [m]}$. Let $r := \text{Lsb}(\mathcal{L}_{\alpha,1}^{(0)})$.

For each $i \in [m]$: Let $c := \text{Lsb}(\mathcal{L}_{\beta,i}^{(0)})$ and define

$$\mathcal{T}_{\beta,i}[r][c] := \text{Enc}_{\mathcal{L}_{\alpha,1}^{(0)}, \mathcal{L}_{\beta,i}^{(0)}}^{(2)}(\tau_{\beta,i,r,c}^{(k)}, \mathcal{L}_{\beta,\text{left},i}^{(0)} \parallel \mathcal{J}_{\beta,i})$$

$$\mathcal{T}_{\beta,i}[r][c \oplus 1] := \text{Enc}_{\mathcal{L}_{\alpha,1}^{(0)}, \mathcal{L}_{\beta,i}^{(1)}}^{(2)}(\tau_{\beta,i,r,c \oplus 1}^{(k)}, \mathcal{L}_{\beta,\text{left},i}^{(1)} \parallel \mathcal{J}_{\beta,i})$$

$$\mathcal{T}_{\beta,i}[r \oplus 1][c] := \text{Enc}_{\mathcal{L}_{\alpha,1}^{(1)}, \mathcal{L}_{\beta,i}^{(0)}}^{(2)}(\tau_{\beta,i,r \oplus 1,c}^{(k)}, \mathcal{J}_{\beta,i} \parallel \mathcal{L}_{\beta,\text{right},i}^{(0)})$$

$$\mathcal{T}_{\beta,i}[r \oplus 1][c \oplus 1] := \text{Enc}_{\mathcal{L}_{\alpha,1}^{(1)}, \mathcal{L}_{\beta,i}^{(1)}}^{(2)}(\tau_{\beta,i,r \oplus 1,c \oplus 1}^{(k)}, \mathcal{J}_{\beta,i} \parallel \mathcal{L}_{\beta,\text{right},i}^{(1)})$$

Figure 7: Redefined $\Pi_{\text{napPRF}}^{(k)}.\text{KeyGen}$ over $\{0, 1\}^k \rightarrow \{0, 1\}^m$.

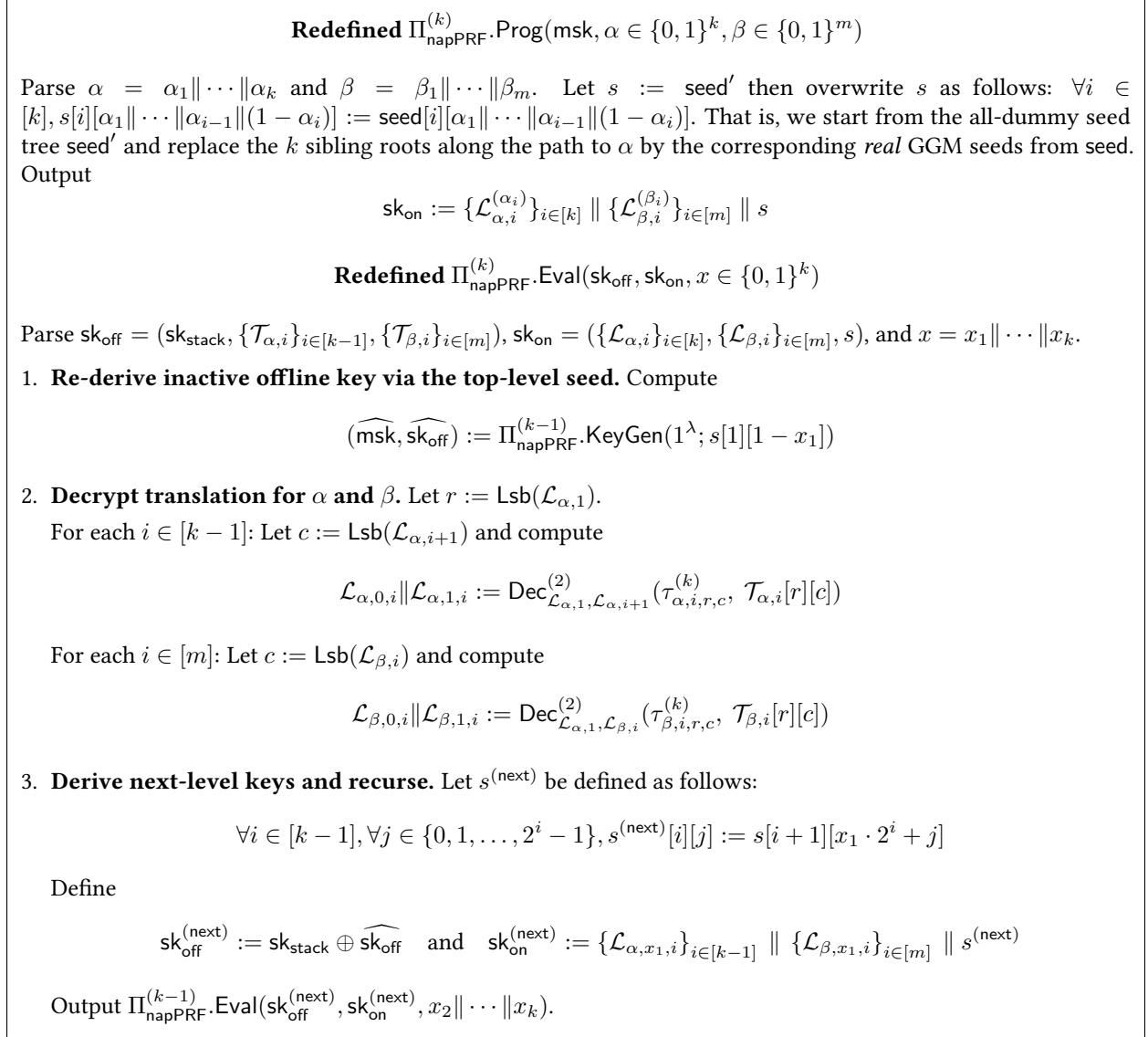


Figure 8: Redefined $\Pi_{\text{napPRF}}^{(k)}.\text{Prog}$ and $\Pi_{\text{napPRF}}^{(k)}.\text{Eval}$ over $\{0, 1\}^k \rightarrow \{0, 1\}^m$.

Let $\text{REAL}_\alpha^{(k)}$ be the distribution of $(\text{sk}_{\text{off}}, \text{sk}_{\text{on}})$ for uniform $\beta \xleftarrow{\$} \{0, 1\}^m$. The simulator $\text{Sim}^{(k)}$ runs $(\text{msk}, \text{sk}_{\text{off}}) \leftarrow \text{KeyGen}^{(k)}(1^\lambda)$, samples $\beta \xleftarrow{\$} \{0, 1\}^m$, and outputs $(\text{sk}_{\text{off}}, \text{Prog}^{(k)}(\text{msk}, 0^k, \beta))$; it suffices to show $\text{REAL}_\alpha^{(k)} \approx_c \text{REAL}_{0^k}^{(k)}$.

We use the same hybrid sequence as Theorem 8. First, replace the top-level master expansion $G(s_{\text{master}})$ by uniform randomness (PRG security). Second, apply a PRF hybrid (using unique tweaks) to replace every non-decryptable translation-table entry by an independent uniform string. Hence, as before, the view can depend on α only through the revealed labels and the unique decryptable cells of the translation tables.

Then, Step 3a (hiding suffix $\alpha_2 \parallel \dots \parallel \alpha_k$) proceeds exactly as in Theorem 8.

Step 3b (hiding α_1) is the same “swap-the-halves” argument: after cleaning/hybridizing down the candidate active path, each opened translation-cell plaintext is indistinguishable from two independent uniform λ -bit strings, whose distribution is invariant under swapping left/right halves.

At this point, the new seed tree s is handled similarly. Although s may be correlated with sk_{off} through the recursive keys, it is α -invariant: it is obtained by starting from the dummy tree seed' and overwriting a fixed number of positions by seeds derived from s_{master} ; after the standard PRG “cleaning” along the hidden active path (as in the base case), the overwritten seeds are indistinguishable from fresh uniform strings, so the marginal distribution of s does not depend on α . $\square$

Suffix invariance. Clearly, the recursive transformation preserves suffix invariance (cf. Section 7.2). Together with Theorem 13, this implies that the final (redefined) scheme $\Pi_{\text{napPRF}}^{(n)}$ is a short-key offline-online sipPRF. When emphasizing this suffix-invariant structure, we denote the same scheme by $\Pi_{\text{sipPRF}}^{(n)}$.

E The Improved LogStack Garbling Scheme

For completeness, following prior SGC schemes [HK20a, HK21], we formalize the improved LogStack — where the demultiplexer is instantiated with our tailored offline-online sipPRFs (see Appendix D) — as a *garbling scheme* [BHR12]. Here, we use a tailored sipPRF variant in which the decoding bits are omitted, so evaluations output GC labels directly. We use $\Pi^{(n)} = (\text{KeyGen}^{(n)}, \text{Prog}^{(n)}, \text{Eval}^{(n)})$ to denote the sipPRF over domain $\{0, 1\}^n$, where each evaluation returns I λ -bit labels. The parameters n and I are fixed by the garbling scheme.

Our garbling scheme ILS is the tuple of procedures defined in Figures 9 to 12:

$$(\text{ILS.ev}, \text{ILS.Ev}, \text{ILS.Gb}, \text{ILS.En}, \text{ILS.De})$$

For simplicity, we omit any subprocedures that are identical to those in [HK21]. We make the following remarks.

- **Circuit syntax and semantics.** The circuit supports three recursive forms:
 1. Netlist: a standard Boolean circuit given as a netlist of AND/XOR gates.
 2. Seq: sequential composition of two components, i.e., $\mathcal{C}_1 \circ \mathcal{C}_0$.
 3. Cond: an n -bit conditional/dispatch over a vector $\vec{C}$ of 2^n subcircuits, where the input is parsed as $\alpha \parallel \vec{x}'$ and α selects the active branch.
- **The base Yao garbling scheme.** We model standard Yao garbling as the Base scheme, instantiated with an OWF-based point-and-permute construction. Alternatively, Base can use *free XOR* [KS08] and *half gates* [ZRE15] under the additional assumption of a *circular correlation-robust hash function* [CKKZ12].

- **Garble a conditional vector.** The improved algorithm GbCond for garbling a conditional vector is defined in Figure 10. Relative to GbCond in LogStack [HK21], the key change is how we implement the demultiplexer: we invoke our tailored offline-online sipPRF to generate the branch input labels. In particular, the offline key sk_{off} is included in the garbling material, while the master key msk contains the garbling encodings (i.e., the wire label pairs).

We introduce a new subprocedure ComputeGarbageInput (defined in Figure 11) that helps G generate all dummy input labels by evaluating the sipPRF to enumerate all off-point outputs, stored in Λ_{in} . Its running time is $\mathcal{O}(B \log^2 B)$, which is lower order compared to evaluating $\mathcal{O}(B \log B)$ dummy GCs.

- **Evaluate a conditional vector.** The improved algorithm EvCond for evaluating a conditional vector is defined in Figure 12. Relative to EvCond in LogStack [HK21], the key change is how we implement the demultiplexer: we invoke our tailored offline-online sipPRF to generate the branch input labels. In particular, it invokes our offline-online sipPRF's FullEval procedure.

Finally, we obtain the following theorem.

Theorem 14 (Improved LogStack). *Assuming one-way functions (via black-box use), the garbling scheme ILS is correct, authentic, oblivious, and private in the sense of standard definitions in [BHR12].*

Proof sketch. The proof follows the analysis of LogStack and reuses its simulators essentially verbatim. The only change is in the demultiplexer: wherever LogStack samples values derived from its PRF-based routing gadget, we instead sample them using the index-hiding simulators for our tailored offline-online sipPRFs. All other components are unchanged, so correctness, authenticity, obliviousness, and privacy reduce to the corresponding guarantees established in [HK21] together with index-hiding of the sipPRF. $\square$


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ILS.ev( $\mathcal{C}, \vec{x}$ ) :
  ▷ What are the circuit semantics?
switch  $\mathcal{C}$  :
  case Netlist( $\cdot$ ) : return Base.ev( $\mathcal{C}, \vec{x}$ )
  case Seq( $\mathcal{C}_0, \mathcal{C}_1$ ) : return ILS.ev( $\mathcal{C}_1$ , ILS.ev( $\mathcal{C}_0, \vec{x}$ ))
  case Cond( $\vec{\mathcal{C}}$ ) :
    ▷ Split branch index from input.
     $\alpha \mid \vec{x}' \leftarrow \vec{x}$ 
    ▷ Run the active branch.
    return ILS.ev( $\vec{\mathcal{C}}[\alpha], \vec{x}'$ )

ILS.Ev( $\mathcal{C}, M, \vec{X}$ ) :
  ▷ How does E evaluate the GC?
switch( $\mathcal{C}$ ) :
  case Netlist( $\cdot$ ) : return Base.Ev( $\mathcal{C}, M, \vec{X}$ )
  case Seq( $\mathcal{C}_0, \mathcal{C}_1$ ) :
     $M_0 \mid M_{tr} \mid M_1 \leftarrow M$ 
    return ILS.Ev( $\mathcal{C}_1, M_1$ , trans.Ev( $M_{tr}$ , ILS.Ev( $\mathcal{C}_0, M_0, \vec{X}$ )))
  case Cond( $\vec{\mathcal{C}}$ ) : return EvCond( $\vec{\mathcal{C}}, M, \vec{X}$ )

ILS.En( $e, \vec{x}$ ) :
  ▷ How do inputs map to labels?
  ▷ This works for all projective schemes:
   $\vec{X} \leftarrow \{\}$ 
  for  $i \in 0..\text{InpSize}(\mathcal{C})-1$  :
    ( $X^{(0)}, X^{(1)}$ )  $\leftarrow e[i]$ 
    if  $\vec{x}[i] = 0$  :  $\{\vec{X}[i] \leftarrow X^{(0)}\}$  else :  $\{\vec{X}[i] \leftarrow X^{(1)}\}$ 
  return  $\vec{X}$ 

ILS.Gb( $1^\lambda, \mathcal{C}, S$ )
  ▷ How does G garble the GC?
  ▷  $S$  is an explicit seed.
switch  $\mathcal{C}$  :
  case Netlist( $\cdot$ ) :
    return Base.Gb( $1^\lambda, \mathcal{C}, S$ )
  case Seq( $\mathcal{C}_0, \mathcal{C}_1$ ) :
    ▷ Derive seeds for two circuits.
     $S_0 \leftarrow F_S(0)$ 
     $S_1 \leftarrow F_S(1)$ 
    ( $M_0, e_0, d_0$ )  $\leftarrow$  ILS.Gb( $1^\lambda, \mathcal{C}_0, S_0$ )
    ( $M_1, e_1, d_1$ )  $\leftarrow$  ILS.Gb( $1^\lambda, \mathcal{C}_1, S_1$ )
    ▷ Labels out of  $\mathcal{C}_0$  must be translated
    ▷ to labels into  $\mathcal{C}_1$ .
     $M_{tr} \leftarrow \text{trans.Gb}(d_0, e_1)$ 
     $M \leftarrow M_0 \mid M_{tr} \mid M_1$ 
    return ( $M, e_0, d_1$ )
  case Cond( $\vec{\mathcal{C}}$ ) : return GbCond( $\vec{\mathcal{C}}, S$ )

ILS.De( $d, \vec{Y}$ ) :
  ▷ How do labels map to outputs?
  ▷ This works for all projective schemes:
   $\vec{y} \leftarrow \{\}$ 
  for  $i \in 0..\text{OutSize}(\mathcal{C})-1$  :
    ( $Y^{(0)}, Y^{(1)}$ )  $\leftarrow d[i]$ 
    if  $\vec{Y}[i] = Y^{(0)}$  :  $\vec{y}[i] \leftarrow 0$ 
    else if  $\vec{Y}[i] = Y^{(1)}$  :  $\vec{y}[i] \leftarrow 1$ 
    else : ABORT
  return  $\vec{y}$ 

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Figure 9: ILS’s garbling scheme. The included algorithms are typical except for the handling of conditionals Ev and Gb delegate the core of our approach: GbCond (Figure 10) and EvCond (Figure 12). This level matches LogStack [HK21].

GbCond($\vec{C}, S$) :

- $B \leftarrow |\vec{C}|, n \leftarrow \log B, I \leftarrow \text{InpSize}(\text{Cond}(\vec{C})), O \leftarrow \text{OutSize}(\text{Cond}(\vec{C}))$
- ▷ Sample output encodings for the conditional.
- $d \leftarrow \text{GenProjection}(S, O)$
- ▷ Derive all ‘good’ and ‘bad’ seeds for the entire tree from tailored sipPRF.
- $s_{\text{master}} \xleftarrow{\$S} \{0, 1\}^\lambda, (\text{msk}, \text{sk}_{\text{off}}) \leftarrow \text{KeyGen}^{(n)}(1^\lambda; s_{\text{master}})$
- ▷ Parse msk into input encodings for the conditional, ‘good’ and ‘bad’ seeds.
- $e \mid \text{seed} \mid \text{seed}' \leftarrow \text{msk}$
- ▷ Parse encoding into encoding of α and encoding of rest of I -bit input.
- $e_\alpha \mid e' \leftarrow e$
- ▷ Garble SortingHat based on the encoding of α , ‘good’ and ‘bad’ seeds.
- $M_{\text{SortingHat}} \leftarrow \text{SortingHat.Gb}(e_\alpha, \text{seed}, \text{seed}')$
- ▷ Construct the stacked material and input encodings for each branch.
- $M_{\text{cond}}, e_{\text{cond}}, d_{\text{cond}} \leftarrow \text{GbSubtreeFromSeed}(\vec{C}, 0, B - 1, s_{\text{master}})$
- ▷ Compute the dummy inputs by evaluating the sipPRF over the entire domain.
- ▷ Here G would try every programmed point α , and evaluate on every $x' \neq \alpha$.
- ▷ The suffix invariance ensures that many cross α evaluations would coincide.
- $\Lambda_{\text{in}} \leftarrow \text{ComputeGarbageInput}(n, I, \text{sk}_{\text{off}}, \text{seed}, \text{seed}', s_{\text{master}})$
- ▷ Compute all possible garbage outputs.
- $\Lambda_{\text{out}} \leftarrow \text{ComputeGarbage}(\vec{C}, M_{\text{cond}}, \Lambda_{\text{in}}, \text{seed}, \text{seed}')$
- ▷ The multiplexer collects garbage outputs.
- $M_{\text{mux}} \leftarrow \text{Mux.Gb}(e_\alpha, d, d_{\text{cond}}, \Lambda_{\text{out}})$
- return ($M_{\text{SortingHat}} \mid \text{sk}_{\text{off}} \mid M_{\text{cond}} \mid M_{\text{mux}}, e, d$)

Figure 10: The improved algorithm for garbling a conditional vector. Relative to GbCond in LogStack [HK21], the key change is how we implement the demultiplexer: we invoke our tailored offline-online sipPRF (over $\{0, 1\}^n \rightarrow \{0, 1\}^I$, with decoding bits omitted so that Eval outputs wire labels) to generate the branch input labels. In particular, the offline key sk_{off} is included in the returned garbled material. W.l.o.g., we assume B is a power of two. The procedures GenProjection, SortingHat, GbSubtreeFromSeed, and Mux are exactly as in LogStack. We introduce a new subroutine, ComputeGarbageInput (defined in Figure 11), which computes the dummy input labels for each branch (i.e., the off-point evaluation outputs of our sipPRF). Consequently, LogStack’s subroutine ComputeGarbage [HK21] requires a minor corresponding modification: when evaluating the dummy garbled circuit, the evaluator supplies the branch-specific dummy labels stored in Λ_{in} . We omit this change from the formalization since it is straightforward. We remark that KeyGen of sipPRF and GbSubtreeFromSeed are invoked using the same master seed s_{master} , ensuring that the sipPRF output at the programmed point coincides with the corresponding real GC labels.

$\text{ComputeGarbageInput}(n, I, \text{sk}_{\text{off}}, \text{seed}, \text{seed}', s) :$
 $\triangleright$ We first define a recursive subprocedure.
 $\text{ComputeGarbageInput}'(\ell, i, j, \text{sk}_{\text{off}}, \text{seed}, \text{seed}', s) :$
 if $i = j :$
 $\triangleright$ Base case: do nothing.
 else :
 $k \leftarrow \lfloor (i + j)/2 \rfloor, \quad d \leftarrow j - i + 1$
 $\triangleright$ Parse the stacked next-level offline key.
 $\text{sk}_{\text{stack}} \mid_{\perp} \leftarrow \text{sk}_{\text{off}}$
 $\triangleright$ Compute the real next-level offline keys.
 $(\omega, \text{sk}_{\text{off}, \text{left}}) \leftarrow \text{KeyGen}^{(n-\ell)}(1^\lambda; \text{seed}[\ell][2i/d])$
 $\text{sk}_{\text{off}, \text{right}} \leftarrow \text{sk}_{\text{stack}} \oplus \text{sk}_{\text{off}, \text{left}}$
 $\triangleright$ Compute the dummy next-level offline keys.
 $(\omega, \text{sk}'_{\text{off}, \text{left}}) \leftarrow \text{KeyGen}^{(n-\ell)}(1^\lambda; \text{seed}'[\ell][2i/d])$
 $(\omega, \text{sk}'_{\text{off}, \text{right}}) \leftarrow \text{KeyGen}^{(n-\ell)}(1^\lambda; \text{seed}'[\ell][2i/d + 1])$
 $\text{sk}''_{\text{off}, \text{left}} \leftarrow \text{sk}_{\text{stack}} \oplus \text{sk}'_{\text{off}, \text{right}}, \quad \text{sk}''_{\text{off}, \text{right}} \leftarrow \text{sk}_{\text{stack}} \oplus \text{sk}'_{\text{off}, \text{left}}$
 $\triangleright$ Re-derive the dummy labels as in sipPRF KeyGen.
 $\vec{\mathcal{J}} \xleftarrow{\$s} \{0, 1\}^{(n-\ell+I)\lambda}$
 $\triangleright$ Setup dummy seeds for sipPRF Eval.
 $\text{seed}''_{\text{left}}[\chi][\gamma] \leftarrow \text{seed}'[\ell + \chi][i/2^{n-\ell-\chi} + \gamma], \forall \chi \in [n - \ell], \gamma \in \{0, 1, \dots, 2^\chi - 1\}$
 $\text{seed}''_{\text{right}}[\chi][\gamma] \leftarrow \text{seed}'[\ell + \chi][(k + 1)/2^{n-\ell-\chi} + \gamma], \forall \chi \in [n - \ell], \gamma \in \{0, 1, \dots, 2^\chi - 1\}$
 $\triangleright$ Fill dummy input labels using sipPRF FullEval.
 $\Lambda_{\text{in}}[i \dots k][n - \ell] \leftarrow \text{FullEval}^{(n-\ell)}(\text{sk}''_{\text{off}, \text{left}}, \vec{\mathcal{J}}, \text{seed}''_{\text{left}})$
 $\Lambda_{\text{in}}[k + 1 \dots j][n - \ell] \leftarrow \text{FullEval}^{(n-\ell)}(\text{sk}''_{\text{off}, \text{right}}, \vec{\mathcal{J}}, \text{seed}''_{\text{right}})$
 $\triangleright$ Recurse into left and right.
 $\text{ComputeGarbageInput}'(\ell + 1, i, k, \text{sk}_{\text{off}, \text{left}}, \text{seed}, \text{seed}', \text{seed}[\ell][2i/d])$
 $\text{ComputeGarbageInput}'(\ell + 1, k + 1, j, \text{sk}_{\text{off}, \text{right}}, \text{seed}, \text{seed}', \text{seed}[\ell][2i/d + 1])$
 $\triangleright$ Start recursive process from the top level.
 $\text{ComputeGarbageInput}'(1, 0, 2^n - 1, \text{sk}_{\text{off}}, \text{seed}, \text{seed}', s_{\text{master}})$
 return Λ_{in}

Figure 11: A new subroutine $\text{ComputeGarbageInput}$. It is invoked within GbCond (Figure 10) to compute the dummy input labels for each branch (i.e., the off-point evaluation outputs of our sipPRF). For each branch $t \in \{0, \dots, 2^n - 1\}$, the resulting dummy labels are stored in $\Lambda_{\text{in}}[t]$; moreover, by suffix invariance, each $\Lambda_{\text{in}}[t]$ falls into one of only n possibilities, denoted $\Lambda_{\text{in}}[t][0], \dots, \Lambda_{\text{in}}[t][n - 1]$. Note that the dummy labels $\vec{\mathcal{J}}$ must be derived from the seed s according to the corresponding $\text{KeyGen}^{(\cdot)}$ procedure; we omit the details for simplicity.

$\text{EvCond}(\vec{C}, M, X) :$
 $B \leftarrow |\vec{C}|, n \leftarrow \log B, I \leftarrow \text{InpSize}(\text{Cond}(\vec{C})), O \leftarrow \text{OutSize}(\text{Cond}(\vec{C}))$
 $\triangleright$ Parse the active branch index from the rest of the input.
 $X_\alpha \mid X' \leftarrow X$
 $\triangleright$ Parse material for gadgets and body of conditional.
 $M_{\text{SortingHat}} \mid \text{sk}_{\text{off}} \mid M_{\text{cond}} \mid M_{\text{mux}} \leftarrow M$
 $\triangleright$ Run SortingHat to compute all of E's seeds.
 $\text{es} \leftarrow \text{SortingHat.Ev}(X_\alpha, M_{\text{SortingHat}})$
 $\triangleright$ Run the sipPRF FullEval to compute input for each branch C_i .
 $\vec{X}_{\text{cond}} \leftarrow \text{FullEval}^{(n)}(\text{sk}_{\text{off}}, X_\alpha, X', \text{es})$

$\triangleright$ We define a recursive subprocedure that evaluates C_i to C_j using material M .
 $\text{EvCond}'(i, j, M) :$
 if $i = j$:
 $\triangleright$ Base case: compute output by evaluating the branch normally.
 $\triangleright$ This base case corresponds to guess = i .
 $\triangleright$ Accumulate output labels into the vector $\vec{Y}_{\text{cond}}$ (for later garbage collection).
 $\vec{Y}_{\text{cond}}[i] \leftarrow \text{Base.Ev}(C_i, M, \vec{X}_{\text{cond}}[i])$
 else :
 $k \leftarrow \lfloor (i + j)/2 \rfloor, \quad d \leftarrow j - i + 1$
 $\triangleright$ Garble the right subtree using the available seed,
 $\triangleright$ unstack, and recursively evaluate the left subtree.
 $M_{k+1,j,\omega,\omega} \leftarrow \text{GbSubtreeFromSeed}(\vec{C}, k+1, j, \text{es}[B - \log(d) + 1][2i/d + 1])$
 $\text{EvCond}'(i, k, M \oplus M_{k+1,j})$
 $\triangleright$ Symmetrically evaluate the right subtree.
 $M_{i,k,\omega,\omega} \leftarrow \text{GbSubtreeFromSeed}(\vec{C}, i, k, \text{es}[B - \log(d) + 1][2i/d])$
 $\text{EvCond}'(k+1, j, M \oplus M_{i,k})$

$\triangleright$ Start recursive process from the top level.
 $\text{EvCond}'(0, B-1, M_{\text{cond}})$
 $\triangleright$ Eliminate garbage and propagate $\vec{Y}_\alpha$ via the multiplexer.
 return $\text{Mux.Ev}(X_\alpha, \vec{Y}_{\text{cond}}, M_{\text{mux}})$

Figure 12: The improved algorithm for evaluating a conditional vector. Relative to EvCond in LogStack [HK21], the key change is how we implement the demultiplexer: we invoke our tailored offline-online sipPRF (over $\{0, 1\}^n \rightarrow \{0, 1\}^I$, with decoding bits omitted so that Eval outputs wire labels) to generate the branch input labels. In particular, the offline key sk_{off} is included in the input garbled material. W.l.o.g., we assume B is a power of two. The procedures SortingHat, Base, GbSubtreeFromSeed, and Mux are exactly as in LogStack.